SMALL-LEVEL CONVERSE THEOREMS

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Abstract. We give the small-level converse arguments accompanying *Towards a twistless converse theorem*. Together with the classical small-level results, they yield membership in the two-trace space $M_k^\circ(\Gamma_0(N))$ and the exact count for $1 \le N \le 36$, $N \notin \{29, 31\}$, in every positive even weight. The new level arguments use Taylor–Fricke identities, finite systems of additive Mellin transforms, exact polynomial certificates, and holomorphic rigidity. The common passage from modularity to the trace conditions and the eigenform count is recorded separately from the individual levels.

Contents

1. The common modularity and counting conclusion 3
2. The thirteen additional levels 4
Deterministic audit of the small-level polynomial certificates 4
3. A closed Taylor–Fricke system 5
4. The converse theorem at level 19 7
4.1. The two cusps of denominator 10 7
4.2. A scalar quotient for the quadratic twist 8
4.3. Continuation of the auxiliary grid 8
4.4. Pole cancellation 10
4.5. Modularity and cuspidality 13
5. Level 21: meromorphic continuation on the 42-grid 14
6. Level 21: pole exclusion on the 42-grid 16
7. Level 22: uniform continuation of the 66-grid 17
8. Level 22: all grid poles are simple cusp poles 18
8.1. Audit of higher-order poles at the exceptional weight 20
9. Excluding the weight pole at level 22 21
10. An extra-prime leading-kernel certificate in weight two 22
11. Upper integer poles on the enlarged exceptional grids 22
12. The quadratic twist modulo five in the exceptional level-21 case 23
13. Removal of the remaining level-21 weight-two pole 24
14. The quadratic twist modulo five in the exceptional level-22 case 25
15. Removal of the remaining level-22 weight-two pole 27
16. An entire quadratic twist for every parameter at levels 21 and 22 27
17. Completion at level 21 when the Hecke-2 eigenvalue is nonzero 29
18. Additional exact reductions when $N = 21$ and $\lambda_2 = 0$ 30
19. The zero Hecke-2 eigenvalue at level 21 30
20. Level 22: the paired denominator-five transformations 31
21. Completion of the level-22 converse theorem 32
22. Level 25: uniform continuation of the 150-grid 33
23. A level-25 first-jet obstruction 35
24. The positive-Fricke denominator-five pole sector at level 25 37

25. A source-free quotient for the level-25 prime-seven extension 38
Independent audit of the level-25 zero-λ_2 certificates 40
26. Exact algebra for level-25 modularity 41
26.1. A local Mellin criterion for the level-25 transformations 42
27. The second positive-sign level-25 polar chart 44
28. The first positive-sign level-25 polar chart 46
28.1. Exact exclusion of the degree-67 exponential locus 47
29. The second negative-sign level-25 polar chart 48
30. Two exceptional strata in the first negative level-25 chart 50
31. The denominator-zero branch in the first negative level-25 chart 51
32. Complete negative-sign pole control at level 25 51
33. Level 26: continuation from a shifted Taylor relation 55
Level 26: a denominator-156 quotient 56
34. Level 26: auxiliary prime 5 and an enlarged closed system 57
35. Level 26: the weight-point obstruction for every even weight 58
35.1. Level 26: exclusion of poles for both bad-13 signs 59
35.2. An exact modulus exclusion for a finite algebraic locus 62
35.3. Level 26: modularity from the simple-pole restriction 63
36. The converse theorem at level 28 66
Known transformations and the exact finite systems 66
The constant odd mode and meromorphic continuation 67
Restriction and removal of poles 68
A genuine new transformation and modularity 69
Cuspidality 69
37. A complete converse proof at level 30 70
37.1. The modulus-six transformation 70
37.2. The modulus-four transformation 71
37.3. Exact generation and cuspidality 71
38. An exact pole-elimination certificate at level 30 72
39. Levels 33 and 34: a modulus-six elliptic argument 74
39.1. Uniform auxiliary continuation at level 33 75
39.2. A closed continuation subsystem at level 34 75
40. Level 34: removal of the quadratic-twist poles using prime 5 76
41. Completion of the level-34 converse theorem for all eigenvalues 78
42. Level 33: the quadratic twist, pole cancellation, and modularity 79
43. Level 35: a closed subsystem and meromorphic character twists 81
43.1. Level 35: the modulus-six elliptic relation 82
43.2. The quadratic twist modulo three 82
43.3. Uniform continuation of the complete 210-grid 83
43.4. Exclusion of shifted poles away from the weight point at level 35 83
43.5. Level 35: the unshifted leading residue 85
43.6. Level 35: the positive-product unshifted case 88
43.7. The exceptional rightmost point at level 35 89
44. Completion of the level-35 converse theorem 90
45. A converse theorem at level 32 92
45.1. The odd Fourier support and the Hecke relation 93
45.2. An infinite elliptic transformation kills the defect 93
45.3. A finite-character modular form at level eight 94
45.4. Holomorphy and vanishing at every cusp 95

45.5. Explicit generation certificate 95
46. A converse theorem at level 36 95
47. The converse theorem at level 27 97
47.1. The precise analytic input and its certificates 98
47.2. Vertical bounds and sectorial cusp asymptotics 100
47.3. The obstruction and its exact affine Hecke relation 101
47.4. Dilation rigidity and cuspidality 102
48. A finite-pole-chain consequence of Taylor–Fricke 103
49. The general modular identification and exact count 104
References 107

1. The common modularity and counting conclusion

Throughout, k is positive and even. The coefficients f_n are normalized:

$$L_f(s) = \sum_{n\geq 1} f_n n^{-s}, \qquad F(z) = \sum_{n\geq 1} f_n n^{(k-1)/2} e^{2\pi i n z}.$$

The completion and Euler factors are exactly those defining $\mathscr{E}(N,k)$ in [3]; the normalized functional equation has centre $1/2$. No additional twist or growth-rate hypothesis is imposed. The level calculations also use Mellin transforms of F; translating their integration variable by $(k-1)/2$ recovers the normalized variable.

Define $M_k^{\mathrm{new}}(\Gamma_0(N))$ by requiring that the traces of both F and $F|W_N$ to $\Gamma_0(N/p)$ vanish for every prime $p \mid N$. Its subspace $M_k^\circ(\Gamma_0(N))$ consists of forms vanishing at 0 and ∞.

Theorem 1.1 (The small-level theorem of the accompanying paper)**.** *If $1 \leq N \leq 36$ and $N \notin \{29, 31\}$, every sequence in $\mathscr{E}(N,k)$ gives a form in $M_k^\circ(\Gamma_0(N))$, and*

$$\#\mathscr{E}(N,k) = \dim M_k^\circ(\Gamma_0(N)).$$

Proposition 1.2 (The common classical reduction)**.** *At every level, $M_k^\circ(\Gamma_0(N))$ has a basis of normalized simultaneous Hecke eigenforms whose coefficient sequences belong to $\mathscr{E}(N,k)$. Conversely, the members of $\mathscr{E}(N,k)$ that are modular are precisely these normalized eigenforms. Consequently modularity of every member of $\mathscr{E}(N,k)$ implies the stated exact count.*

Proof. Use the cuspidal newform decomposition and the primitive Eisenstein packets. The two trace conditions are imposed packet by packet: traces, Fricke and Hecke operators preserve the decomposition, and distinct primitive packets have distinct Hecke systems away from the level. The local trace calculation, including its weight-two exceptional case, is given in Lemma 49.1 below. It identifies the simultaneous trace kernels with the cuspidal newspace and the Eisenstein eigenlines described there. These lines have nonzero first coefficient, so normalize it to one. Their good Euler factors are $1 - f_p p^{-s} + p^{-2s}$; their ramified factors satisfy $f_p^2 = p^{-1}$ for $p \parallel N$ and $f_p = 0$ for $p^2 \mid N$. They are Fricke eigenforms. Vanishing at both indicated cusps gives an entire finite-order completion by splitting the Mellin integral and using the Fricke transformation. Thus their sequences lie in $\mathscr{E}(N,k)$.

Conversely, the Euler recurrences make a modular member a simultaneous Hecke eigenform, including the ramified operators, and its functional equation gives its Fricke eigenvalue. Decompose it into primitive packets and apply the same local trace calculation to these eigenvalues. The prescribed ramified factors and the Fricke condition select the two-trace kernels; the Fourier expansion and its Fricke image have zero constant terms. In weight two one must retain the exceptional Eisenstein eigenlines rather than replace the trace kernel by the cuspidal newspace. The packet calculation

therefore places the form in M_k°. Distinct normalized eigenlines have distinct coefficient sequences, which proves both the intersection assertion and the count. □

For $N \le 4$ use Hecke's theorem [6]; for $N \in \{5, \dots, 18, 20, 23, 24\}$ use [4, 5, 1]. The remaining thirteen levels are treated below. Thus the level-specific task is modularity; the trace conditions and the exact count follow from the common reduction at every level.

2. The thirteen additional levels

Level	Result established here
19	Complete converse: $f \in S_k(\Gamma_0(19))$ for all permitted parameters.
21	Complete converse: $f \in S_k(\Gamma_0(21))$ for all permitted parameters.
22	Complete converse: $f \in S_k(\Gamma_0(22))$ for all permitted parameters.
25	Complete converse: $f \in M_k(\Gamma_0(25))$ for all permitted parameters; no cuspidality assertion.
26	Complete converse: $f \in S_k(\Gamma_0(26))$ for all permitted parameters.
27	Complete converse: $f \in S_k(\Gamma_0(27))$ for all permitted parameters.
28	Complete converse: $f \in S_k(\Gamma_0(28))$ for all permitted parameters.
30	Complete converse: $f \in S_k(\Gamma_0(30))$ for all permitted weights and signs.
32	Complete converse: $f \in S_k(\Gamma_0(32))$ for all permitted parameters.
35	Complete converse: $f \in S_k(\Gamma_0(35))$ for all permitted parameters.
33	Complete converse: $f \in S_k(\Gamma_0(33))$ for all permitted parameters.
34	Complete converse: $f \in S_k(\Gamma_0(34))$ for all permitted parameters.
36	Complete converse: $f \in S_k(\Gamma_0(36))$ for all permitted parameters.

The target conclusion is $f \in M_k$ at level 25, and cuspidality at each of the other twelve levels. In particular, possible cusp-constant terms at level 25 must not be discarded merely to force entireness.

Theorem 2.1 (The thirteen small-level converses)**.** *Let k be a positive even integer and let*

$$N \in \{19, 21, 22, 25, 26, 27, 28, 30, 32, 33, 34, 35, 36\}.$$

Suppose (f_n) satisfies exactly the Euler-product, ramified-coefficient, entire finite-order continuation, and functional-equation hypotheses defining $\mathscr{E}(N,k)$ in [3]*. Then*

$$\sum_{n\ge 1} f_n n^{(k-1)/2} e^{2\pi i n z} \in \begin{cases} M_k(\Gamma_0(25)), & N = 25, \\ S_k(\Gamma_0(N)), & N \ne 25. \end{cases}$$

No reality, Ramanujan, nonzero unramified-eigenvalue, or additional twisted-functional-equation hypothesis is imposed.

The sections below prove the assertion separately at every level, including the exceptional sign, eigenvalue, and weight cases.

Exact symbolic certificates are proofs of their stated finite algebraic identities. A numerical rank at a sampled eigenvalue is only exploratory. Neither kind of calculation replaces the analytic continuation argument or the required modular transformation laws.

Deterministic audit of the small-level polynomial certificates

This audit covers the final computational dependencies of levels $19, 21, 22, 28, 30, 33, 34, 35$. The later level-25 and level-26 computations have their own independent source audits, recorded in `level25_complete_audit.json` and `level26_complete_audit.json`. The installed Magma documentation (`text1313.htm`, lines 327–338 and 556–564) states that the default rational global-modular Gröbner algorithm is Monte Carlo. Setting `SetGBGlobalModular(false)` changes to

step-by-step modular F4 over the rationals and affects internal ideal computations as well. All reruns described below set this flag globally and assert its value before constructing any ideals. Original proof sources are unchanged.

The affected final dependencies are exactly the four level-19 checks (equal signs, generic and gap zero; opposite signs, generic and $G = 0$) and the two level-35 unshifted-residue checks. The other listed levels use exact SymPy polynomial arithmetic, rational matrix calculations, exact root isolation, or finite group enumeration in their final proofs. The level-35 positive-gap-one Magma check uses polynomial gcd and determinants, not Gröbner bases. Exploratory scripts that were not used in the proofs are outside this audit.

All six final dependencies now have deterministic certificates. Five direct replays completed: the level-19 equal-sign gap-zero ideal is the unit ideal; the opposite-sign generic ideal has the exact recorded basis; the opposite-sign $G = 0$ ideal is the unit ideal; and both level-35 scripts pass every residue-branch ideal membership and saturation assertion. The equal-sign gap-zero rerun adds an explicit unit-ideal assertion to the original script, which only printed its computed basis. Machine-readable source hashes and result records are in `deterministic_gb_audit_results.json`.

The level-19 equal-sign generic dependency is replaced by a smaller exact certificate, rather than a replay of its former large basis. The source `level19_exact_membership_sparse_prepare.py` constructs polynomial four-row syzygies from the original Taylor matrix, checks their annihilation identities, and verifies the compactification by inverse substitution. A fresh independent source reconstruction checks all these identities and the original fifteen proportional-row constraints; its manifest is `level19_rootind_sparse_manifest.json`.

The standalone verifier `level19_sparse_poles_final_certificate.m` uses those fifteen constraints and six sparse syzygies. With global modular reconstruction disabled, its unsaturated rational ideal has an explicitly asserted six-element basis. Two basis elements are

$$(Y + 1)(X - 1)(X - 4), \qquad (Y + 1)(A + X - 1).$$

Here $X = x^2$, $Y = y^2$, and $A = ax$. The original generic assumptions give $Y + 1 \neq 0$ and $X - 1 \neq 0$, hence $X = 4$ and $A = -3$. The gap relation $A = -X/L - L$, $L = 2^n > 0$, then gives $3 = 4/L + L \geq 4$, a contradiction. No additional J or D nonvanishing assumptions, saturation computations, or finite-field inferences are needed in this final certificate. Its standalone run completed in 1.98 seconds. The failed large-basis replays remain recorded as historical unsuccessful attempts and are not proof dependencies.

Each batch is noninteractive, uses one thread and at most 500 MB, and writes its own completion marker after all assertions. Runtime bounds are 45 seconds for the five completed checks and source-export phase; the isolated equal-sign generic full-basis replay has a 120-second bound. Unsuccessful exploratory compact-ideal calculations are not proof dependencies and are not counted as verification successes.

3. A closed Taylor–Fricke system

Throughout, k is a positive even integer, $a_n = n^{(k-1)/2} f_n$, and

$$f(z) = \sum_{n \geq 1} a_n e^{2\pi i n z}, \qquad F(s, r) = (2\pi)^{-s} \Gamma(s) \sum_{n \geq 1} a_n e^{2\pi i n r} n^{-s}.$$

The variable s is classical, with functional-equation center $k/2$. The Euler-product and functional-equation hypotheses are precisely those of [3]. In particular the coefficients have polynomial growth. Put $\ell_q = \operatorname{lcm}(N, q^2)$ and

$$f_{(q)}(z) = \sum_{a \bmod q}^{*} f(z + a/q), \quad F_q(s, r) = \sum_{a \bmod q}^{*} F(s, r + a/q), \quad \eta_q = i^k \epsilon_f \prod_{p \| N,\ p | q} (-\sqrt{p} f_p).$$

The Ramanujan-twist lemma of the manuscript gives $f_{(q)}|_k W_{\ell_q} = \eta_q f_{(q)}$ and entireness of $F_q(s,0)$.

Lemma 3.1 (Taylor–Fricke, with a movable holomorphic remainder)**.** *Fix q and $r \neq 0$, and set $\beta = -1/(\ell_q r)$. There is a constant B such that, for every positive integer M,*

$$F_q(s,r) - \eta_q(\ell_q r^2)^{s-k/2} \sum_{m=0}^{M-1} (i\ell_q r)^m \binom{s+m-k}{m} F_q(s+m,\beta) \tag{1}$$

has a holomorphic continuation to $\Re s > B - M$. Here the binomial coefficient is its polynomial definition. The assertion concerns the continuation of the displayed combination; it does not initially assume continuation of any individual additive twist.

Proof. Write $\ell = \ell_q$ and $g = f_{(q)}$. Its exact transformation law is

$$g(r+iy) = \eta_q \ell^{-k/2} r^{-k} (1+iy/r)^{-k} g\left(\beta + i\frac{y}{\ell r^2(1+iy/r)}\right).$$

Choose a real B in a half-plane of absolute convergence for all the Dirichlet series in question. Mellin inversion of the last factor on $\Re w = B$ makes its integrand, apart from the constant prefactor,

$$F_q(w,\beta)(\ell r^2)^w y^{-w}(1+iy/r)^{w-k}.$$

Expand the last factor through degree $M-1$ in y. Stirling's bound for $\Gamma(B+it)$ gives exponential decay in $|t|$; for sufficiently small fixed $y > 0$, Taylor's integral remainder is bounded by a constant times $y^M(1+|t|)^M e^{cy|t|}$. Choosing the cutoff small enough therefore bounds the integrated remainder by $O(y^{M-B})$. All constants may depend on M, q, r, which is harmless.

Taking the cutoff Mellin transform in y yields a remainder holomorphic on $\Re s > B - M$. Removing the cutoff from each coefficient adds an entire function: the original Fourier series and its derivatives decay exponentially for y bounded away from zero. In the degree-m term set $w = s + m$. Its scalar coefficient is

$$\eta_q \ell^{-k/2} r^{-k} (\ell r^2)^{s+m} (i/r)^m \binom{s+m-k}{m} = \eta_q (\ell r^2)^{s-k/2} (i\ell r)^m \binom{s+m-k}{m}.$$

The equality uses that k is even. This proves the formula. It is the local Mellin version of the Taylor argument in Booker–Krishnamurthy, *Weil's converse theorem with poles*, Lemma 2.3. □

For a positive integer Q, write $V_j(s) = F(s, j/Q)$. When $q \mid Q$, $\ell_q \mid Q^2$, and $t \neq 0$ divides Q^2/ℓ_q, both $r = t/Q$ and its Fricke partner belong to the same Q-grid. Every shifted term in (1) consequently uses only coordinates of $V(s+m)$. The exact Hecke rows on this grid are

$$\sum_{a=0}^{p-1} V_{j+aQ/p}(s) - \lambda_p p^{k/2-s} V_{pj}(s) + \chi_0(p) p^{k-2s} V_{p^2 j}(s) = 0, \quad \lambda_p = \sqrt{p} f_p,$$

for primes $p \mid Q$ and $0 \le j < Q/p$.

Lemma 3.2 (Closed-system continuation)**.** *Suppose a square matrix $A(s)$ consists of leading Taylor rows, entire Ramanujan rows, and exact Hecke rows on the same finite grid, and $\det A(s)$ is not identically zero. Then every coordinate of the grid has meromorphic continuation to $\mathbb{C}$.*

Proof. For arbitrary M the equations have the form

$$A(s)V(s) = \sum_{m=1}^{M-1} A_m(s) V(s+m) + H_M(s),$$

with H_M holomorphic on $\Re s > B - M$. If V is already meromorphic on $\Re s > b$, take M so large that $B - M < b - 1$. The right side is meromorphic on $\Re s > b - 1$, so inversion of $A(s)$ continues V one strip to the left. Iteration starts with the convergent Dirichlet series and proves the assertion.

The same reasoning applies to additional known matrix-invariance Taylor rows, provided all their source and target coordinates lie on the same grid. □

This lemma does not confine poles to a bounded vertical strip: shifted terms can propagate them indefinitely to the left. A separate pole-cancellation argument is essential. Likewise entireness, by itself, is not a functional equation for each individual multiplicative twist.

4. The converse theorem at level 19

Theorem 4.1. *Under the hypotheses of* [3] *at level $N = 19$, the function f belongs to $S_k(\Gamma_0(19))$. This holds for every positive even weight, every allowed root number and ramified sign, and all complex unramified Hecke eigenvalues, including zero.*

The proof first obtains the meromorphic quadratic twist modulo 5, then removes its poles using a finite grid of additive twists and their first Taylor–Fricke relations. Exact rational certificates cover every exceptional specialization separately. No modularity is assumed while constructing these relations. Write

$$a_n = n^{(k-1)/2} f_n, \qquad f(z) = \sum_{n\geq 1} a_n e(nz), \qquad F(s,r) = (2\pi)^{-s}\Gamma(s) \sum_{n\geq 1} a_n e(nr) n^{-s}.$$

Here s is the unnormalized Mellin variable, with center $k/2$. Let $\epsilon = \epsilon_f$, $\eta = i^k\epsilon$, and use the determinant-normalized weight-k slash operator. Thus $f|W_{19} = \eta f$ and $f|T = f$.

4.1. **The two cusps of denominator 10.** The c_2 twist is just $f(z + 1/2)$. Its Fricke relation at conductor 76 and that of f at conductor 19 imply

$$f \left| \begin{pmatrix} 2 & -1 \\ -19 & 10 \end{pmatrix} = f. \right.$$

Indeed, with $t_r = \left(\begin{smallmatrix} 1 & r \\ 0 & 1 \end{smallmatrix}\right)$, $W_{19}^{-1} t_{1/2} W_{76} t_{-1/2}$ is twice this matrix. Likewise, using $t_{-1/2}$ instead of $t_{1/2}$ gives

$$f|E = f, \qquad E = \begin{pmatrix} 2 & 1 \\ 19 & 10 \end{pmatrix}.$$

More explicitly, periodicity identifies the two half-translates, and

$$W_{19}^{-1} t_{-1/2} W_{76} t_{1/2} = 2 \begin{pmatrix} 2 & 1 \\ 19 & 10 \end{pmatrix}.$$

This is the asserted invariance under E without any word calculation. Now

$$EW_{19} = \begin{pmatrix} 19 & -2 \\ 190 & -19 \end{pmatrix}, \qquad EW_{19} : z \longmapsto \frac{1}{10} - \frac{1}{1900(z - 1/10)}.$$

Thus $g_+(z) = f(z + 1/10)$ satisfies $g_+|W_{1900} = \eta g_+$. For $g_-(z) = f(z - 1/10)$ use the first invariant matrix and W_{19} in exactly the same way. Both functions are exponentially decreasing at ∞, so their Fricke identities make them exponentially decreasing, after multiplication by y^{-k}, as $y \downarrow 0$ on the imaginary axis. Their Mellin transforms are consequently entire and rapidly decreasing in fixed vertical strips. In particular

$$B(s) = F(s, 1/10) + F(s, -1/10)$$

is entire and satisfies

$$B(s) = \epsilon\, 1900^{k/2-s} B(k - s). \tag{2}$$

These statements follow by splitting the Mellin integral at a fixed height; no modularity of f on $\Gamma_0(19)$ is used.

4.2. **A scalar quotient for the quadratic twist.** Let

$$S_1(s) = F(s,1/5) + F(s,-1/5), \qquad S_2(s) = F(s,2/5) + F(s,-2/5), \qquad R(s) = S_1(s) + S_2(s).$$

The Ramanujan-twist lemma proves that R is entire and

$$R(s) = \epsilon\, 475^{k/2-s} R(k-s).$$

The Hecke-2 identity is

$$\frac{1}{2}\left(f(z/2) + f((z+1)/2)\right) + 2^{k-1} f(2z) = a_2 f(z).$$

Taking Mellin transforms at $z = \pm 1/5 + iy$ and adding yields

$$a_2 S_1(s) = C(s) S_2(s) + D(s), \qquad C(s) = 2^{s-1} + 2^{k-s-1}, \quad D(s) = 2^{s-1} B(s). \tag{3}$$

Consequently

$$S_1(s) = \frac{C(s)R(s) + D(s)}{a_2 + C(s)}, \qquad S_2(s) = \frac{a_2 R(s) - D(s)}{a_2 + C(s)}. \tag{4}$$

These identities first hold in an absolutely convergent right half-plane, and then furnish their meromorphic continuation. For the even quadratic character χ_5 modulo 5, the Gauss identity is

$$\sqrt{5}\, F(s,\chi_5) = S_1(s) - S_2(s) = \frac{(C(s) - a_2)R(s) + 2D(s)}{a_2 + C(s)}.$$

Here $F(s,\chi_5) = (2\pi)^{-s}\Gamma(s)\sum a_n \chi_5(n) n^{-s}$. Equation (2) implies $D(s) = \epsilon 475^{k/2-s} D(k-s)$, and $C(s) = C(k-s)$. It follows that

$$F(s,\chi_5) = \epsilon 475^{k/2-s} F(k-s,\chi_5) \tag{5}$$

as a meromorphic identity. In particular the sign is ϵ, since $\chi_5(19) = \chi_5(4) = 1$.

The possible poles lie among the zeros of $a_2 + C(s)$. Writing $t = s - k/2$ and $\lambda = \sqrt{2} f_2 = 2^{1-k/2} a_2$, these zeros satisfy

$$2^t + 2^{-t} = -\lambda.$$

Thus they lie on at most two vertical arithmetic progressions, are contained in a bounded vertical strip, and have order at most two. The double-zero case is exactly $\lambda = \pm 2$. More generally two possible poles can differ by a positive integer m only if $\lambda = \pm(2^{m/2} + 2^{-m/2})$. This last condition follows by imposing the same quadratic on 2^t and 2^{t+m}; it is only a necessary condition for actual poles.

4.3. **Continuation of the auxiliary grid.** Let $F(s,r)$ be the unnormalized Mellin transform defined above. Under the hypotheses of [3],

$$F(s, j/114) \quad (0 \le j < 114)$$

all continue meromorphically to $\mathbb{C}$, for every allowed weight, root number, and choice of Hecke eigenvalues. Consequently all four individual additive twists of denominator 5 continue meromorphically, including their odd parts. The proof is a small exact symbolic rank certificate combined with Taylor–Fricke continuation.

Put $h = s - k/2$, $x = 2^h$, $y = 3^h$, $z = 19^h$, $a = \sqrt{2} f_2$, $b = \sqrt{3} f_3$, $\eta = i^k \epsilon_f \in \{1, -1\}$, and $\delta = \sqrt{19} f_{19} \in \{1, -1\}$. Let $Q = 114$. The accompanying self-contained Python script constructs the following leading-coefficient rows over $\mathbb{Q}(x, y, z, a, b)$.

(1) For each word $\gamma = \left(\begin{smallmatrix} A & B \\ C & D \end{smallmatrix}\right)$ of length at most three in

$$T, \quad U = \begin{pmatrix} 1 & 0 \\ 19 & 1 \end{pmatrix}, \quad E_- = \begin{pmatrix} 2 & -1 \\ -19 & 10 \end{pmatrix}$$

and their inverses, include

$$e_j - |Cj/Q + D|^{2h} e_v, \qquad v \equiv Q\gamma(j/Q) \pmod{Q},$$

whenever the second point is on the grid. If $Cj/Q + D = 0$, include e_j: this cusp is equivalent to infinity and its Mellin transform is entire. All displayed generators fix f; the last follows from the modulus-2 Ramanujan functional equation.

(2) Include each entire Ramanujan row $R_{q,0}$ for $q \mid Q$ with $\ell(q) = \operatorname{lcm}(19, q^2) \mid Q^2$. For each positive divisor t of $Q^2/\ell(q)$, and also its negative, put $u = -Q^2/(\ell(q)t)$ and include

$$R_{q,t} - \eta_q \left(\frac{\ell(q)t^2}{Q^2}\right)^h R_{q,u}, \qquad \eta_q = \begin{cases} \eta, & 19 \nmid q, \\ -\delta\eta, & 19 \mid q. \end{cases}$$

All indices are reduced modulo Q.

(3) For $p = 2, 3, 19$ and $0 \le j < Q/p$, include the exact Hecke row

$$\sum_{r=0}^{p-1} e_{j+rQ/p} - \lambda_p p^{-h} e_{pj} + \mathbf{1}_{p \ne 19} p^{-2h} e_{p^2 j}, \quad (\lambda_2, \lambda_3, \lambda_{19}) = (a, b, \delta).$$

All positive rational bases here have prime factors in $\{2, 3, 19\}$, so the rows are Laurent-rational expressions in x, y, z, a, b with no ambiguity about complex exponentiation.

Exact rank and uniform specialization. The script `small_levels/level19_uniform_continuation.py` performs Gaussian elimination with exact rational-function arithmetic. It checks eight cases: the four independent choices (η, δ), and, for each, $a \ne 0$ with a retained as a symbol, or $a = 0$ specialized before elimination. Every case has rank 114.

A generic-field rank computation alone would be insufficient here. Accordingly, the script audits every pivot numerator and denominator. View each as a polynomial in x, y, z with coefficients in $\mathbb{Q}[a, b]$. For the $a \ne 0$ case, it verifies that at least one coefficient is a single nonzero rational multiple of a power of a, independent of b. For the $a = 0$ case, it verifies that at least one coefficient is a nonzero rational constant, independent of b. Therefore every pivot remains a nonzero exponential polynomial after *every* permitted fixed specialization of a, b: the distinct real exponents $\log(2^i 3^j 19^l)$ are distinct by unique factorization, and their exponentials are linearly independent. This proves that an appropriate 114-row minor is not identically zero as a meromorphic function of s, without excluding any eigenvalue.

For additional transparency, when $a \ne 0$ the only nonmonomial pivot factors are $x^2 + ax + 1$, $x^4 + (a^2 - 2)x^2 + 1$, and a polynomial whose $x^{10}y^2$ coefficient is 1, together with nonzero powers of a. When $a = 0$, the new factors are $x^4 - 1$, $x^2 - 1$, and a polynomial whose $x^8 y^2$ coefficient is -1. The complete exact expressions are emitted by the verifier. These assertions hold for all four sign choices.

Why leading rank proves continuation. For each of the finitely many rows above, Taylor expansion gives, for arbitrarily large M, an identity

$$A_0(s)\mathbf{F}(s) + \sum_{m=1}^{M-1} A_m(s)\mathbf{F}(s+m) = \mathbf{H}_M(s),$$

where $\mathbf{H}_M$ is holomorphic on $\Re s > B - M$ for a fixed initial absolute-convergence bound B. The matrices A_m are entire in s; A_0 has the rows just verified. Exact Hecke rows have no higher terms.

For a determinant-one invariant matrix γ, writing $t = Cr + D \neq 0$, the row expansion is specifically

$$F(s,r) - |t|^{2s-k} \sum_{m=0}^{M-1} (iCt)^m \binom{s+m-k}{m} F(s+m, \gamma r),$$

which is holomorphic on $\Re s > B - M$. The Fricke-sum rows have the same form with the appropriate Fricke eigenvalue.

Choose a non-identically-zero square minor. Its inverse is meromorphic on $\mathbb{C}$. Initially all components of $\mathbf{F}$ are holomorphic on $\Re s > B$. If they have been continued to $\Re s > B - n$, take $M = n+1$: every shifted component on the left is already meromorphic on $\Re s > B - n - 1$. Inverting the selected leading minor continues all components one strip further. Overlap agrees with the original functions by the identity theorem. Induction proves the claim.

Finally the original Fricke Taylor relation pairs r with $-1/(19r)$. For $r = \pm 1/5, \pm 2/5$ these partners are $\mp 5/19, \mp 5/38$, respectively, and are in the 114-grid. Taking arbitrarily long expansions therefore continues each individual denominator-5 twist meromorphically.

4.4. **Pole cancellation.** Suppose the quadratic twist has a pole at s_χ. By its functional equation, choose it with $\Re s_\chi \geq k/2$. Then some additive twist on the 1/114 grid has a pole at $s_\chi + n$ for an integer $n \geq 0$. Indeed, the Taylor–Fricke identities for the four twists of denominator 5 express them, to arbitrarily large Taylor order, in terms of grid arguments $\mp 5/19$ and $\mp 5/38$ at nonnegative integer shifts of s. All grid twists are holomorphic in their original right half-plane. Thus there is a rightmost grid pole in this discrete integer-translate class. Choose such a pole s_0; then $s_0 = s_\chi + n$ for some $n \geq 0$. Put

$$h = s_0 - k/2, \qquad x = 2^h, \quad y = 3^h, \quad z = 19^h, \quad a = \sqrt{2} f_2, \quad b = \sqrt{3} f_3.$$

The scalar quotient for the quadratic twist gives

$$a = -\big(2^{-n} x + 2^n / x\big). \tag{6}$$

For a nonzero top Laurent coefficient vector v, the leading grid relations give $A_0(s_0)v = 0$. A row syzygy among the leading relations at $s_0 - 1$ gives a first-Taylor relation on v with scalar factor $i(s_0 - k)$. In particular, after assuming $s_0 \neq k$, this factor can be removed. All higher Taylor terms are holomorphic at $s_0 - 1$ because s_0 was chosen rightmost in its class. This argument does not assume a uniform bound on orders of poles to the left.

For $a \neq 0$, monomial pivots reduce the leading system to three coordinates, denoted $w = v_{18}$, $u = v_{42}$, and $v' = v_{72}$. Among the resulting equations are

$$(x^2 + ax + 1)(u + v') = 0, \qquad (x^4 + yb)(u + v') = 0, \qquad (yb + 1)(u + v') = 0.$$

Consequently $x^4 \neq 1$ implies $v' = -u$. Another leading equation then gives

$$xa(y^2 + 1)w + \big(x^4 + (a^2 - 2)x^2 + 1\big)u = 0.$$

On the additional stratum $y^2 + 1 \neq 0$, this eliminates w. If $u = 0$, all three coordinates, and hence the full top coefficient vector, vanish. Thus a putative pole on this stratum gives polynomial constraints after setting $u = 1$.

The generators, Hecke rows, and Ramanujan rows described above give all relations used below. Gaussian elimination uses only monomial pivots (which are nonzero when the separately treated eigenvalue a is nonzero). Every polynomial row syzygy is checked as an exact identity; whole syzygies, rather than only their scalar outputs, have their denominators cleared. Thus no specialization is lost through a canceled pivot. The simultaneous change $(\eta, \delta, z) \mapsto (-\eta, -\delta, -z)$ preserves the leading and Taylor rows. It suffices to check $(\eta, \delta) = (-1, -1)$ and $(-1, 1)$.

Generic poles: $a \neq 0$, $x^4 \neq 1$, $y^2 \neq -1$, $s_0 \neq k$. Put $X = x^2$, $Y = y^2$, $A = ax$, and $G = X - 2A + 4$. The gap condition gives $A = -X/L - L$, where $L = 2^n$ and $|X| \geq L^2$. Consequently $G = (1+2/L)(X+2L)$. If $G = 0$, then $n \leq 1$. For $n = 1$ this forces $a = 0$, already excluded in this paragraph. For $n = 0$ we have $X = -2$, $A = 1$. For equal signs the gap-zero certificate of `level19_pole_probe.py` gives the rational unit ideal; for opposite signs `level19_opposite_gzero_check.py` does so. Both use only the audited nonzero factors $xyza(y^2+1)(x^4-1)$. Hence $G \neq 0$ throughout the remaining generic stratum.

For equal signs, an exact polynomial certificate now gives $X = 4$, $A = -3$. Here is its construction, including its specialization safeguards. Start with the original 60×6 first-Taylor matrix, whose lower and upper columns are $(18, 42, 72 \mid 132, 156, 186)$. Clear the denominators of each whole row; every denominator divides x^7y^4a. For each $j \in \{34, 40, 42, 44, 46, 54\}$, take rows $(1, 37, 43, j)$ and form the four signed 3×3 minors of their lower block. Divide the common polynomial gcd from all four weights and check the resulting polynomial syzygy directly. Thus no nonvanishing assumption on that gcd or any minor is used at a pole. Applying the syzygy to the entire meromorphic germ equation eliminates the lower germs before taking Laurent coefficients.

Contract the upper block with $(-J, xa(y^2+1), -xa(y^2+1))^{\mathsf{T}}$, where $J = x^4 + (a^2-2)x^2 + 1$. Remove only factors of $xya(y^2+1)(x^4-1)G$, already nonzero in this case. Passing to $(B, A, Z, Y, X) = (by, ax, z, y^2, x^2)$ removes only nonzero Laurent monomials, with an exact inverse-substitution check. Together with the fifteen original proportional-row constraints, these are twenty-one necessary polynomials over $\mathbb{Q}$. The source audit regenerates all original rows, checks that their listed columns omit no unknowns, and checks each compact polynomial against its original expression.

With global modular reconstruction disabled, the exact reduced Gröbner basis of these twenty-one polynomials is

$$\begin{aligned}
g_1 &= (B+5)(Y+1)(X-1),\\
g_2 &= (Z-17)(Y+1)(X-1),\\
g_3 &= (Y-9)(Y+1)(X-1),\\
g_4 &= (X-4)(Y+1)(X-1),\\
g_5 &= A^2 - \tfrac{3}{5}YX + X^2 + \tfrac{3}{5}Y - \tfrac{13}{5}X + \tfrac{8}{5},\\
g_6 &= (A+X-1)(Y+1).
\end{aligned}$$

Set $q = (Y+1)(X-1) \neq 0$. The exact identities

$$q(X-4) = g_4, \qquad q(A+3) = (X-1)g_6 - g_4$$

prove the required consequences. This calculation uses no saturation or additional exceptional-locus assumption. The independent source certificate is `level19_rootind_sparse_manifest.json`; the independent final rational replay is `level19_cusp_exact_final_replay.m`, with its source hashes and successful result recorded in the matching audit manifest.

For opposite signs, the corrected reduced first-jet matrix has 56 rows and columns $(30, 42, 76 \mid 132, 156, 186)$. The three sparse polynomial syzygies in `level19_opposite_syzygies.json` annihilate the first three columns identically. Contract their remaining columns with $(-J, xa(y^2+1), -xa(y^2+1))^{\mathsf{T}}$, where $J = x^4 + (a^2-2)x^2 + 1$. Together with two leading constraints this gives five polynomials with $36, 77, 46, 209, 88$ monomials. Passing to $X = x^2, Y = y^2, A = ax, B = by$ removes only nonzero monomial factors. After adjoining $TXYA(Y+1)(X^2-1)G - 1$, their exact reduced basis over $\mathbb{Q}$ is

$$B+5, \quad A+3, \quad T+1/226800, \quad Y-9, \quad X-4.$$

The verifier `level19_opposite_compact_check.py` checks the syzygies against the regenerated matrix and asserts this precise basis. In either sign class the gap condition would now imply $3 = 4/L + L \geq 4$, which is impossible.

The central exceptional condition $y^2 = -1$. This condition forces $\Re(s_0 - k/2) = 0$, hence $n = 0$ for the chosen quadratic-twist pole. Thus $a = -x - x^{-1}$. Simultaneously having $x^4 = 1$ would force $\log 2/\log 3$ rational, unless $s_0 - k/2 = 0$; the latter is incompatible with $y^2 = -1$. Therefore $x^4 \neq 1$ and $a \neq 0$. The leading equations give $u + v = 0$ and $Ju = 0$, where now $J = 2(x^4 + 1)$. The equality $J = 0$ would again make both $2^{s_0 - k/2}$ and $3^{s_0 - k/2}$ roots of unity, contradicting $\log 2/\log 3 \notin \mathbb{Q}$. Hence $u = v = 0$. In the monomially reduced leading system, the row labelled $(H, 3, 5)$ has a nonzero monomial coefficient of the remaining coordinate w, after substituting $a = -x - x^{-1}$ and $y^2 = -1$. This is checked exactly by `level19_central_y_check.py`: after multiplying the numerator by x^{20} the result is $-x^{22}$. Thus $w = 0$, a contradiction. This removes $y^2 = -1$ in both sign classes without using a first Taylor coefficient.

The central exceptional condition $x^4 = 1$ with $a \neq 0$. Again $n = 0$, and $a = -x - x^{-1}$. Hence $x^2 = 1$ and $a = -2x$. The simultaneous change $(x, a) \mapsto (-x, -a)$ preserves the relations, so take $x = 1, a = -2$. Put $Y = y^2$, $B = by$, and $Z = z$. The case $Y = -1$ was excluded above. The reduced row $(H, 2, 5)$ gives $(Y + 1)w - u + v = 0$. Substitute $w = (u - v)/(Y + 1)$ into the other leading rows, clear the displayed nonzero denominators, and form the 2×2 minors of the resulting two-column matrix. A nonzero residue vector requires all these minors to vanish. There are forty-six distinct normalized minors. Exact rational elimination, with $TY(Y + 1) = 1$, gives the necessary equation

$$4Y^3 + 7Y^2 + 1 = 0.$$

This computation is independently executable with SymPy in `level19_central_x_check.py`. The same script checks that this cubic is coprime to its reciprocal $Y^3 + 7Y + 4$. Because its coefficients are real, a root on the unit circle would be a common root of the two polynomials. There are therefore no such roots, contradicting $|Y| = 1$. This removes the remaining central exception for $a \neq 0$ in both sign classes.

The zero-eigenvalue case. Suppose $a = 0$. The quadratic-twist pole relation gives $x^2 = -4^n$, so $\Im(s_0) \neq 0$ and in particular $s_0 \neq k$. Redoing the unit-pivot elimination with $a = 0$ leaves the three top coordinates $(w, u, v) = (v_{18}, v_{42}, v_{84})$. For $n \geq 1$, the row $(H, 2, 42)$ gives $(x^4 - 1)u = 0$, hence $u = 0$. The row $(F, 19, 36)$ then gives $(y^2 + 1)w + (x^2 - 1)v = 0$. Here $y^2 \neq -1$ by $|y^2| = 9^n > 1$, so a nonzero vector is proportional to $(1 - x^2, 0, y^2 + 1)$.

The corresponding unit-reduced first-jet system has eighty rows, with lower columns $(42, 76, 84)$ and top columns $(132, 156, 198)$. Index these rows from zero, in the deterministic order of the script. The combination

$$R_{59} - \frac{x^2 + 4}{x^2} R_{66} + \frac{3b}{y} R_{17}$$

has all lower coordinates zero. Contracting its top coordinates against $(1 - x^2, 0, y^2 + 1)$ gives

$$-\frac{361}{6x^2y^2}(3x^6 + 12x^4 + x^2y^2 - 18x^2 - 4y^2).$$

Only monomial denominators occur in the full row combination, so the identity remains valid when $x^2 = -4$. The executable exact check is `level19_zero2_firstjet_check.py`. Writing $X = x^2 = -4^n$ and $Y = y^2$ gives

$$Y = -\frac{3X(X^2 + 4X - 6)}{X - 4} \in \mathbb{R}.$$

But X is negative real, whereas Y being real would force $2\Im(s_0)\log 2$ and $2\Im(s_0)\log 3$ to be integer multiples of π. The first multiple is odd, contradicting the irrationality of $\log 3/\log 2$. Thus all $n \geq 1$ are excluded.

For $n = 0$, set $x = i$ (the choice $-i$ gives the same equations). The leading relation is $(Y + 1)w - 2u - 2v = 0$. The case $Y = -1$ already contradicts irrationality of $\log 3/\log 2$. Eliminating w and taking the two-column minors gives twenty-nine distinct polynomials. Exact rational elimination, checked in `level19_zero2_central_check.py`, yields

$$(Y - 1)(Y^2 + 1) = 0.$$

Thus Y is a root of unity, again impossible together with $x = i$. This completes the zero-eigenvalue pole exclusion for both sign classes: for $(-1, 1)$ the row indices $(59, 66, 17)$ are replaced by $(55, 62, 17)$, and the checked scalar identity is unchanged. The central rational basis is also unchanged.
Uniform exclusion of the real exceptional position $s_0 = k$. This step uses only the leading equations, and applies to both sign classes. Put $m = k/2 \geq 1$. If $s_0 = k$ and the quadratic-twist pole is $s_\chi = s_0 - n$, the choice $\Re s_\chi \geq k/2$ gives $0 \leq n \leq m$. Write $\ell = m - n$. Then

$$X = x^2 = 4^{n+\ell}, \quad Y = y^2 = 9^{n+\ell}, \quad A = ax = -2^n(4^\ell + 1).$$

In particular $a \neq 0$, $x^4 \neq 1$, and $y^2 + 1 \neq 0$; the leading residue vector is proportional to $(-J, xa(y^2 + 1), -xa(y^2 + 1))$.

Two of the scalar leading equations, expressed in $(X, Y, A, B) = (x^2, y^2, ax, by)$, have respectively thirty-six and seventy-seven terms and degrees one and two in B. These equations do not depend on z or on the sign class. Their resultant in B is a nonzero rational constant times $(Y + 1)R(X, Y, A)$, where R has 243 terms. Thus a pole would force $R(X, Y, A) = 0$. The following finite exact certificate proves that this is impossible for all $n, \ell \geq 0$.

Expand

$$R(4^{n+\ell}, 9^{n+\ell}, -2^n(4^\ell + 1)) = \sum_{\alpha,\beta} c_{\alpha,\beta}\alpha^n\beta^\ell.$$

There are 487 nonzero rational coefficients, ninety distinct positive integer bases α, and fifty-eight distinct positive integer bases β. After replacing $c_{\alpha,\beta}$ by $c_{\alpha,\beta}\alpha^2\beta^2$, all $90 \cdot 58 = 5220$ upper-right cumulative sums are strictly negative. Two applications of finite Abel summation therefore show that the displayed expression is strictly negative for $n, \ell \geq 2$. For clarity, if the bases are increasingly ordered, write $C_{ij} = \sum_{r \geq i, s \geq j} c_{rs}\alpha_r^2\beta_s^2$. Then for $u, v \geq 0$,

$$\sum c_{rs}\alpha_r^{u+2}\beta_s^{v+2} = \sum_{i,j} C_{ij}(\alpha_i^u - \alpha_{i-1}^u)(\beta_j^v - \beta_{j-1}^v),$$

where the preceding weight at the first index is defined to be zero. All differences are nonnegative and at least one product is positive.

The four boundary strips have equally short one-dimensional certificates: for fixed $n = 0, 1$, all upper cumulative sums are negative after shifting ℓ by one; for fixed $\ell = 0$, shift n by three; for fixed $\ell = 1$, shift n by two. Every earlier value on these strips is nonzero by exact integer evaluation. The complete reconstruction of the two leading equations, the resultant, the exponential expansion, and all cumulative-sum assertions is in `level19_real_top_check.py`. It has been executed over $\mathbb{Q}$ for $(\eta, \delta) = (-1, -1)$ and $(-1, 1)$. This excludes $s_0 = k$ uniformly for every even positive weight; no numerical approximation or finite-weight inference is used.

These cases exhaust all possible poles. Therefore $F(s, \chi_5)$ is entire. The argument never requires a common bound for pole orders of grid twists to the left of s_0: first-jet syzygies are identities of meromorphic germs and eliminate the entire lower layer before taking the top Laurent coefficient.

4.5. **Modularity and cuspidality.** The quotient above then has rapid decay on every fixed vertical strip. Indeed its numerator does, and its denominator is periodic in the imaginary direction. Away from fixed small disks about its zeros it is bounded below; inside each disk the maximum-modulus principle applied to the now holomorphic quotient gives the same bounds. Mellin inversion therefore converts (5) into the genuine Fricke transformation of the quadratic Fourier twist at conductor 475.

Set

$$A = \begin{pmatrix} 5 & -1 \\ -19 & 4 \end{pmatrix}, \quad B = \begin{pmatrix} 5 & 1 \\ 19 & 4 \end{pmatrix}, \quad t_r = \begin{pmatrix} 1 & r \\ 0 & 1 \end{pmatrix}.$$

Combining the c_5 and χ_5 Fricke identities separates the two quadratic-residue classes and gives

$$f|(A-1)t_{1/5} + f|(B-1)t_{-1/5} = 0. \tag{7}$$

For example the matrix identity behind this equation, and all the analogous Ramanujan identities, is

$$p^{-1}W_{19}^{-1}t_{a/p}W_{19p^2}t_{-b/p} = \begin{pmatrix} p & -b \\ -19a & (19ab+1)/p \end{pmatrix}, \qquad 19ab \equiv -1 \pmod{p}.$$

For $p = 5$ the residue pair has $a = b = \pm 1$. For the modulus 4 Ramanujan identity, $a = b = \pm 1$ gives instead B^{-1} and A^{-1}. Hence

$$f|(B^{-1}-1)t_{1/4} + f|(A^{-1}-1)t_{-1/4} = 0. \tag{8}$$

Put $h = f|(A-1)$. Equation (7) says $f|(B-1) = -h|t_{2/5}$. Substituting this into (8) and multiplying on the right by $t_{1/4}A$ gives

$$h|M = h, \qquad M = t_{2/5}B^{-1}t_{1/2}A = \begin{pmatrix} -14/5 & 2/5 \\ -19/2 & 1 \end{pmatrix}.$$

The matrix M has determinant 1 and trace $-9/5$. It is elliptic of infinite order: a finite-order matrix with rational trace would have integral trace, whereas $-9/5$ is not an integer. A holomorphic function on $\mathbb{H}$ of positive weight k invariant under an infinite-order elliptic matrix is zero. One direct proof conjugates its fixed point to the disk origin: the Taylor coefficients transform with rotation exponents $k + 2n > 0$, none of which is fixed by an irrational rotation. Thus $h = 0$, and then (7) yields $f|B = f$ as well.

Finally T, $U = \left(\begin{smallmatrix} 1 & 0 \\ 19 & 1 \end{smallmatrix}\right)$, E, A, and B generate $\Gamma_0(19)$ modulo its center. An exact, bounded Todd–Coxeter verification is supplied in `small_levels/level19_generation.py`. It uses the presentation $\mathrm{PSL}_2(\mathbb{Z}) = \langle S, T \mid S^2 = (ST)^3 = 1 \rangle$, checks the matrix words exactly, and finds index 20 for the displayed subgroup. Containment in $\Gamma_0(19)$ and the familiar index 20 then prove equality. Both cusps of $\Gamma_0(19)$ are cuspidal for f, since f vanishes at ∞ and satisfies its original Fricke relation. We conclude that $f \in S_k(\Gamma_0(19))$, as claimed.

5. Level 21: meromorphic continuation on the 42-grid

This section proves a continuation result, not the converse theorem. Use unnormalized coefficients $a_n = f_n n^{(k-1)/2}$ and write

$$F_j(s) = (2\pi)^{-s}\Gamma(s)\sum_{n \geq 1} a_n e(nj/42)n^{-s} \quad (j \bmod 42).$$

Set

$$E = i^k\epsilon_f, \quad u = a_3 3^{1-k/2}, \quad v = a_7 7^{1-k/2}, \quad L = a_2 2^{1-k/2}.$$

Here $E, u, v \in \{\pm 1\}$, whereas $L \in \mathbb{C}$ is arbitrary. Let $h = s - k/2$ and $X = 2^{-h}$, $Y = 3^{-h}$, $Z = 7^{-h}$.

Rows of the leading system. Write e_j for the coordinate row indexed by j mod 42. For $q \mid 42$, put

$$\ell_q = \operatorname{lcm}(21, q^2), \qquad d_q = 42^2/\ell_q, \qquad A_q = \{42a/q : 0 \le a < q,\ (a,q) = 1\},$$

and $E_q = E(-u)^{[3|q]}(-v)^{[7|q]}$. The following rows are explicit Laurent-polynomial rows in X, Y, Z, L:

$$R_q = \sum_{a \in A_q} e_a,$$

$$T_{q,t}(s) = \sum_{a \in A_q} \left(e_{a+t} - E_q \left(\frac{\ell_q t^2}{42^2} \right)^h e_{a - d_q/t} \right) \qquad (t \mid d_q, t \neq 0),$$

$$H_{2,j}(s) = e_j + e_{j+21} - LXe_{2j} + X^2 e_{4j}.$$

All indices are reduced modulo 42. The rows R_q applied to F are entire by the Ramanujan-twist functional equations. The rows $H_{2,j}$ applied to F vanish identically by the 2-Hecke relation. The rows $T_{q,t}$ are the leading ($m = 0$) parts of the Taylor–Fricke expansions for $f_{(q)}$ at $t/42$.

Take the following 42 rows, in the displayed order:

family	indices, in order
R_1	
$T_{1,t}$	$t = 1, -1, 2, -2, 3, -3, 4, -4, 6, -6, 7, -7$
R_2	
$T_{2,t}$	$t = 1, -1, 3, -3$
R_3	
$T_{3,t}$	$t = 1, -1, 2, -2, 4, -4$
R_6	
$T_{6,t}$	$t = 1, -1$
$T_{7,t}$	$t = 1, -1, 2, 3$
$H_{2,j}$	$j = 2, 3, 4, 5, 6, 8, 9, 10, 11, 16.$

Let $A(s)$ be the resulting square matrix, with columns indexed $0, 1, \ldots, 41$.

Proposition 5.1 (Exact determinant certificate)**.** *For every choice $E, u, v \in \{\pm 1\}$,*

$$\det A(s) = \frac{uX^2}{Z} \left((LX+1)Z(Y-u) + (Euv-1)X^2YZ - EuX^2Y - uX^2Z \right). \tag{9}$$

In particular, $\det A(s)$ *is not identically zero for any $L \in \mathbb{C}$.*

The identity is an exact finite algebraic certificate. The supplied `level21_certificate.py` builds the rows as above and computes the determinant over the rational function field for all eight sign choices, verifying (9) by exact subtraction. The corresponding output and row lists are recorded in `level21_certificate.json`. No numerical eigenvalue specialization is used in this determinant identity.

There are two elementary ways to see the final assertion. As a Laurent polynomial, the bracket has nonzero constant-in-X part $Z(Y-u)$, independent of L. Distinct Laurent monomials in X, Y, Z remain distinct exponentials after substituting p^{-h}, by unique factorization of rational numbers. Alternatively, directly as $\Re s \to +\infty$,

$$\det A(s) \sim -X^2;$$

indeed the potentially non-obvious ratio $X^2Y/Z = (7/12)^h$ tends to zero. There are therefore no exceptional Hecke eigenvalues for this continuation certificate.

Corollary 5.2. *Under the hypotheses of* [3] *at level* 21*, all $F_j(s)$, j* mod 42*, continue meromorphically to $\mathbb{C}$. Consequently so does every multiplicative twist whose modulus divides* 42*.*

Proof. For each integer $M > 0$, the Taylor–Fricke argument gives a system

$$A(s)F(s) = \sum_{m=1}^{M-1} A_m(s)F(s+m) + H_M(s),$$

where H_M is holomorphic on $\Re s > K - M$ for a fixed K, and the entries of A_m are entire functions of s. Here one may enlarge K to cover the finite collection of rows. The exact Hecke and entire Ramanujan rows fit into the same system. Crucially, every higher-shift source coordinate also belongs to the same complete 42-grid; this is a square, closed system.

The initial Dirichlet series give a right half-plane of holomorphy. If meromorphic continuation is known on $\Re s > b$, choose M sufficiently large that $K - M < b - 1$. Every shifted term on the right is then meromorphic on $\Re s > b - 1$. The nonzero determinant permits inversion of $A(s)$ as a meromorphic matrix, continuing every coordinate one unit to the left. Iteration proves the assertion. Fourier inversion of any character of modulus dividing 42 gives the final claim. □

What this does not prove. No functional equation for an individual multiplicative twist is asserted here. In particular, the recurrence may propagate poles to successively smaller real parts, so this argument gives neither a bounded strip of poles nor entireness. These, and the missing modular transformations, remain separate steps in a level-21 converse proof.

6. Level 21: pole exclusion on the 42-grid

Write $V(s) = (F(s, j/42))_{j \bmod 42}$ and $x = 2^{s-k/2}$, $y = 3^{s-k/2}$, $z = 7^{s-k/2}$, $a = \lambda_2$. Let η be the Fricke sign and $u = \lambda_3$, $v = \lambda_7$, so $\eta, u, v \in \{\pm 1\}$. The already proved continuation of V is used here.

Proposition 6.1. *At level 21, every component of V is entire unless*

$$k = 2, \qquad (\eta, u, v) = (-1, -1, 1), \qquad a = -3/2.$$

In this exceptional case its only possible pole is a simple pole at $s = 2$. This assertion alone does not establish modularity.

Proof. Augment the leading grid system with the known transformations generated by $T, U_{21}, \left(\begin{smallmatrix} 2 & -1 \\ -21 & 11 \end{smallmatrix}\right)$ and W_{21}, using words of length at most three. The third matrix has multiplier one and W_{21} has multiplier η. All signs are retained when words are formed. The exact builder is `level21_augmented_builder.py`.

Elimination using only nonzero rational multiples of Laurent monomials in x, y, z gives 41 pivots, leaving coordinate 6 free. Thus there are no exceptional eigenvalue divisions. The residual rows are supported only on this coordinate. Apply the same elimination, shifted by one, to the first Taylor source. The zero rows eliminate every unknown Laurent coefficient at the lower point. Combining their numerators with the residual leading rows gives, for every sign triple, the exact ideal

$$\left(ax + x^2 - 1, \quad y + 3u, \quad z - \eta u(6 + \eta uv)\right). \tag{21.1}$$

The verifier `level21_augmented_first_jet.py` asserts this Groebner basis over the rationals, including $a = 0$.

Choose a rightmost pole s_0 in a class modulo integer translation. Such a choice is possible because all components are holomorphic in a common right half-plane. At its largest pole order the leading residue is a nonzero multiple of the computed kernel vector. If $s_0 \neq k$, the first Taylor factor $s_0 - k$ is nonzero, so (21.1) holds. The equation $|y| = 3$ gives $\Re(s_0 - k/2) = 1$. Thus $|z| = 7$; the four cases with $|z| = 5$ are impossible. In the others, both $3^{i\Im s_0}$ and $7^{i\Im s_0}$ are signs. A nonzero imaginary part

would make $\log 3/\log 7$ rational, contrary to unique factorization. Consequently $y = 3$, $z = 7$, and the only surviving parameters are

$$(\eta, u, v) = (-1, -1, 1), \qquad s_0 = k/2 + 1, \qquad a = -3/2. \tag{21.2}$$

For (21.2), evaluate the full shifted Taylor recurrence, including all intermediate unknown pole vectors. Put w_0 equal to the leading kernel vector normalized by coordinate 6. After the standard falling-factorial normalization, its equations are

$$A(s_0 - n)w_n = \sum_{m=1}^{n} \frac{C_m(s_0 - n)}{m!} w_{n-m}. \tag{21.3}$$

Here C_m includes the mth power of the rational slope in each Fricke or known-matrix row, and is zero for Hecke and unshifted Ramanujan rows. At $n = 1, 2, 3$, the matrices $A(s_0 - n)$ have full column rank 42. The first two systems are consistent; at order three the Ramanujan row for $q = 7$ has residual -1029. These exact rational statements are asserted in `level21_exceptional_jet.py`. If $s_0 \neq k$, then $k \geq 4$ and $s_0 - k = 1 - k/2$ is a negative integer. Every falling factorial used to normalize (21.3) is therefore nonzero. This excludes (21.2) away from k.

At $s_0 = k$, a pole of order at least two also forces (21.1): apply the exact first-row syzygy at the coefficient one order below the top, since its source has the simple factor $s - k$. Hence only $k = 2$ and (21.2) can survive. Put $e = s - 2$ and $P_n(e) = \prod_{j=0}^{n-1}(e - j)$. Full column rank at the three lower points shows that their pole orders are at most one less than the top order. Normalizing those leading coefficients by $i^n P_n'(0)$ gives exactly (21.3), since

$$\binom{e - n + m}{m} P_{n-m}(e) = P_n(e)/m!.$$

Thus the same order-three inconsistency excludes every higher-order pole at k; no lower-point Laurent coefficients were assumed to vanish. A simple pole at k contributes holomorphically to every lower shifted Taylor equation because the relevant binomial coefficient has a zero. Repeating the rightmost-pole argument therefore excludes all lower poles.

It remains to test simple poles at k. The leading residual rows alone, with $x = 2^n, y = 3^n, z = 7^n$ and $n = k/2$, suffice. Eliminating a between two such rows gives $x^2 P(x, y, z)$, where P has largest exponential base $252 = 2^2 3^2 7$. In each of the eight cases its coefficient has absolute value one, and the sum of the absolute values of all other terms is strictly smaller at $n = 2$. Since their bases are smaller, the inequality persists for every $n \geq 2$. In the lexicographic sign order $---, --+, -+-, -++, +--, +-+, ++-, +++$, the eight values $P(2, 3, 7)$ are

$$-92,\ 0,\ -120,\ -72,\ 420,\ 384,\ 488,\ 384.$$

The zero case forces $a = -3/2$ in a linear residual row. The verifier `level21_weight_pole.py` recomputes these resultants and the exact dominance inequalities. This proves the stated exception list. □

All eight first-jet checks, all eight weight checks, and the exceptional third-jet check are recorded with source hashes in `level21_poles.json`. The preceding proposition leaves a possible simple weight-2 pole. The later quadratic-twist argument removes it; the additional identities proving modularity are given in Sections 17 and 19.

7. Level 22: uniform continuation of the 66-grid

Use $D_j(s) = (2\pi)^{-s}\Gamma(s)\sum a_n e(nj/66)n^{-s}$, $h = s - k/2$, and $x = 2^h, y = 3^h, z = 11^h$. Write $\eta = i^k \epsilon_f$, $\sigma_2 = a_2 2^{1-k/2}$, $\sigma_{11} = a_{11} 11^{1-k/2}$ and $a = a_3 3^{1-k/2}$. The first three parameters are independent signs, and a is an arbitrary complex number.

Proposition 7.1. *All 66 additive Mellin transforms D_j continue meromorphically to $\mathbb{C}$, for every allowed choice of these parameters.*

Here is a finite exact certificate, using the same Taylor–Mellin continuation argument as for levels 21 and 25. The exact transformations used are

$$T = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad U = \begin{pmatrix} 1 & 0 \\ 22 & 1 \end{pmatrix}, \quad E = \begin{pmatrix} 2 & -1 \\ -22 & 12 \end{pmatrix}, \quad W = \begin{pmatrix} 0 & -1 \\ 22 & 0 \end{pmatrix}.$$

Their weight-k slash eigenvalues are respectively $1, 1, -\sigma_2, \eta$. In particular, E has determinant 2, and treating its eigenvalue as always 1 would be incorrect. The identity for E follows from the c_2-twisted Fricke involution at conductor 44 and the original Fricke involution; the twisted eigenvalue is $-\eta\sigma_2$. Use every word of length at most three in these matrices and their inverses, tracking the product of their eigenvalues.

For a resulting pair (g, μ), with $g = \left(\begin{smallmatrix} A & B \\ C & D \end{smallmatrix}\right)$, and $\alpha = j/66$ whose image $\beta = g\alpha$ lies on the same grid, the leading row is

$$e_j - \mu \left(\frac{(C\alpha + D)^2}{\det g} \right)^h e_{66\beta}.$$

If $g\alpha = \infty$, use the entire row e_j. Add all Ramanujan and Taylor–Fricke rows for $q \mid 66$ and signed $t \mid 66^2/\operatorname{lcm}(22, q^2)$, with signs

$$\eta_q = \eta(-\sigma_2)^{[2|q]}(-\sigma_{11})^{[11|q]},$$

and add the exact Hecke rows for $p = 2, 3, 11$ with normalized coefficients σ_2, a, σ_{11}. The quadratic local term is present only for $p = 3$.

The script `level22_uniform_continuation.py` constructs these rows and performs exact sparse elimination over $\mathbb{Q}(x, y, z, a)$. For each of the eight sign triples it checks both $a \neq 0$ and $a = 0$, giving sixteen cases. Every case reaches rank 66. Each pivot numerator and denominator is checked to contain a coefficient, when grouped by exponential monomials in x, y, z, that is a nonzero rational multiple of a power of a (power zero in the $a = 0$ case). Thus the elimination is valid as meromorphic functions for every permitted a, not merely for generic a. Distinct monomials remain distinct exponentials by unique factorization. The source hash and sixteen verification results are recorded in `level22_uniform_continuation.json`.

Taylor expansion to arbitrary order gives the closed shifted system on the complete 66-grid. Its full leading rank allows successive continuation by strips of width one, beginning with the absolutely convergent Dirichlet series. This proves the proposition. It also gives meromorphic continuation of multiplicative twists of every modulus dividing 66. No bounded pole strip, pole cancellation, additional twisted functional equation, or modularity is asserted by this certificate alone.

8. Level 22: all grid poles are simple cusp poles

This section strengthens the continuation certificate, but does not claim modularity. Use the notation of Section 7.

Proposition 8.1. *Every additive Mellin transform $D_j(s)$, $j \bmod 66$, is holomorphic except possibly for a simple pole at $s = k$. The assertion holds for all three independent signs and every complex Hecke eigenvalue $a = a_3 3^{1-k/2}$.*

First Taylor coefficients. For a leading transformation row

$$e_\alpha - \mu \left(\frac{(C\alpha + D)^2}{\delta} \right)^h e_\beta, \qquad \beta = g\alpha, \quad \delta = \det g,$$

its mth Taylor source row, apart from the common factor $i^m\binom{s+m-k}{m}$, is

$$C_m(s) = \mu\left(\frac{(C\alpha+D)^2}{\delta}\right)^h\left(\frac{C(C\alpha+D)}{\delta}\right)^m e_\beta.$$

For a Ramanujan–Fricke row at $\alpha = t/66$, sum over the corresponding translated destinations and replace the last parenthesis by $\ell_q t/66$. Exact Hecke, entire Ramanujan, and cusp-to-infinity rows have zero Taylor source. These formulas follow by expanding $g(\alpha+iy) = \beta + i\delta y/((C\alpha+D)^2(1+iCy/(C\alpha+D)))$.

A division-safe first-jet certificate. The script `level22_first_taylor.py` eliminates the leading system using only pivots whose numerator and denominator are monomials in x, y, z with nonzero rational coefficient. In particular, no eigenvalue or exponential polynomial is inverted. For every sign triple this gives exactly 65 pivots and one free coordinate. Back substitution gives a leading kernel vector $v(x, y, z, a)$ normalized to have that free coordinate equal to 1. Its entries have only monomial denominators. The remaining leading rows give scalar polynomial constraints.

All rows reduced to zero are retained as exact left syzygies. Applying the same row operations at $s-1$ to the first Taylor source gives further scalar constraints on $v(s)$, after removing the common factor $i(s-k)$. Thus the unknown vector at $s-1$ is eliminated identically; it is never assumed holomorphic.

Set $d = \eta\sigma_2$ and $t = d\sigma_{11}$. The exact reduced Gröbner basis of the leading constraints and these zero-row first-jet constraints is

$$\begin{aligned}
&2529ay^2 + 1416dy^3z - (15025 - 1416t)y^3,\\
&2529x - 240\eta y^2z + \sigma_2(3442 - 240t)y^2,\\
&281y^4 - 405dy^2z + (2331 - 405t)y^2,\\
&40y^2z^2 - d(679 - 80t)y^2z + (2428 - 679t)y^2.
\end{aligned}$$

The script verifies these four polynomial identities for all eight sign triples by exact rational arithmetic. No exceptional parameter stratum is omitted.

Since x, y, z are nonzero, the last three equations force one of

$$(x, y^2, z) = (-2\sigma_2,\ 9,\ d(12-t)) \quad \text{or} \quad \left(\sigma_2,\ -\frac{9}{8},\ \frac{d(199-40t)}{40}\right).$$

The second possibility is incompatible with $x = 2^h, y = 3^h$: $|x| = 1$ implies $\Re h = 0$, whereas $|y^2| = 9/8$. In the first possibility, $|x| = 2$ gives $\Re h = 1$. The phase conditions from $x = \pm 2$ and $y^2 = 9$ force $\Im h = 0$, by multiplicative independence of 2 and 3. Hence $x = 2, y = 3, z = 11$, and the signs must be

$$(\eta, \sigma_2, \sigma_{11}) = (-1, -1, +1).$$

The first polynomial then gives $a = -7/3$. Thus the sole possible exponential-curve point satisfying all these constraints is

$$h = 1, \quad a = -7/3, \quad (\eta, \sigma_2, \sigma_{11}) = (-1, -1, +1). \tag{10}$$

Exact third-jet exclusion of the remaining point. The script `level22_exceptional_jet.py` specializes to (10), verifies the nonzero leading kernel vector against every leading row, and solves the complete shifted systems over $\mathbb{Q}$. If s_0 is the upper pole and $\tau = s_0 - k$, normalize its successive leading Laurent-coefficient vectors by

$$r_n = i^n\tau(\tau-1)\cdots(\tau-n+1)w_n.$$

The equations solved are

$$A(s_0-n)w_n=\sum_{m=1}^{n}\frac{C_m(s_0-n)}{m!}w_{n-m}.$$

The matrices at orders one, two and three each have rank 66. Orders one and two are consistent. At order three the row $H_{2,2}$ has nonzero residual -1248478 with the script's fixed kernel normalization. Other rows also give nonzero residuals. This is an exact inconsistency, with all intermediate vectors w_1, w_2 explicitly retained.

Proof of the proposition. Fix a class modulo integer shifts containing a pole. There is a rightmost pole in that class, because the Dirichlet series are holomorphic in a right half-plane. Let it be s_0, with maximal pole order r among the finite vector of grid transforms, and let $v_0 \neq 0$ be its top Laurent coefficient. The leading system at s_0 gives its scalar leading constraints.

At s_0-1, the zero-row first-jet syzygies eliminate the entire unknown lower vector. All sources shifted by at least two are holomorphic, by the choice of s_0. If $s_0 \neq k$, the coefficient of the pole of order r therefore gives the first-jet constraints above. We must be at (10), so $s_0 = k/2+1$ and $\tau = 1-k/2$. When $k \geq 4$, the factors $\tau, \tau-1, \tau-2$ are all nonzero; the third-jet inconsistency then excludes the pole. When $k=2$, this point is $s_0=k$. Thus the only possible rightmost pole is $s_0=k$.

Suppose its order r is at least two. The first Taylor factor $s+1-k$ has a simple zero at $s=k-1$, so the pole of order $r-1$ in the first-jet syzygy imposes the same first-jet constraints on the top coefficient v_0. Except at (10) this is already impossible. At that exception necessarily $k=2$. The lower shifted matrices have full rank. The coefficients of order $r-1$ in their successive Laurent expansions are obtained by differentiating the falling factor at $\tau=0$; this derivative is nonzero at each positive order. Consequently the same normalized w_1, w_2, w_3 recurrence and the same third-order inconsistency apply. Higher-order poles at k are therefore excluded.

Finally a simple pole at k contributes holomorphically to every positive Taylor shift: its multiplier has the zero $\binom{k-k}{m}=0$ for $m \geq 1$. If a lower pole remains in the integer class of k, take the rightmost such lower pole. The preceding argument applies unchanged, since the contribution from the simple pole at k is holomorphic. It excludes that lower pole. This proves the proposition in every class modulo integer shifts. □

Verification and continuation of the proof. The eight symbolic first-jet cases and the exact exceptional third-jet calculation passed; source hashes and results are in `level22_poles.json`. Possible simple poles at k are treated below. The additional transformations and the proof of cuspidality are given in Section 21.

8.1. **Audit of higher-order poles at the exceptional weight.** Here are details of the derivative step in the preceding level-22 argument. They ensure that poles of higher order at the lower shifted points are not implicitly assumed away. At the exceptional point $k=s_0=2$, put $e=s-s_0$ and

$$P_n(e)=\prod_{j=0}^{n-1}(e-j), \qquad P_0(e)=1.$$

Suppose $V(s)$ has a pole of order $r \geq 2$ at s_0, with top coefficient v_0. The matrices $A(s_0-n)$ for $n=1,2,3$ have full column rank, as verified in the exceptional certificate. At s_0-1 the only nonholomorphic source has a factor e, so $V(s-1)$ has order at most $r-1$. Inductively the same bound holds at s_0-2 and s_0-3: the direct source from s_0 has the simple factor $P_n(e)$ and all intermediate sources already have order at most $r-1$. A holomorphic left inverse of A near each of these three points justifies these order bounds.

Let u_n be the coefficient of $e^{-(r-1)}$ in $V(s_0-n+e)$. Since $P_n'(0)=(-1)^{n-1}(n-1)! \neq 0$, write

$$u_n=i^n P_n'(0)w_n, \qquad w_0=v_0.$$

The coefficient of $e^{-(r-1)}$ in the shifted Taylor system gives

$$A(s_0-n)w_n=\sum_{m=1}^{n}\frac{C_m(s_0-n)}{m!}w_{n-m}.$$

For $m<n$, this follows by differentiating at zero the polynomial identity

$$\binom{e-n+m}{m}P_{n-m}(e)=\frac{P_n(e)}{m!}.$$

For $m=n$ the same identity is used directly on the order-r term v_0e^{-r}. Derivatives of matrix coefficients do not enter: they multiply strictly lower Laurent orders. Terms shifted to the right of s_0 are holomorphic. This is exactly the recurrence in the exceptional verifier, so its order-three inconsistency rules out every $r\ge 2$. This supplies the missing detail behind the simple-pole conclusion without any assumption about the unknown Laurent coefficients at intermediate points.

9. Excluding the weight pole at level 22

This strengthens the pole conclusion of Section 8; it does not supply the missing modular transformation law. Retain its notation $h=s-k/2$, $x=2^h$, $y=3^h$, $z=11^h$, $a=\lambda_3$, and signs $\eta,\sigma_2,\sigma_{11}$. Every additive Mellin transform on the 66-grid is entire unless

$$k=2,\qquad(\eta,\sigma_2,\sigma_{11})=(-1,-1,1),\qquad a=-7/3.$$

In this exceptional case the previously established conclusion remains: only a simple pole at $s=2$ is possible.

Here is an exact finite verification. The monomial elimination used in `level22_pole_kernel.py` has rank 65 without any exceptional specialization of $x,y,z\neq 0$. Every nonzero residue vector therefore has nonzero free coordinate 42 and must annihilate the residual scalar rows. Eliminate a from two independent such rows. Up to a nonzero rational monomial and positive multiplicities, their resultant has factors $x+\sigma_2$ and the polynomial below. Write $e=\eta$, $u=\sigma_2$, $t=\eta\sigma_2\sigma_{11}$:

$$\begin{aligned}P=\;&2x^8+ux^7y^2+ux^7+x^6-2ux^5y^2-3ux^5+x^4y^4-eux^4y^2z+(2-t)x^4y^2-x^4\\&+ux^3y^6+2ux^3y^4-ex^3y^2z+u(1-t)x^3y^2-ux^3+3x^2y^6-eux^2y^2z+(3-t)x^2y^2+x^2\\&+uxy^6-3uxy^4-2uxy^2-euy^4z-ty^4.\end{aligned}$$

At $s=k$ put $n=k/2\ge 1$, so $(x,y,z)=(2^n,3^n,11^n)$. The factor $x+\sigma_2$ is nonzero. The term ux^3y^6 of P becomes $u\,5832^n$, and every other exponential base is strictly less than 5832. For each of the eight sign triples, the sum of absolute values of those other terms is strictly less than 5832^n at $n=3$. Dividing by 5832^n proves the same strict inequality for every $n\ge 3$. The values at $n=1,2$, with signs ordered as $(\eta,\sigma_2,\sigma_{11})$, are

$(-,-,-)$	378	−13518072
$(-,-,+)$	0	−13564890
$(-,+,-)$	23070	69500376
$(-,+,+)$	23736	69567930
$(+,-,-)$	4158	−7899912
$(+,-,+)$	4536	−7853094
$(+,+,-)$	16410	61393896
$(+,+,+)$	15744	61326342

The sole zero is the stated exceptional weight-2 case. Substitution into the original scalar rows then forces $a=-7/3$. The script `level22_weight_pole.py` verifies the resultant factorization, strict exponential inequalities, both small integer values, and this exceptional eigenvalue directly from the

matrix rows. All eight runs passed; source hashes and results are recorded in `level22_weight_pole.json`.

10. An extra-prime leading-kernel certificate in weight two

This intermediate lemma establishes continuation and a conditional pole constraint. The following sections remove the condition and the final pole; modularity remains a separate step.

At level 21, specialize to the exceptional signs $(\eta, \sigma_3, \sigma_7) = (-1, -1, 1)$ and $\lambda_2 = -3/2$; at level 22, specialize to $(\eta, \sigma_2, \sigma_{11}) = (-1, -1, 1)$ and $\lambda_3 = -7/3$. Adjoin the prime 5 to the grid, giving $Q = 210$ and $Q = 330$ respectively. Include all the same known-map and Ramanujan Taylor rows, together with the exact Hecke-5 rows, and leave $t = \lambda_5$ as a free parameter.

At $h = s - k/2 = 1$, exact elimination using only nonzero rational constant pivots has rank $Q - 1$. There is one remaining column. The greatest common divisor of its residual scalar numerators is, up to a nonzero rational factor,

$$5t + 18 \quad (N = 21), \qquad 5t + 19 \quad (N = 22).$$

Thus the enlarged leading matrix is invertible at $h = 1$ except at $t = -18/5$ and $t = -19/5$, respectively. At that exceptional value it is invertible at $h = 2$: exact reduction modulo 1000003 gives rank 210, respectively 330, and all rational denominators are nonzero modulo that prime. Consequently the leading matrix is not identically singular for any complex λ_5. The closed Taylor-system argument therefore proves meromorphic continuation of the entire enlarged grid.

For a nonzero pole vector to lie in the leading kernel at $s = 2$, the same calculation forces the displayed exceptional value of λ_5. However this is a *conditional* pole constraint: the original 42- or 66-grid pole bound does not yet bound the poles of the newly continued coordinates. New poles at $s = 2 + m$ could contribute to the Taylor equations at $s = 2$. Therefore it is not justified to claim that the original simple pole forces these eigenvalues without first excluding those higher-pole source terms.

The exact verifier is `level21_22_extra_prime5.py`. Both level runs passed. Source hashes and successful results are recorded in `level21_22_extra_prime5.json`.

11. Upper integer poles on the enlarged exceptional grids

The gap identified in Section 10 can now be closed. Retain its exceptional weight-two signs and original eigenvalue. On the enlarged $Q = 210$ or 330 grid, there are no poles at any integer $s \geq 3$. Consequently, if the original grid has a pole at $s = 2$, then

$$\lambda_5 = -18/5 \quad (N = 21), \qquad \lambda_5 = -19/5 \quad (N = 22).$$

This still does not exclude the pole at 2 or prove modularity.

Here is the exact rank certificate. At an integer $s \geq 3$, the embedded old-grid coordinates, whose indices are multiples of 5, are holomorphic. Add unit rows for those coordinates to the enlarged leading matrix. The Hecke-5 equations then reduce to the sum of the five preimage coordinates: both eigenvalue terms lie on the old grid and vanish. Thus these additional rows involve no assumption on λ_5. Specialize the original eigenvalue to $-3/2$ or $-7/3$, but retain the nonzero variables $x = 2^h, y = 3^h, z = p^h, b = 5^h$, where $h = s - 1$ and $p = 7$ or 11.

Elimination using only nonzero rational monomial pivots leaves two coordinates: $(44, 148)$ at level 21 and $(138, 192)$ at level 22. There are respectively 208 and 328 pivots. Two remaining rows give nonzero two-by-two determinants for every integer $h \geq 2$. At level 21, the selected rows are the reduced rows $F_{7,-2}$ and $F_{7,2}$. The determinant is

$$-\frac{(x^2 - 1)}{3x^3 y}(4x^4 + 3x^3 y + 3x^3 - 4x^2 y + 4x^2 - 3xy + 3x - 4y).$$

At level 22, the selected rows are $F_{33,-1}$ and $F_{22,-1}$. Its numerator, with the denominator $9x^4y^6$ removed, has 24 monomials; the exact polynomial is printed and verified by the script. After substitution $(x,y)=(2^h,3^h)$, the largest exponential base is 5832 with coefficient 27. The sum of the absolute values of all other terms is strictly smaller at $h=3$. This proves nonvanishing for every $h\geq 3$ by division by 5832^h. At $h=2$ the numerator is $431003448\neq 0$. The same automated dominance check at level 21 has largest base 96, coefficient -3, cutoff 3, and value -34500 at $h=2$. All these are exact integer checks.

For the analytic implication, if an integer pole at $s\geq 3$ existed, choose the largest such integer, which is possible by convergence in a right half-plane. Its highest Laurent coefficient satisfies the leading equations because all positive Taylor shifts are holomorphic. The old-grid coordinates of that vector vanish. The certified full rank therefore contradicts its being nonzero. At $s=2$ there are now no positive-shift pole sources. The earlier eigenvalue-dependent leading-rank certificate hence makes the displayed λ_5 constraints unconditional whenever a pole exists on the old grid.

Verifier: `level21_22_extra_prime5_poles.py`, with arguments 21 and 22. Both runs pass, and source hashes are recorded in the companion JSON manifest. This argument bounds integer poles only; no assertion about other shift classes is needed or made here.

12. The quadratic twist modulo five in the exceptional level-21 case

Assume $N=21$, $k=2$, $(\eta,\sigma_3,\sigma_7)=(-1,-1,1)$ and $\lambda_2=-3/2$. These are precisely the parameters at which the 42-grid pole theorem leaves a possible simple pole at $s=2$. Write $h=s-1$ and $F(s,r)$ for the completed, unnormalized additive Mellin transform. We prove that $F(s,\chi_5)$ is entire and satisfies

$$F(s,\chi_5)=\epsilon_f 525^{-h}F(2-s,\chi_5).$$

This is a new twisted functional equation. It does not by itself prove modularity or remove the possible pole at 2 on the original grid.

Put $E=\left(\begin{smallmatrix}2 & -1\\ -21 & 11\end{smallmatrix}\right)$, $T=\left(\begin{smallmatrix}1 & 1\\ 0 & 1\end{smallmatrix}\right)$. Both fix f. A direct identity is

$$ETW_{21}=\begin{pmatrix}21 & -2\\ -210 & 21\end{pmatrix}=-\frac{1}{10}t_{-1/10}W_{2100}t_{-1/10}.$$

Consequently the translates at $\pm 1/10$ are exchanged by Fricke at conductor 2100, with multiplier η. Both are exponentially small at the two Mellin endpoints, so

$$B(s)=F(s,1/10)+F(s,-1/10)$$

is entire, rapidly decreasing on fixed vertical strips, and obeys $B(s)=\epsilon_f 2100^{-h}B(2-s)$. Set $S_j=F(s,j/5)+F(s,-j/5)$, $R=S_1+S_2$ and $C(s)=2^h+2^{-h}$, $D(s)=2^hB(s)$. The exact Hecke relation is

$$-\tfrac{3}{2}S_1=C(s)S_2+D(s).$$

The Ramanujan-5 theorem makes R entire with functional equation $R(s)=\epsilon_f 525^{-h}R(2-s)$, and D has the same equation. Thus

$$\sqrt{5}F(s,\chi_5)=S_1-S_2=\frac{(C(s)+3/2)R(s)+2D(s)}{C(s)-3/2}$$

has the asserted meromorphic functional equation. Its only possible poles have $2^h=u$ where $u+u^{-1}=3/2$. The two roots $u=(3\pm i\sqrt{7})/4$ have absolute value one and are not ± 1.

We now exclude every such pole, including possible sources at higher integer shifts in the larger additive grid. The previously verified continuation theorem makes all 210-grid twists meromorphic, for every λ_5. If a possible twist pole at s_* survives, choose a rightmost pole $s_0=s_*+n$ of that grid in its integer-shift class, where $n\geq 0$. This exists by absolute convergence in a common right half-plane. At its highest pole order, all positive Taylor shifts vanish. The embedded 42-grid coordinates also

vanish: their only possible pole is the real point 2, whereas this shift class is nonreal. The Hecke-5 equations therefore reduce to their five-term sums, independently of λ_5.

The exact old-zero leading elimination in `level21_22_extra_prime5_poles.py` has 208 nonzero Laurent-monomial pivots and two remaining coordinates. Its selected minor is

$$-\frac{x^2-1}{3x^3y}P(x,y), \qquad P = 4x^4+3x^3y+3x^3-4x^2y+4x^2-3xy+3x-4y,$$

where $x = 2^{s_0-1}$ and $y = 3^{s_0-1}$. Thus a nonzero residue vector requires $(x^2-1)P(x,y) = 0$. Here $x = 2^n u$, so $x^2 \neq 1$ and $|y| = 3^n$.

Write $P = A(x)y + B(x)$ with

$$A = 3x^3-4x^2-3x-4, \qquad B = 4x^4+3x^3+4x^2+3x.$$

If $P = 0$, then $|B|^2 - 9^n|A|^2 = 0$, including when $A = 0$. Since $\overline{u} = u^{-1} = 3/2 - u$, exact reduction modulo $2u^2-3u+2$ gives four times this difference as

$$\begin{aligned} H(r,t) = {} & 64r^8+72r^7-36r^6t^2+52r^6+72r^5t^2+18r^5-55r^4t^2+73r^4 \\ & -126r^3t^2+72r^3-52r^2t^2+36r^2-72rt^2-64t^2, \qquad (r,t) = (2^n,3^n). \end{aligned}$$

For $n = 0,1,2$ its values are $54, 10656, -2232288$. For $n \geq 3$, its largest exponential base is 576, with coefficient -36. At $n = 3$ this term exceeds the sum of the absolute values of all other terms by $3706333056 > 0$; all their bases are strictly smaller, so the strict inequality persists for every $n \geq 3$. Therefore $H \neq 0$ for all $n \geq 0$, a contradiction.

This proves entireness. Rapid vertical decay follows from the quotient: outside fixed small disks about the periodic denominator zeros it is immediate, and inside those disks the maximum-modulus principle applies after cancellation. Mellin inversion therefore also gives the genuine Fricke transformation of the quadratic Fourier twist at conductor 525. The verifier `level21_22_weight2_chi5.py` recomputes the 208-pivot determinant, center-exchange identity, exact polynomial reduction and all uniform inequalities; its companion JSON records source hashes and the successful certificate values.

13. Removal of the remaining level-21 weight-two pole

The 42-grid is now entire for every allowed weight, Fricke sign, ramified sign and unramified eigenvalue at level 21. The earlier pole theorem already proves this except when

$$k = 2, \quad (\eta, \sigma_3, \sigma_7) = (-1,-1,1), \quad \lambda_2 = -3/2.$$

We treat that exception below. This statement concerns pole cancellation; full modularity at level 21 is not asserted here.

Use the entire quadratic twist and its conductor-525 Fricke identity proved in Section 12. Combining it with the Ramanujan-5 identity separates the two even pairs. Set

$$A = \begin{pmatrix} 5 & 1 \\ -21 & -4 \end{pmatrix}, \quad B = \begin{pmatrix} 5 & -1 \\ 21 & -4 \end{pmatrix}, \quad C = \begin{pmatrix} 5 & -2 \\ -42 & 17 \end{pmatrix}, \quad D = \begin{pmatrix} 5 & 2 \\ 42 & 17 \end{pmatrix}.$$

They have determinant one, and the separated identities are

$$f|(A-1)t_{-1/5} + f|(B-1)t_{1/5} = 0, \qquad f|(C-1)t_{2/5} + f|(D-1)t_{-2/5} = 0. \tag{21.4}$$

Indeed the elementary matrix formula is

$$\frac{1}{5}W_{21}^{-1}t_{a/5}W_{525}t_{-b/5} = \begin{pmatrix} 5 & -b \\ -21a & (21ab+1)/5 \end{pmatrix}, \qquad ab \equiv -1 \pmod 5.$$

The pairs $(a,b) = (1,-1), (-1,1), (2,2), (-2,-2)$ give the displayed matrices and the translations in (21.4).

Let $E = \left(\begin{smallmatrix} 2 & -1 \\ -21 & 11 \end{smallmatrix}\right)$, $T = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$ and $U = \left(\begin{smallmatrix} 1 & 0 \\ 21 & 1 \end{smallmatrix}\right)$. These already fix f. Direct multiplication gives

$$C = ETD^{-1}UE, \qquad D = UEC^{-1}ET.$$

For $c = f|(C-1)$ it follows that $f|(D-1) = -c|C^{-1}ET$. The second relation (21.4) now says

$$c = c|C^{-1}Et_{1/5}, \qquad C^{-1}Et_{1/5} = \begin{pmatrix} -8 & 17/5 \\ -21 & 44/5 \end{pmatrix}.$$

This matrix has determinant one and trace $4/5$, so is elliptic of infinite order. The positive-weight holomorphic elliptic lemma gives $c = 0$, and hence $f|C = f|D = f$.

For the three Hecke-2 matrices

$$M_0 = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}, \quad M_1 = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}, \quad M_2 = \begin{pmatrix} 1 & 1 \\ 0 & 2 \end{pmatrix}$$

the exact identities are

$$M_0A = -T^{-1}E^{-1}M_1, \qquad M_1A = CM_2, \qquad M_2A = E^{-1}U^{-1}DM_0.$$

At even weight the minus sign is immaterial. Since all the left factors on the right fix f, the Hecke identity gives $\lambda_2 f|A = \lambda_2 f$. Here $\lambda_2 = -3/2 \neq 0$, so A fixes f, and the first relation (21.4) gives the same conclusion for B.

Adjoin the four newly proved matrices to the known maps and construct the closed Taylor system on $Q = 126$. All map rows send a grid point to another grid point; all Ramanujan conductors divide Q^2. The same exact Hecke-$2, 3, 7$ rows are included, with the fixed exceptional parameters. At $h = s - 1 = 1$, its leading matrix has rank 126 over $\mathbb{Q}$: reduction modulo 1000003 has rank 126, with every specialized denominator nonzero modulo that prime. Hence the leading matrix is not identically singular. The closed Taylor continuation lemma proves meromorphic continuation on the entire 126-grid.

Next add unit rows for the embedded old coordinates $3j$, $0 \leq j < 42$. Elimination of this augmented symbolic leading matrix has 126 pivots, each a nonzero rational Laurent monomial in $2^h, 3^h, 7^h$. Thus it has full rank for every complex h. If an integer pole above 2 existed on the enlarged grid, choose the largest such integer and its highest pole order. All positive Taylor shifts are holomorphic; the old-grid coordinates are holomorphic there by the earlier pole bound. The residue vector satisfies the augmented leading system, a contradiction. Consequently all 126-grid coordinates are holomorphic at every integer $s \geq 3$.

At $s = 2$ the full, unaugmented leading matrix is invertible, by the rank-126 certificate above. There are now no higher-shift pole sources, so its residue equation excludes every pole at 2. In particular the old-grid exception is removed. The earlier pole theorem therefore makes every 42-grid coordinate entire. Applying the same rightmost-pole argument in an arbitrary integer-shift class, with the uniformly invertible old-zero augmented matrix, then proves that the whole 126-grid is entire as well.

The verifier `level21_22_weight2_level21.py` checks all matrix identities, the rational rank certificate at $h = 1$, and every nonvanishing monomial pivot of the augmented system. Its companion JSON records the source hashes and successful ranks. Together with the earlier eight-sign pole theorem and the independently verified quadratic-twist certificate, this removes the last level-21 pole case without a Ramanujan bound, integrality assumption, or an assumed modularity theorem.

14. The quadratic twist modulo five in the exceptional level-22 case

Assume $N = 22$, $k = 2$, $(\eta, \sigma_2, \sigma_{11}) = (-1, -1, 1)$ and $\lambda_3 = -7/3$. We prove that the quadratic twist $F(s, \chi_5)$ is entire, with functional equation

$$F(s, \chi_5) = -\epsilon_f 550^{1-s} F(2 - s, \chi_5). \tag{22.5}$$

No value of λ_5 is needed in this step. The already proved continuation of all 330-grid twists holds for every λ_5.

The known matrix $E = \left(\begin{smallmatrix} 2 & -1 \\ -22 & 12 \end{smallmatrix}\right)$ has determinant 2 and eigenvalue $-\sigma_2 = 1$. With $T = t_1$,

$$ETW_{22} = -\frac{1}{5} t_{-1/10} W_{1100} t_{-1/10}.$$

Thus the translates at $\pm 1/10$ are exchanged under Fricke, and their even Mellin sum B is entire, rapidly decreasing in fixed vertical strips, with $B(s) = \epsilon_f 1100^{1-s} B(2-s)$. Set $h = s-1$, $x = 2^h$, $S_j = F(s, j/5) + F(s, -j/5)$ and $R = S_1 + S_2$. The ramified Hecke-2 relation, with $\lambda_2 = \sigma_2 = -1$, gives $-S_1 = x(S_2 + B)$. Therefore

$$\sqrt{5} F(s, \chi_5) = S_1 - S_2 = \frac{(x+1)R + 2xB}{x-1}. \tag{22.6}$$

The Ramanujan theorem makes R entire with functional equation of sign ϵ_f and conductor 550. Substitution of these two functional equations into (22.6) gives (22.5) meromorphically. Its only possible poles have $2^h = 1$.

We need a uniform old-zero rank statement. In the 330-grid leading system, set all embedded 66-grid coordinates to zero. The Hecke-5 rows then reduce to sums, independently of λ_5. Exact elimination leaves two coordinates after 328 nonvanishing monomial pivots. Form the 2×2 minors between the first residual row and the other rows. Remove only nonzero rational factors and powers of $x = 2^h, y = 3^h, b = 5^h$. There are nine distinct remaining polynomials in x, y. The greatest common divisor of their nonzero resultants against the first polynomial, eliminating y, is, up to a nonzero rational scalar,

$$\begin{aligned} &x^6 (x+1)^4 (49x^4 + 36)^2 (49x^4 - 183x^2 + 49) \\ &\cdot (49x^5 + 58x^4 + 18x^3 + 45x^2 + 36x + 36)\, Q(x), \end{aligned}$$

where

$$\begin{aligned} Q(x) = {} & 13024x^{12} + 15159x^{11} + 90542x^{10} + 214759x^9 + 211371x^8 + 60397x^7 \\ & - 64811x^6 - 75366x^5 - 22023x^4 + 5004x^3 + 4860x^2 + 1944x + 324. \end{aligned}$$

This is nonzero whenever $x = 2^n$, $n \geq 0$. All factors other than the quartic and Q are positive for $x \geq 1$. The quartic equals -85 at 1 and 101 at 2, and is strictly increasing for $x \geq 2$. For $x \geq 1$, the three negative terms of Q have total absolute value at most $162200x^6$, whereas $211371x^8$ alone is larger. Thus $Q > 0$. A nonzero common residue vector would force all these minors, and consequently all their resultants, to vanish. We conclude that the old-zero leading matrix has rank 330 whenever $x = 2^n$, $n \geq 0$, for arbitrary nonzero values of y, z, b.

Consider a possible pole s_* in (22.6). If s_* is nonreal, choose a rightmost 330-grid pole $s_0 = s_* + n$, $n \geq 0$, in its integer-shift class. Its old-grid coordinates are holomorphic, since their only possible pole is the real point 2. Its highest Laurent coefficient satisfies the old-zero leading system with $x = 2^n$. The full rank just proved is a contradiction.

The real candidate is $s_* = 1$. The earlier enlarged-grid certificate already excludes all integer poles at $s \geq 3$. At $s = 2$, a pole of order at least two would have vanishing highest coefficient on every old-grid coordinate, because the original pole theorem permits only a simple pole there. The same old-zero rank statement, now with $x = 2$, excludes such higher orders. Thus every 330-grid pole at 2 is at most simple. In each first Taylor term contributing from $s+1 = 2$ to $s = 1$, the binomial factor is $s + 1 - k = s - 1$. It cancels that simple pole. Higher shifts are holomorphic. Hence any pole vector at $s = 1$ again satisfies the old-zero leading system, now with $x = 1$, and must vanish. This excludes the last candidate in (22.6).

Entireness and the maximum-modulus principle around the periodic zeros of $x - 1$ imply rapid vertical decay, so Mellin inversion yields the genuine Fourier-twist Fricke identity. The exact verifier

`level21_22_weight2_chi5_22.py` recomputes the 328-pivot reduction, nine minors, resultant factorization, positivity bounds and center-exchange matrix. Its companion JSON records source hashes.

15. Removal of the remaining level-22 weight-two pole

Together with the earlier all-sign pole theorem, the following argument proves that all 66-grid twists are entire at level 22, for every allowed weight and parameter choice. Full modularity is a separate step.

It suffices to treat the exceptional case

$$k = 2, \quad (\eta, \sigma_2, \sigma_{11}) = (-1, -1, 1), \quad \lambda_3 = -7/3.$$

Suppose an original-grid pole at $s = 2$ exists. The independently verified extra-prime certificate then forces $\lambda_5 = -19/5$ and excludes all integer poles above 2 on the meromorphically continued 330-grid. The preceding quadratic-twist argument is independent of this value of λ_5 and proves that $F(s, \chi_5)$ is entire with sign $-\epsilon_f$ and conductor 550.

Write $S_1 = F(s, 1/5) + F(s, -1/5)$ and $S_2 = F(s, 2/5) + F(s, -2/5)$. Their sum is the entire Ramanujan-5 twist, with sign ϵ_f, and their difference is the entire quadratic twist, with sign $-\epsilon_f$. Therefore

$$S_1(s) = \epsilon_f 550^{1-s} S_2(2-s), \qquad S_2(s) = \epsilon_f 550^{1-s} S_1(2-s).$$

The rapid vertical bounds established above make these genuine cross-Fricke identities for the two even sums of Fourier translates.

Add their leading Taylor rows to the existing 330-grid system. Explicitly, put $Q = 330$, $\ell = 550$, $d = Q^2/\ell = 198$ and let (a, b) be $(1, 2)$ or $(2, 1)$. For every positive divisor t_0 of d and $t = \pm t_0$, set $u = -d/t$. The new leading row is

$$\sum_{v=\pm a} e_{vQ/5+t} - \eta \left(\frac{\ell t^2}{Q^2}\right)^h \sum_{v=\pm b} e_{vQ/5+u}, \qquad h = s - 1,$$

with indices taken modulo Q. The entire sums also give rows $e_{aQ/5} + e_{-aQ/5}$. These are precisely the ordinary Ramanujan Taylor rows with their source and target residue pairs separated; no individual additive functional equation is assumed.

At $h = 1$, $\lambda_3 = -7/3$, $\lambda_5 = -19/5$, this enlarged leading matrix has rank 330. The exact verifier `level21_22_weight2_level22.py` reduces it modulo 1000003, checks every specialized rational denominator is nonzero in that field, and obtains 330 pivots. This certifies a nonzero rational maximal minor. Since all positive integer shifts of 2 are holomorphic, any highest Laurent coefficient at 2 must satisfy this full-rank homogeneous system and is zero, a contradiction.

The last exceptional pole has therefore been excluded. The preceding all-sign, all-weight pole theorem proves entireness of every original 66-grid twist without any exceptional case. The companion JSON records the exact parameter specialization, rank, finite field and source hashes.

16. An entire quadratic twist for every parameter at levels 21 and 22

The completed pole arguments now show that every 42-grid additive twist at level 21, and every 66-grid additive twist at level 22, is entire. This removes the parameter restrictions from the auxiliary quadratic functional equations previously used only in the exceptional weight-two cases. The conclusion below is uniform in positive even weight, all Fricke and ramified signs, and every complex unramified eigenvalue. The further transformations needed for modularity are proved in the following sections.

Write $h = s - k/2$, $x = 2^h$, $S_j(s) = F(s, j/5) + F(s, -j/5)$, $R = S_1 + S_2$, and $B(s) = F(s, 1/10) + F(s, -1/10)$. The elementary center-exchange identities are

$$E_N T W_N = -c_N t_{-1/10} W_{L_N} t_{-1/10}, \qquad (c_{21}, L_{21}) = (1/10, 2100), \quad (c_{22}, L_{22}) = (1/5, 1100),$$

where $E_{21} = (2, -1; -21, 11)$ fixes f, whereas $E_{22} = (2, -1; -22, 12)$ acts by $-\sigma_2$. Their inverses exchange the opposite centers as well. Thus B is entire, rapidly decreasing in fixed vertical strips, with functional equation sign ϵ_f at level 21 and $-\sigma_2\epsilon_f$ at level 22. The respective conductors are L_N. The Ramanujan-5 twist R is entire and has sign ϵ_f and conductor $25N$ in both cases.

At level 21, Taylor–Fricke expansions pair the four additive twists at $\pm 1/5, \pm 2/5$ with twists at $\mp 5/21, \mp 5/42$. All partners are on the entire 42-grid. Taking arbitrarily long Taylor expansions therefore makes all four denominator-five twists entire. The good-prime Hecke-2 relation is

$$\lambda_2 S_1 = (x + x^{-1}) S_2 + xB.$$

Consequently

$$\sqrt{5} F(s, \chi_5) = S_1 - S_2 = \frac{(x + x^{-1} - \lambda_2) R + 2xB}{x + x^{-1} + \lambda_2}, \qquad F(s, \chi_5) = \epsilon_f 525^{-h} F(k - s, \chi_5).$$

The quotient identity initially holds in a right half-plane and the functional equation follows meromorphically from those of R and B. Its left side has just been proved entire, including when $\lambda_2 = 0$ or the denominator has double zeros.

At level 22, the two twists at $\pm 1/5$ are entire because their Fricke partners $\mp 5/22$ belong to the entire 66-grid. The ramified Hecke-2 relation gives

$$\sigma_2 S_1 = x(S_2 + B).$$

It follows that S_2 is entire as well, without needing continuation of the individual partners $\mp 5/44$. Here the quotient and its functional equation are

$$\sqrt{5} F(s, \chi_5) = \frac{(x - \sigma_2) R + 2xB}{x + \sigma_2}, \qquad F(s, \chi_5) = -\epsilon_f 550^{-h} F(k - s, \chi_5).$$

The calculation is valid for either ramified sign and does not involve λ_3 or λ_5.

In each case the quotient has rapid decay on every fixed vertical strip away from small disks around the periodic denominator zeros. After pole cancellation, the maximum-modulus principle transfers those bounds inside the disks. Mellin inversion therefore gives genuine Fricke transformations of the quadratic Fourier twist. In particular, at level 21 the two even sums S_1, S_2 have separate conductor-525 Fricke laws; at level 22 they are exchanged by Fricke at conductor 550.

Uniform additional transformations at level 21. Set

$$A = \begin{pmatrix} 5 & 1 \\ -21 & -4 \end{pmatrix}, \quad A' = \begin{pmatrix} 5 & -1 \\ 21 & -4 \end{pmatrix}, \quad C = \begin{pmatrix} 5 & -2 \\ -42 & 17 \end{pmatrix}, \quad D = \begin{pmatrix} 5 & 2 \\ 42 & 17 \end{pmatrix}.$$

The separated even functional equations give

$$f|(A - 1)t_{-1/5} + f|(A' - 1)t_{1/5} = 0, \qquad f|(C - 1)t_{2/5} + f|(D - 1)t_{-2/5} = 0.$$

The identities $C = ETD^{-1}UE$ and $D = UEC^{-1}ET$, with $E = E_{21}$ and $U = (1, 0; 21, 1)$, are independent of the parameters. Hence $g = f|(C - 1)$ satisfies

$$g = g|C^{-1}Et_{1/5}, \qquad C^{-1}Et_{1/5} = \begin{pmatrix} -8 & 17/5 \\ -21 & 44/5 \end{pmatrix}.$$

The matrix has determinant one and nonintegral trace $4/5$, so is elliptic of infinite order. The positive-weight holomorphic elliptic lemma gives $g = 0$, and then D fixes f as well.

Let $M_0 = (2, 0; 0, 1)$, $M_1 = (1, 0; 0, 2)$, and $M_2 = (1, 1; 0, 2)$. The checked identities

$$M_0 A = -T^{-1}E^{-1}M_1, \qquad M_1 A = CM_2, \qquad M_2 A = E^{-1}U^{-1}DM_0$$

now imply $\lambda_2(f|A - f) = 0$. Thus A and A' also fix f when $\lambda_2 \neq 0$. No division by λ_2 is made in the zero case. The following section treats the nonzero-eigenvalue case; the later zero-eigenvalue argument completes the other case.

The exact checker `level21_22_modularity_chi5.py` verifies the center matrices, both uniform scalar functional equations, the elliptic matrix, and all three Hecke coset identities. Entireness is the analytic input proved by the earlier all-parameter grid pole theorems.

17. Completion at level 21 when the Hecke-2 eigenvalue is nonzero

Assume the all-parameter original-grid entireness theorem and the transformations established in Section 16. In this section $\lambda_2 \neq 0$, so T, U, E, A, A', C, D all fix f. Put $\delta = -\sigma_3$ and

$$P = \begin{pmatrix} 3 & 1 \\ -21 & -6 \end{pmatrix}, \qquad Q = \begin{pmatrix} 3 & -1 \\ 21 & -6 \end{pmatrix}.$$

Matrices differing by a nonzero scalar act identically in even weight. The Ramanujan-3 functional equation, justified by the entire 42-grid, gives

$$(f|P - \delta f)|t_{-1/3} + (f|Q - \delta f)|t_{1/3} = 0.$$

The exact identity $PT^{-1}Q^{-1} = CE^{-1}U^{-1}$ holds projectively. Thus, with $F = f|P$, we have $f|Q = F|T^{-1}$. Writing $h = F - \delta f$ and using the period one of f, the preceding relation becomes $h + h|t_{-1/3} = 0$. In particular

$$h|t_{1/3} = -h, \qquad h|t_{2/3} = h, \qquad h|T = -h, \qquad F|T^{-1} = 2\delta f - F.$$

The following three identities are exact, up to scalar:

$$M_0P = APM_1, \qquad M_1P = EC^{-1}PT^{-1}M_2, \qquad M_2P = C^{-1}PM_0.$$

Applying the Hecke-2 law to $f|P$ therefore gives

$$\lambda_2 F = F|(M_0 + M_1 - M_2) + 2\delta f|M_2, \qquad \lambda_2 h = h|(M_0 + M_1 - M_2).$$

Right translation by $2/3$ fixes the left side and $h|M_0$; it changes the signs of $h|M_1$ and $h|M_2$. Subtracting yields $h|M_1 = h|M_2$. Since the two slash factors are equal, this says that h has period $1/2$, and hence period one. Together with $h|T = -h$, it proves $h = 0$. We have established

$$f|P = \delta f, \qquad f|Q = \delta f.$$

The matrix P normalizes $\Gamma_0(21)$, as does W_{21}. Take the seven matrices T, U, E, C, D, A, A' and conjugate each by each of $1, P, W_{21}, PW_{21}$. Every resulting matrix is integral of determinant one, with lower-left entry divisible by 21, and fixes f. There are 26 distinct resulting matrices. Exact Todd–Coxeter enumeration in $\mathrm{PSL}_2(\mathbf{Z}) = \langle S, T \mid S^2 = (ST)^3 = 1 \rangle$ gives their subgroup index 32. Since $[\mathrm{PSL}_2(\mathbf{Z}) : \overline{\Gamma_0(21)}] = 32$, they generate $\overline{\Gamma_0(21)}$. Consequently f is modular for $\Gamma_0(21)$.

The polynomial coefficient bound gives polynomial growth at all cusps; periodicity after a cusp scaling therefore excludes negative Fourier powers. There are four cusp classes, represented by $1/d$ for $d \mid 21$. All these points are on the entire 42-grid. A nonzero constant term at any of these cusps would give a pole at $s = k$ of its additive Mellin transform, whereas that transform is entire. All constant terms vanish. Thus $f \in S_k(\Gamma_0(21))$ whenever $\lambda_2 \neq 0$.

The exact script `level21_22_modularity_generic21.py` verifies the five matrix identities used above and exports the subgroup to the bounded Magma input `level21_22_modularity_generic.m`. The exported input completes with index 32. The case $\lambda_2 = 0$ is not covered by this argument: the first Hecke identity uses A invariance, which was obtained by division by λ_2.

18. Additional exact reductions when $N = 21$ and $\lambda_2 = 0$

This section supplies the preliminary identities for the complete proof in Section 19. Retain P, Q, δ from the generic level-21 proof and set $F = f|P$, $G = f|P^{-1}$, $h = F - \delta f$, $g = G - \delta f$. The Ramanujan-3 argument does not use $\lambda_2 \neq 0$, so $h|t_{1/3} = -h$ still holds. Here $A = P^{-2}$ projectively. The same Hecke coset identities, without assuming that A fixes f, give

$$0 = G|M_1 + F|M_0 + (2\delta f - F)|M_2.$$

Subtracting the zero Hecke eigenrelation for δf gives

$$g|M_1 + h|M_0 - h|M_2 = 0, \qquad g = h|t_{1/2} - h|\begin{pmatrix} 4 & 0 \\ 0 & 1 \end{pmatrix}.$$

Consequently g has period $2/3$ and hence period two. Both F and G have period two, proving two additional integral invariances:

$$f|PT^2P^{-1} = f, \quad PT^2P^{-1} = \begin{pmatrix} 43 & 6 \\ -294 & -41 \end{pmatrix}, \qquad f|P^{-1}T^2P = f, \quad P^{-1}T^2P = \begin{pmatrix} 85 & 24 \\ -294 & -83 \end{pmatrix}.$$

The completion note combines these invariances with four zero-Hecke-2 steps and one Hecke-5 step to force $f|A = f$.

19. The zero Hecke-2 eigenvalue at level 21

The five finite Hecke steps below are certified by explicit words in previously known generators. The independent checker uses only integer matrix products and rational arithmetic; subgroup-search software and negative membership answers are not used in the verification.

Assume $\lambda_2 = 0$ and retain P, Q, δ, F, G, h, g from Section 18. We already know

$$h|t_{1/3} = -h, \qquad g = h|t_{1/2} - h|\begin{pmatrix} 4 & 0 \\ 0 & 1 \end{pmatrix}, \qquad h|P^{-1} = -\delta g.$$

In particular g has period $2/3$. The following two matrices fix f:

$$K_+ = PT^2P^{-1} = (43, 6; -294, -41), \qquad K_- = P^{-1}T^2P = (85, 24; -294, -83).$$

Let $\mathcal{H}_0$ be generated by T, U, E, C, D, K_+, K_- and their W_{21} conjugates. Every element of $\mathcal{H}_0$ fixes f.

Finite Hecke certificates. For a good prime p, write

$$M_{p,0} = (p, 0; 0, 1), \qquad M_{p,j+1} = (1, j; 0, p) \quad (0 \le j < p).$$

For each matrix B below the checker gives a permutation π and integral matrices γ_i satisfying, projectively, $M_{p,i}B = \gamma_i M_{p,\pi(i)}$. At $p = 2$, the equation $f|\sum_i M_{2,i} = 0$ can be multiplied on the right by *any* B. Hence, if two of its three γ_i already fix f, the third fixes f as well. At $p = 5$, the same cancellation works when B fixes f, with no condition on λ_5.

Apply this rule chronologically to the following five rows. After each row, adjoin its new invariant matrix and its W_{21} conjugate.

p	B	new invariant matrix
2	$(-337, 563; 252, -421)$	$(85, -71; -504, 421)$
2	$(-379, 498; 672, -883)$	$(293, -339; 1344, -1555)$
2	$(-293, 494; 672, -1133)$	$(293, -247; -1344, 1133)$
2	$(379, -539; 1092, -1553)$	$(379, -459; 2184, -2645)$
5	$(-391, 242; 21, -13)$	$(391, -283; -105, 76)$

For the first four rows, both other branches belong to the previously generated subgroup. For the last row, B and all five other branches belong to that subgroup. The explicit words in the certificate verify these assertions by integer matrix multiplication. The final subgroup contains

$$J = P(TU)P^{-1} = (-146, -25; 987, 169).$$

Thus $f|J = f$.

The final elliptic matrix. Since both J and TU fix f, the defect $h = f|P - \delta f$ fixes $L = TU = (22, 1; 21, 1)$:

$$h|L = f|PL - \delta f|L = f|JP - \delta f = h.$$

Since $h|P^{-1} = -\delta g$ and g has period $2/3$, h also fixes

$$K = P^{-1}t_{2/3}P = (29, 8; -98, -27).$$

Its anti-period $1/3$ implies that $t_{-4/3}$ fixes it. Therefore

$$h|M = h, \qquad M = LKt_{-4/3} = \begin{pmatrix} 540 & -571 \\ 511 & -1621/3 \end{pmatrix}, \qquad \det M = 1, \quad \operatorname{tr} M = -1/3.$$

This is elliptic of infinite order. The positive-weight holomorphic elliptic lemma gives $h = 0$, so $f|P = \delta f$. Because $P^2/3 = A^{-1}$ and $\delta^2 = 1$, A fixes f; the separated even functional equation then shows that A' fixes f as well. The exact index-32 generation and cusp argument from the generic level-21 proof now apply. Thus $f \in S_k(\Gamma_0(21))$ also when $\lambda_2 = 0$, for every allowed weight and every choice of signs.

The source `level21_22_modularity_zero21_certificate.py` checks all 18 Hecke branches, all 15 claimed memberships, and the final elliptic matrix. Its two small input files are `level21_22_modularity_zero21_closure.json` and `level21_22_modularity_zero21_words.json`. The generator order starts with T, U, E, C, D, K_+, K_-, followed by their seven W_{21} conjugates; each successive new matrix and its Fricke conjugate are appended. Each word is a list of generator-index/exponent pairs. In this ordering the final membership has the particularly short certificate

$$J = G_6G_{11}^{-1}G_{10}G_{24}^{-1}G_2^4 \quad \text{projectively.}$$

20. Level 22: the paired denominator-five transformations

For the completion below, put

$$T = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad U = \begin{pmatrix} 1 & 0 \\ 22 & 1 \end{pmatrix}, \quad E = \begin{pmatrix} 2 & -1 \\ -22 & 12 \end{pmatrix}, \quad W = \begin{pmatrix} 0 & -1 \\ 22 & 0 \end{pmatrix}, \quad \delta = -\sigma_2.$$

The known scalar laws are $f|T = f$, $f|U = f$, $f|E = \delta f$, and $f|W = \eta f$. Define

$$A = \begin{pmatrix} 5 & -2 \\ -22 & 9 \end{pmatrix}, \quad B = \begin{pmatrix} 5 & 2 \\ 22 & 9 \end{pmatrix}, \quad C = \begin{pmatrix} 5 & -1 \\ -44 & 9 \end{pmatrix},$$

$$D = \begin{pmatrix} 5 & 1 \\ 44 & 9 \end{pmatrix}, \quad P = \begin{pmatrix} 4 & 1 \\ 22 & 6 \end{pmatrix}, \quad R = T^{-1}WE.$$

The entire quadratic-5 and Ramanujan-5 functional equations proved above give the exchanged even-sum equations. Factoring their matrices by $W^{-1}t_{a/5}W_{550}t_{b/5}$, for $(a, b) = (1, -2), (-1, 2), (2, -1), (-2, 1)$, gives respectively A, B, C, D. Consequently, if $g_G = f|G - f$,

$$g_B = -g_A|t_{4/5}, \qquad g_C = -g_D|t_{-2/5}.$$

These exact factorizations are checked in `level22_finish_defect.py`. No individual denominator-five transformation is assumed here.

21. Completion of the level-22 converse theorem

Assume the all-parameter entire 66-grid theorem and the entire quadratic-5 functional equation proved earlier. We prove modularity and cuspidality for every allowed weight and sign, without any restriction on an unramified Hecke eigenvalue. All slash operators are normalized by the determinant; matrices differing by nonzero scalar are identified. Write $\delta = -\sigma_2$ and use $T, U, E, W, A, B, C, D, P, R$ defined in the preceding section. Thus $f|E = \delta f$, $f|W = \eta f$, $R = T^{-1}WE$, and $h = f|P - \delta f$.

Introduce an auxiliary defect

$$X = \begin{pmatrix} 3 & -1 \\ 22 & -7 \end{pmatrix}, \qquad Y = \begin{pmatrix} 3 & 1 \\ -22 & -7 \end{pmatrix}, \qquad u = f|Y - f.$$

The entire Ramanujan-3 functional equation gives

$$(f|Y - f)|t_{-1/3} + (f|X - f)|t_{1/3} = 0, \qquad f|X - f = -u|t_{-2/3}.$$

Indeed its two terms are factored by $W^{-1}t_{1/3}W_{198}t_{1/3} = Y$ and the analogous identity for X. The identities

$$X = W^{-1}Y^{-1}W^{-1} = (WT^{-1}E^{-1})Y(T^{-1}WE)$$

therefore imply

$$u|K = \eta u, \quad u|L = -\delta\eta u, \qquad K = \begin{pmatrix} 66 & 23 \\ -198 & -66 \end{pmatrix}, \quad L = \begin{pmatrix} 33 & 7 \\ -66 & -11 \end{pmatrix}.$$

Here $K = Y^{-1}W^{-1}t_{2/3}$ and $L = T^{-1}WEt_{2/3}$.

Set $M_0 = (1, 0; 0, 2)$ and $M_1 = (1, 1; 0, 2)$. The ramified Hecke law is $f|(M_0 + M_1) = \sigma_2 f = -\delta f$. Its four branch factorizations are

$$M_0X = WT^{-1}WYT^{-1}M_1, \qquad M_1X = E^{-1}T^{-1}E^{-1}CM_0,$$
$$M_0Y = WTWXM_1, \qquad M_1Y = WETWBT^{-1}E^{-1}M_0.$$

The earlier quadratic-5 pair and the exact identities $A = WPR$, $C = ETWPW$ show

$$f|C - f = \delta\eta h|W, \qquad f|B - f = -\eta h|Rt_{4/5}.$$

Subtracting the untwisted ramified Hecke law from the two branch equations thus gives

$$\delta u|t_{-2/3} = u|T^{-1}M_1 + \delta\eta h|WM_0,$$
$$-\delta u = -u|t_{-2/3}M_1 - \delta\eta h|Rt_{4/5}T^{-1}E^{-1}M_0.$$

In particular the first equation expresses h entirely in terms of u:

$$h = \eta u|J_1 - \delta\eta u|J_2, \qquad J_1 = \begin{pmatrix} 22 & 3 \\ -33 & 0 \end{pmatrix}, \quad J_2 = \begin{pmatrix} 11 & 1 \\ -22 & 0 \end{pmatrix}.$$

Substitution in the second equation yields the four-term relation

$$u|(\delta I - A_0 - \delta B_0 + C_0) = 0,$$

where

$$A_0 = \begin{pmatrix} 3 & -1 \\ 0 & 6 \end{pmatrix}, \quad B_0 = \begin{pmatrix} 81 & 22 \\ -198 & -51 \end{pmatrix}, \quad C_0 = \begin{pmatrix} 49 & 13 \\ -132 & -34 \end{pmatrix}.$$

This relation has an infinite elliptic factor. Namely, put

$$Z = KC_0 = \begin{pmatrix} 99 & 38 \\ -495 & -165 \end{pmatrix}.$$

The exact identity $A_0Z = KB_0$ and the law $u|K = \eta u$ give

$$(\delta u - u|A_0)|(1 + \delta\eta Z) = 0.$$

Since $(\operatorname{tr} Z)^2/\det Z = 44/25$, Z is elliptic of infinite order. The positive-weight holomorphic elliptic lemma, applied to the eigenvalue $-\delta\eta \in \{1,-1\}$, forces $u|A_0 = \delta u$. Finally

$$V = KLA_0 = \begin{pmatrix} 10 & 3 \\ -33 & -9 \end{pmatrix}, \qquad \det V = 9, \quad \operatorname{tr} V = 1.$$

The three established laws give $u|V = -u$. Again V is elliptic of infinite order, so $u = 0$. The expression for h now gives $h = 0$. Thus Y fixes f and P acts by δ.

For completeness, the elliptic lemma used twice here allows either sign: if g is holomorphic of positive even weight and $g|M = \pm g$ for an infinite-order elliptic M, expansion at its fixed point gives a nonzero coefficient only if a positive integer multiple of the irrational rotation angle is an integer multiple of π. This is impossible. Infinite order in these two rational-matrix cases follows also because the rational number $(\operatorname{tr} M)^2/\det M$ is not an integer, whereas a finite-order elliptic matrix would make it an algebraic integer.

Generation and cuspidality. Take determinant-squareclass representatives $1, E, W, EW$, labelled by $(a,b) \in (\mathbf{Z}/2\mathbf{Z})^2$. Each acts by $\delta^a\eta^b$. For each representative r and each $G \in \{T, E, W, P, Y\}$, form rGr'^{-1}, where r' has the product squareclass. These matrices are integral of determinant one, have lower-left entry divisible by 22, and fix f. The exact checker converts them to words in $\mathrm{PSL}_2(\mathbf{Z})$ and completes Todd–Coxeter enumeration with index 36. This equals the index of $\Gamma_0(22)$, so f is modular for $\Gamma_0(22)$.

The polynomial coefficient bound gives polynomial growth at each cusp; modularity then excludes negative Fourier powers. As 22 is squarefree, its four cusp classes have representatives $1/d$ for $d = 1, 2, 11, 22$. Every d divides 66, so every corresponding additive Mellin transform is entire by the proved grid theorem. A nonzero cusp constant would produce a pole at $s = k$, so all constants vanish. Therefore

$$f \in S_k(\Gamma_0(22)).$$

The exact checker is `level22_finish_complete.py`; it checks all matrix identities, both elliptic invariants, all sign choices, and the complete index-36 coset enumeration.

22. Level 25: uniform continuation of the 150-grid

This section proves meromorphic continuation, not pole cancellation or the level-25 converse theorem. Write $a_n = f_n n^{(k-1)/2}$ and

$$D_j(s) = (2\pi)^{-s}\Gamma(s)\sum_{n\geq 1} a_n e(nj/150)n^{-s}, \qquad j \bmod 150.$$

Set $h = s - k/2$, $x = 2^h$, $y = 3^h$, $z = 5^h$, and

$$\eta = i^k\epsilon_f \in \{\pm 1\}, \quad a = a_2 2^{1-k/2}, \quad b = a_3 3^{1-k/2}.$$

The level-25 hypotheses imply $a_5 = 0$. Both a and b are arbitrary complex numbers in the following argument.

Exact transformations. The untwisted and c_2-twisted functional equations imply that f is invariant under

$$T = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \qquad U = \begin{pmatrix} 1 & 0 \\ 25 & 1 \end{pmatrix}, \qquad E = \begin{pmatrix} 2 & -1 \\ -25 & 13 \end{pmatrix}.$$

Here U follows by conjugating a translation by the Fricke involution. For E, use $f_{(2)}(z) = f(z+1/2)$, its Fricke involution W_{100}, and that both Fricke eigenvalues are η. The product $W_{25}^{-1}T_{1/2}W_{100}T_{-1/2}$ is projectively E, so its eigenvalue is 1. Every word of length at most three in T, U, E and their inverses is therefore an exact transformation with eigenvalue 1. No additional modularity is assumed.

The finite leading system. Let e_j denote a row vector on $\mathbb{Z}/150\mathbb{Z}$. The certificate uses the following rows.

(1) For every indicated word $g = \left(\begin{smallmatrix} A & B \\ C & D \end{smallmatrix}\right)$ and j mod 150, if $\beta = g(j/150)$ belongs modulo one to the same grid, use
$$e_j - |Cj/150 + D|^{2h} e_{150\beta}.$$
If $Cj/150 + D = 0$, use e_j: the Mellin transform is entire because the exact transformation sends this cusp to infinity.

(2) For $q \mid 150$ with $\ell_q = \operatorname{lcm}(25, q^2)$ dividing 150^2, put $A_q = \{150a/q : 0 \le a < q, (a,q) = 1\}$ and $d_q = 150^2/\ell_q$. Use the entire Ramanujan row $R_q = \sum_{j \in A_q} e_j$ and, for every nonzero signed divisor t of d_q, the leading Taylor–Fricke row
$$\sum_{j \in A_q} \left(e_{j+t} - \eta \left(\frac{\ell_q t^2}{150^2} \right)^h e_{j - d_q/t} \right).$$
The sign is always η, including $5 \mid q$, since $5^2 \mid 25$.

(3) For $p = 2, 3, 5$ and $0 \le j < 150/p$, use the exact Hecke rows
$$\sum_{t=0}^{p-1} e_{j+150t/p} - \frac{\lambda_p}{p^h} e_{pj} + \frac{[p \ne 5]}{p^{2h}} e_{p^2 j}, \quad (\lambda_2, \lambda_3, \lambda_5) = (a, b, 0).$$

All indices are reduced modulo 150. The row entries are Laurent polynomials in x, y, z, a, b. In the first family this uses the elementary fact that if both a rational cusp and its image under an integral determinant-one matrix belong to the grid, the prime factors of $|Cj/150 + D|$ belong to $\{2, 3, 5\}$. The script checks this fact for every included row.

Proposition 22.1 (Uniform leading rank)**.** *For each $\eta \in \{\pm 1\}$ and every $(a, b) \in \mathbb{C}^2$, the displayed leading system has rank* 150 *over the field of meromorphic functions of s.*

The executable certificate is `level25_uniform_continuation.py`. It performs exact sparse elimination over $\mathbb{Q}(x, y, z, a, b)$, separately on the four strata
$$(a, b) \in (\mathbb{C}^\times)^2, \quad \mathbb{C}^\times \times \{0\}, \quad \{0\} \times \mathbb{C}^\times, \quad \{(0, 0)\}.$$
Together with the two Fricke signs, these are eight cases. Every case reaches rank 150.

To ensure this is uniform rather than merely generic, every numerator and denominator of every pivot is audited as follows. Collect its polynomial terms by their powers of x, y, z. At least one resulting coefficient is a single nonzero rational multiple of $a^r b^t$, with $r = 0$ whenever a was specialized to zero, and $t = 0$ whenever b was specialized to zero. That coefficient is nonzero at every allowed parameter value on the stratum. Thus the pivot numerator and denominator do not become the zero polynomial at any allowed specialization. Distinct monomials $x^i y^j z^\ell$ remain distinct exponentials after substituting $x = 2^h, y = 3^h, z = 5^h$, by unique factorization. The audited pivots therefore remain nonzero meromorphic functions of s for every allowed a, b. This proves the proposition without excluding exceptional Hecke eigenvalues. The source hash and all eight verification results are recorded in `level25_uniform_continuation.json`.

Corollary 22.2. *For every sequence satisfying the hypotheses of* [3] *at level* 25*, all $D_j(s)$, j* mod 150*, have meromorphic continuation to $\mathbb{C}$. Consequently every multiplicative twist whose modulus divides* 150 *has meromorphic continuation.*

Proof. Taylor expansion of the exact transformations and the Ramanujan-twist Fricke identities gives, to every finite order M, the leading system applied to $D(s)$, a finite sum of entire-coefficient matrices applied to $D(s + m)$ for $1 \le m < M$, and a remainder holomorphic on $\Re s > K - M$. The

constant K can be fixed uniformly for this finite set of rows. Every source coordinate remains on the complete 150-grid. The Hecke identities are exact and the unshifted Ramanujan rows are entire.

Absolute convergence starts the continuation in a right half-plane. At each step the full leading rank supplies a meromorphic left inverse and continues all coordinates one unit farther left; M is chosen large enough for the remainder to be holomorphic there. Iterating proves the claim. Finite Fourier inversion gives the statement about multiplicative twists. □

Remaining limitations. The continuation recurrence may propagate poles to the left. This certificate gives neither a bounded strip for their locations, entireness, nor the individual twisted functional equations required by a converse theorem. Those obligations remain explicit.

23. A level-25 first-jet obstruction

Use the meromorphic 150-grid and notation from Section 22. Assume $\eta = -1$. Let s_0 be a pole with every grid coordinate holomorphic at $s_0 + m$ for all integers $m \geq 1$, and let r_j be its highest-order Laurent-coefficient vector on the whole grid. The following statement applies only when at least one of $r_{30}, r_{60}, r_{90}, r_{120}$ is nonzero.

Proposition 23.1. *If $s_0 \neq k$, then*

$$s_0 = k/2 + 1, \qquad \lambda_2 = -\tfrac{1}{2}, \qquad \lambda_3 = -\tfrac{7}{3}.$$

In particular, this case is impossible when $k = 2$.

Proof. Write $x = 2^{s_0-k/2}$, $y = 3^{s_0-k/2}$, and $z = 5^{s_0-k/2}$. The exact leading rows admit 145 pivots which are nonzero Laurent monomials in x, y, z, independent of the two arbitrary Hecke eigenvalues $a = \lambda_2, b = \lambda_3$. The five free coordinates can be taken to be $r_{42}, r_{60}, r_{108}, r_{120}, r_{126}$; in particular $r_{30} = -r_{120}$ and $r_{90} = -r_{60}$.

For an exact matrix transformation $g = \left(\begin{smallmatrix} A & B \\ C & D \end{smallmatrix}\right)$, put $d = Cj/150 + D$ and let $\beta = g(j/150)$. Its first Taylor coefficient at $s_0 - 1$, after removing the common scalar $i(s_0 - k)$, is its leading coefficient at s_0 times C/d on coordinate 150β. For a Ramanujan–Fricke row translated by $t/150$, the corresponding factor is $150/t$ on every source coordinate. Apply the same leading elimination at $s_0 - 1$ to these first coefficients. Every identically zero leading row gives a necessary first-jet relation on r. This remains valid for poles of any order: one takes the highest Laurent coefficient after eliminating the unknown lower-point twists. No simplicity hypothesis is imposed.

These relations first give $r_{60} = -r_{120}/2$. Thus the assumed nonzero denominator-five component implies $r_{120} \neq 0$. Three remaining equations are $F_i r_{120} = 0$, where

$$\begin{aligned} F_1 &= 2ax + x^2 - 2, \\ F_2 &= -9ax + 18x^2 - 10y^2 + 9, \\ F_3 &= -3ax + 6by + 6x^2 + 2y^2 - 3. \end{aligned}$$

All multipliers suppressed here are nonzero Laurent monomials. The exact identities

$$2F_2 + 9F_1 = 5(9x^2 - 4y^2), \qquad 3F_3 - F_2 = 2(9by + 8y^2 - 9)$$

give $x^2 = 4y^2/9$, $a = x^{-1} - x/2$, and $b = y^{-1} - 8y/9$. In particular $\Re(s_0 - k/2) = 1$ and $|x| = 2$.

It remains to use the nonzero residual leading rows at $s_0 - 1$, not just their zero-row consequences. Normalize $r_{120} = 1$, substitute $y = \pm 3x/2$ and the displayed values of a, b, and retain the three unknown old-grid coefficients and all five unknown lower-point coefficients. Exact augmented minors imply the following two algebraic facts. First, an eight-column coefficient minor, up to a nonzero

Laurent monomial, is

$$\begin{aligned} D(U) = (U-8)(U^2-2U-40)(2U^2-11U-17)(11U^2-20U-76) \\ \cdot (U^4+24U^3-290U^2+380U-144) \\ \cdot (19U^4+180U^3-868U^2+606U-72), \qquad U=x^2. \end{aligned}$$

The gcd of $D(U)$ and $U^{\deg D}D(16/U)$ is one. Thus this minor never vanishes on $|U|=4$. Its five lower-point columns are consequently independent, so the full leading matrix at s_0-1 has rank 150. A holomorphic local left inverse bounds the pole order there by the pole order at s_0. This justifies taking the same Laurent order in the inhomogeneous first-jet equations; higher-order lower-point poles are not silently excluded.

Second, the augmented minors now imply that U is a zero of

$$\begin{aligned} P(U) = (U-4)(U-8)(2U^2-11U-17)(11U^2-20U-76) \\ \cdot (U^4+24U^3-290U^2+380U-144) \\ \cdot (63U^8+950U^7-7151U^6+55992U^5+19064U^4 \\ -617786U^3+712800U^2-857640U+431136). \end{aligned}$$

The two signs of y give identical equations: every initial coefficient is invariant under $(y,b)\mapsto(-y,-b)$. These are actual augmented minors, not rank assertions made after discarding possibly zero parameter denominators. Since $|U|=4$ and the coefficients are real, U must be a common zero of $P(U)$ and $U^{\deg P}P(16/U)$. Their monic greatest common divisor is exactly $U-4$. Consequently $x^2=4$ and $y^2=9$. The irrationality of $\log 2/\log 3$, by unique factorization, then forces $s_0-k/2=1$. Substitution gives the stated eigenvalues.

All row eliminations, the three scalar identities, the augmented minors and the reciprocal-polynomial greatest common divisor are checked by `level25_finish_eta_minus_cusp_poles.py`. Its manifest records source hashes and the precise limited claim. □

Corollary 23.2. *For $\eta=-1$, a rightmost pole $s_0\neq k$ has zero highest-order denominator-five component. The same holds at $s_0=k$ if the maximum pole order on the grid is at least two.*

Proof. The proposition reduces a nonzero component to $h=1$, $a=-1/2, b=-7/3$. The weight $k=2$ is already excluded. For $k\geq 4$, put $\beta=s_0-k=1-k/2$, a negative integer. Normalize the coefficient at the nth lower point by $i^n\beta(\beta-1)\cdots(\beta-n+1)$. All these factors are nonzero. For each known-map or translated Fricke row let L_n denote its leading scalar at s_0-n, let B denote its target coordinate or target-coordinate sum, and put respectively $c=C(Cj/150+D)/\det g$ or $c=\ell_q t/150$. The normalized systems are

$$A(s_0-n)v_n=\left(L_n\sum_{m=1}^{n}\frac{c^m}{m!}Bv_{n-m}\right)_{\text{rows}}.$$

The initial system, including $v_{0,120}=1$ and $v_{0,60}=-1/2$, has rank 150 and a unique solution. At shifts $n=1,2,3$ the coefficient matrices all have rank 150; therefore their holomorphic local left inverses bound the pole orders recursively by the initial order. This justifies comparing the same Laurent order throughout. The first two inhomogeneous systems are consistent, but at the third shift the entire Ramanujan-5 row has residual $-25/2$, a contradiction. All ranks, normalizations and the exact obstruction are checked by `level25_finish_minus_higherjet.py`, which is called by the main certificate.

At $s_0=k$ with maximum pole order $r\geq 2$, the zero-row first-jet relations remain applicable one Laurent order below the top: their common factor $s-k+1$ at the first lower point has a simple zero, and identically zero leading rows eliminate all lower-point coordinates as meromorphic germs. Thus

the same three F_i and cusp relation force $k = 2$, $a = -1/2, b = -7/3$. The full ranks at the lower points bound their pole orders by $r - 1$, since every positive Taylor coefficient has a simple zero at this weight point. For order $r - 1$, divide the coefficient at the nth lower point by $i^n(-1)^{n-1}(n-1)!$. These are the derivatives at $\beta = 0$ of the previous falling factorials. The resulting equations are exactly the same normalized systems, so the third-shift obstruction excludes this case too. □

Remaining obligations. This proposition does not control highest-order vectors supported away from denominator five or simple poles at the weight point. The companion positive-sign certificate treats $\eta = +1$. It does not prove new modular transformations or holomorphy at every cusp. The level-25 target remains $f \in M_k$, with cusp constants allowed.

Rightmost active poles. The statements remain valid if "rightmost" ignores simple poles at $s = k$. Indeed, a positive Taylor shift of a simple pole at k is holomorphic: at $s = k - m$, $m \geq 1$, its coefficient contains the factor $s + m - k$, which cancels that pole. Such poles therefore contribute no principal part to any lower-point equation used above. Equivalently, one may subtract their principal parts locally before taking Laurent coefficients. This extension is needed when descending through an integer-shift class containing an allowed cusp-constant pole.

24. The positive-Fricke denominator-five pole sector at level 25

Use the meromorphic 150-grid $D_j(s)$ and put $h = s - k/2$, $x = 2^h$, $y = 3^h$, $z = 5^h$, $a = \lambda_2$, $b = \lambda_3$. Assume $\eta = +1$ and even $k \geq 2$. Let s_0 be a rightmost pole in its integer-shift class, and let r be the maximum order of a pole there among the 150 coordinates.

Proposition 24.1. *If $s_0 \neq k$, or if $s_0 = k$ and $r \geq 2$, the coefficient of order r vanishes in all four denominator-five coordinates* $30, 60, 90, 120$.

Proof. Suppose otherwise. Uniform Laurent-monomial elimination of the leading matrix gives 146 pivots and free coordinates $48, 54, 102, 120$. The denominator-five coordinates depend only on coordinate 120, so normalize its highest coefficient to one. Two leading Hecke rows immediately give

$$a = -x - 2/x, \qquad b = -y - 3/y. \tag{25+.1}$$

We first treat $s_0 \neq k$.

Apply the same 146 safe pivots to the first-Taylor systems at $s_0 - 1$. Work over the original field $\mathbb{Q}(x, y, z, a, b)$, before making substitution (25+.1). There are four remaining lower-point coordinates. Eliminate these by determinant identities, clearing denominators only by nonzero Laurent monomials. This gives 39 exact meromorphic identities among the top-point coordinates. They are valid without any assumption on pole orders at the lower point: all its unknown germs have been eliminated identically.

Now impose (25+.1), combine these identities with the leading rows, and eliminate the other three top coefficients. Two resulting augmented minors are

$$25G(x^2 - 4), \qquad 25G(y^2 - 9), \qquad G = 9x^2 - 4y^2.$$

Their linear combination with coefficients $9, -4$ is $25G^2$. Hence $G = 0$, and in particular $\Re h = 1$ and $|x| = 2$. Both signs $y = \pm 3x/2$ give the same equations, since every original coefficient is invariant under $(y, b) \mapsto (-y, -b)$.

Put $U = x^2$, so $|U| = 4$. We use the full normalized Taylor systems, with falling-factorial normalization as in the negative-sign calculation. The initial system, including the normalization at coordinate 120, and the first two lower-point systems have coefficient minors, up to nonzero Laurent

monomials,

$$D_0(U) = (4U^2 + 3U - 5)(36U^4 + 91U^3 + 155U^2 - 125U - 84),$$
$$D_1(U) = (U - 4)(U^2 + 4U - 8)(7U^4 - 12U^3 - 64U^2 - 576U - 768),$$
$$D_2(U) = (U - 8)(9U^2 + 146U - 544)(531U^4 - 17892U^3 - 160736U^2 - 1050112U - 2451456).$$

The monic gcd of $D_i(U)$ with $U^{\deg D_i} D_i(16/U)$ is, respectively, $1, U - 4, 1$. Thus all three minors are nonzero on this circle unless $U = 4$. For $U \neq 4$, the lower-point full ranks bound their pole orders by r. The first lower-point system is consistent, but the second has the exact residual

$$-\frac{125(9U - 104)}{4U}$$

in the Hecke-2 row indexed by 20. It is nonzero on $|U| = 4$, since $|9U| = 36 < 104$. This is a contradiction.

It remains to treat $U = 4$. Together with $y^2 = 9$, unique factorization and the irrationality of $\log 2/\log 3$ give $h = 1$. Consequently $a = -3, b = -4$. If $s_0 \neq k$, then $k \geq 4$ and $\beta = s_0 - k = 1 - k/2$ is a negative integer. All falling-factorial normalizations are nonzero. Instead of assuming a rank statement at each lower point, form one block system containing all 150 coordinates at each of $s_0, s_0 - 1, s_0 - 2$. Its 450 columns have rank 450. This excludes poles rigorously even when a lower point might otherwise have larger pole order: take the largest order anywhere in this finite block. Its normalized coefficient vector solves the homogeneous full-rank block system and hence vanishes. All sources from points to the right of s_0 are holomorphic. This contradiction removes the exceptional branch.

At $s_0 = k$ and $r \geq 2$, the first-source factor has a simple zero. The same exact germ identities, read at order $r - 1$, give $G = 0$, therefore $k = 2$, $a = -3, b = -4$. The homogeneous lower-point block at the top Laurent order has zero top coordinate; full block rank excludes higher or equal orders at lower points. At the next order normalize lower-point coefficients by $i^n(-1)^{n-1}(n-1)!$, the derivative at zero of the falling factorial. The resulting block equations are identical to those just checked, with the top coefficient still of order r. Their full rank again gives a contradiction.

The complete exact computation is reproduced by `level25_finish_eta_plus_cusp_poles.py`. It checks the meromorphic-germ eliminations, the two augmented minors and their scalar identity, reciprocal-polynomial gcds, second-order obstruction, and rank-450 exceptional block. The manifest records all source hashes. □

Scope. The proposition leaves highest pole vectors supported entirely in the other grid coordinates. It does not prove holomorphy at all cusps or new modular transformations. Simple poles at $s = k$ are allowed: the level-25 conclusion is $f \in M_k$.

Rightmost active poles. The statements remain valid if "rightmost" ignores simple poles at $s = k$. Indeed, a positive Taylor shift of a simple pole at k is holomorphic: at $s = k - m$, $m \geq 1$, its coefficient contains the factor $s + m - k$, which cancels that pole. Such poles therefore contribute no principal part to any lower-point equation used above. Equivalently, one may subtract their principal parts locally before taking Laurent coefficients. This extension is needed when descending through an integer-shift class containing an allowed cusp-constant pole.

25. A source-free quotient for the level-25 prime-seven extension

This proves a relative continuation and pole-control statement used in the level-25 converse. The conclusion at this level is $f \in M_k$, with cusp constants permitted.

Use the already continued 150-grid, and enlarge to $Q = 1050$. Put $h = s - k/2$, $x = 2^h, y = 3^h, z = 5^h, w = 7^h$ and $a = \lambda_2, b = \lambda_3, c = \lambda_7$. Include the known transformation words of length at most three generated by T, U_{25}, E_2, W_{25}, with their tracked multipliers, all closed Ramanujan–Fricke

rows, and the exact Hecke rows for $2, 3, 5, 7$. No further transformation is assumed. Denote the genuine leading matrix by $A_0(s)$.

Let P select the 150 coordinates with indices divisible by 7. These are precisely the embedded old grid. Exact elimination of

$$\begin{pmatrix} A_0(s) \\ P \end{pmatrix}$$

using only nonzero rational Laurent-monomial pivots gives 982 pivots, 68 free coordinates, and no residual rows, for each Fricke sign. Every pivot is independent of the arbitrary eigenvalues a, b, c. Consequently the kernel has an explicitly reconstructed basis $W(s)$ with 68 columns, normalized to the identity on its free coordinate set J. Its entries are Laurent polynomials in x, y, z, w with polynomial dependence on a, b, c; in particular they are entire functions of s for every fixed choice of eigenvalues. We have, identically,

$$A_0(s)W(s) = 0, \qquad PW(s) = 0, \qquad W(s)|_J = I_{68}. \tag{25.7}$$

The verifier `level25_cusp_aux_prime7_symbolic.py` reconstructs every basis vector and checks the following stronger coordinate identities over the exact rational function field. For every known-map row with $C \neq 0$, its target coordinate in each column of W is zero. For every Ramanujan–Fricke row, the sum of its target coordinates in each column is zero. Therefore, if $A_m(s)$ is any positive-order Taylor matrix,

$$A_m(s)W(s+m) = 0 \qquad (m \geq 1). \tag{25.8}$$

Indeed, in a known-map row a positive Taylor term is a scalar times the target coordinate, with a positive power of C; in a Fricke-sum row it is a scalar times the same target sum. These checks prove (25.8) at all orders. Constancy of W is not required: the coordinate identities hold identically at every argument. The matrices W here do depend on the eigenvalues and exponentials.

In the initial common right half-plane define

$$\widetilde{\mathbf{F}}(s) = \mathbf{F}(s) - W(s)\mathbf{F}(s)|_J.$$

Then $\widetilde{\mathbf{F}}|_J = 0$ and $P\widetilde{\mathbf{F}} = P\mathbf{F}$, so the old 150-grid is unchanged. Equations (25.7)–(25.8) imply that every finite Taylor identity has exactly the same holomorphic remainder after this replacement. The gauged leading equations, together with the now known meromorphic old coordinates, have full column rank and an entire left inverse: the preceding monomial elimination inverts them without any non-monomial denominator. Taylor strip induction therefore continues every quotient coordinate meromorphically to the plane, for both signs and all a, b, c. This makes no assertion about continuation of the 68 discarded auxiliary coordinate functions individually.

Proposition 25.1 (No new rightmost old-zero polar vector)**.** *Let s_0 be a rightmost pole, in a class modulo integer translation, of the continued quotient family. Its highest nonzero Laurent vector has a nonzero component on the original 150-grid.*

Proof. All positive shifts are holomorphic, so the leading polar vector v satisfies $A_0(s_0)v = 0$ and $v|_J = 0$. If $Pv = 0$ also, the everywhere invertible augmented gauged system gives $v = 0$, a contradiction. The statement is valid for any pole order and every parameter specialization because all pivots are monomials. □

Thus the prime-seven extension introduces no extra highest-order old-zero pole obstruction. It does not assume that the three original denominator-25 modes vanish when the denominator-five components vanish. Those modes are treated in the following zero-λ_2 calculation and, for nonzero λ_2, in Sections 27, 28, and 32.

Independent audit of the level-25 zero-λ_2 certificates. The independent runner executes the original source files without changing their mathematical contents. It redirects only scratch filenames under `/tmp`, and records hashes of every source read. The stages are `prime7_firstjet`, `prime7_jetrank`, and `prime7_jetideals`, each with arguments η and `zero2`. The original `prime7_symbolic` verifier is also run with `zero2old150` to check the specialized source-free kernel directly, rather than relying only on specialization of a previous generic certificate. Run results are recorded in the companion `level25_zero2_independent_manifest.json`.

η	augmented rank/free	gauged leading rank/free	combined constraints	normalized final rows
$+1$	982/68	1047/3	540	95
-1	982/68	1046/4	852	68

The plus-sign free coordinates are $294,630,672$; one safe jet pivot removes 630. The minus-sign free coordinates are $420,714,840,882$; two safe jet pivots remove $420,840$. Both remaining two-coordinate systems have unit ideals in both projective charts.

Algebraic scope. The common field retains arbitrary λ_3 and λ_7. Every leading or jet pivot is a nonzero rational Laurent monomial in x, y, z, w; no eigenvalue-dependent polynomial is inverted. The first jet is transported through the original elimination operations evaluated at $s-1$, and only identically zero lower-point rows supply first-source constraints. Nonzero residual leading rows are retained separately. Thus no bound on the order of a lower-point pole is assumed. The later normalization removes only rational constants and Laurent monomials. The two projective charts cover every nonzero remaining residue vector, and the saturation equation $txyzw-1=0$ uses only the nonvanishing of the four exponentials.

The comparison verifier checks the complete freshly exported polynomial lists against the stable worker certificates, not merely their printed Groebner basis. Fresh assertion-based Magma checks then test $1 \in I$ for all four chart ideals. Hence the common zero set on the exponential torus is empty for arbitrary complex remaining eigenvalues.

The first-source normalization. For a known-map row let $d = Cr + D$ and $\rho = \det(g)/d^2$. Expanding the transformed Mellin integral to first order and integrating the derivative term by parts gives

$$i(s+1-k)\,\mu\rho^{-(s+1-k/2)}(C/d)D_{gr}(s+1).$$

At the lower point s_0-1, after suppressing $i(s_0-k)$, this is exactly the source implemented in the original checker: its exponential coefficient is evaluated at s_0, while the row elimination coefficients are evaluated at s_0-1. The analogous translated Ramanujan row has factor Q/t. The original source performs precisely these two different evaluations.

Allowed simple poles. Call a pole active unless it is a simple pole at $s=k$. Choose a rightmost active pole in one integer-translation class; the original right half-plane of convergence guarantees its existence whenever an active pole exists. A simple pole at k in a positive Taylor source is canceled by the factor $s+m-k$ in $\binom{s+m-k}{m}$. It contributes no principal part to any equation used below. Consequently the leading polar vector and all zero-row first-source compatibility equations remain valid at a rightmost active pole. At $s_0 \neq k$ divide by s_0-k. At $s_0=k$ of order $r \geq 2$, take the coefficient of order $r-1$ in $(s-k)D(s)$ instead. Derivatives of holomorphic coefficient matrices affect only lower orders. The entire lower-point germs have already been canceled by the zero-row syzygies, so this step does not presume any lower-point order bound.

The augmented old150-zero matrix has monomial full rank after its 68 gauges are appended. Therefore the highest nonzero Laurent vector of the quotient cannot vanish on the embedded old150 coordinates. This also rules out an auxiliary pole of strictly larger order than every old-coordinate pole at the same point.

For $\lambda_2 = 0$, the independently established denominator-five sector exclusions simplify substantially. For $\eta = -1$, their first-source equations give

$$x^2 = 2, \qquad 9x^2 = 4y^2.$$

For $\eta = +1$, the leading equation and first germ-elimination identities give

$$x^2 = -2, \qquad 9x^2 = 4y^2.$$

In either case the second equality, with $x = 2^h, y = 3^h$, forces $\Re h = 1$, hence $|x^2| = 4$, contradicting the first equality. The same argument works one Laurent order down at $s_0 = k$ of order at least two. Thus the older higher-jet exceptional branches are unnecessary for this zero-λ_2 audit. It still uses their already proved first-source sector identities, which are explicit dependencies rather than consequences of the new chart checks.

The highest quotient pole vector therefore has zero denominator-five components. The newly rebuilt projective chart identities force that entire vector to vanish, a contradiction. The recorded source and ideal checks all passed. This proves that all continued quotient coordinates, in particular the old150 twists, have only possible simple poles at k, for both Fricke signs and $\lambda_2 = 0$. It does not address $\lambda_2 \neq 0$ or claim cuspidality at level25.

26. Exact algebra for level-25 modularity

We first give the algebraic reduction. The local Mellin criterion in Section 26.1 supplies the analytic step once the simple-pole restriction has been proved.

Put

$$A = \begin{pmatrix} 3 & 1 \\ -25 & -8 \end{pmatrix}, \quad A' = \begin{pmatrix} 3 & -1 \\ 25 & -8 \end{pmatrix}, \quad B = \begin{pmatrix} 4 & 1 \\ -25 & -6 \end{pmatrix}, \quad B' = \begin{pmatrix} 4 & -1 \\ 25 & -6 \end{pmatrix}.$$

The established scalar laws are $f|T = f|U = f|E = f$ and $f|W = \eta f$, where $U = (1, 0; 25, 1)$, $E = (2, -1; -25, 13)$ and $W = W_{25}$. Write $u = f|A - f$ and $v = f|B - f$. The Ramanujan functional equations at $3, 4, 6$ give paired defect laws for (A, A'), (B, B'), and (B^{-1}, B'^{-1}), respectively. For each pair (G, G') at denominator q, the precise equation is

$$(f|G - f)|t_{-1/q} + (f|G' - f)|t_{1/q} = 0;$$

the checked center identity is $W^{-1}t_{1/q}W_{25q^2}t_{1/q} = G$.

The $q = 4, 6$ equations imply

$$v|M = v, \qquad M = \begin{pmatrix} 6 & 3 \\ -50 & -19 \end{pmatrix}.$$

This matrix has determinant 36 and fixed points $-1/5, -3/10$. The matrix B is parabolic, with trace -2, determinant one, and fixed point $-1/5$. Thus a local principal part proportional to $(z + 1/5)^{-k}$ is preserved by B and cancels in v. The local Mellin argument in Section 26.1 proves $v = 0$ under the simple-pole restriction.

Assume now $v = 0$. Set

$$M_\infty = (2, 0; 0, 1), \quad M_0 = (1, 0; 0, 2), \quad M_1 = (1, 1; 0, 2), \qquad f|(M_\infty + M_0 + M_1) = \tau_2 f.$$

The three exact Hecke factorizations are

$$M_\infty A = B^{-1}M_0, \quad M_0 A = G_1 M_1, \quad M_1 A = G_2 M_\infty,$$

where

$$G_1 = U^{-1}WA^{-1}W = \begin{pmatrix} 3 & -1 \\ -50 & 17 \end{pmatrix}, \qquad G_2 = E^{-1}U^{-1}WA^{-1}WT = \begin{pmatrix} 11 & 7 \\ 25 & 16 \end{pmatrix}.$$

Both outer determinant-one words fix f. The $q = 3$ pair also gives

$$u|K = \eta u, \qquad K = A^{-1}W^{-1}t_{2/3} = \begin{pmatrix} 75 & 26 \\ -225 & -75 \end{pmatrix}.$$

Consequently the two unknown Hecke terms reduce to $-u|t_{-2/3}M_1$ and $-u|t_{1/3}M_\infty$. We obtain, with no division by τ_2,

$$u|(\tau_2 + D + D^{-1}) = 0, \qquad D = \begin{pmatrix} 3 & -1 \\ 0 & 6 \end{pmatrix}.$$

The local Mellin argument uses this fixed second-order dilation recurrence at $A\infty = -3/25$. Its analytic continuation is justified below without assuming rapid vertical-strip bounds.

Once the local Mellin criterion gives $u = v = 0$, the matrices T, U, E, A, B and their W conjugates fix f. The exact checker `level25_aux_modularity_matrix.py` verifies all the displayed matrix identities and completes Todd–Coxeter enumeration with index 30 in $\mathrm{PSL}_2(\mathbf{Z})$. This is the index of $\Gamma_0(25)$, so these invariances suffice for modularity. The cusp-growth argument in Section 26.1 then gives $f \in M_k(\Gamma_0(25))$; cuspidality is not required or asserted.

26.1. **A local Mellin criterion for the level-25 transformations.** Throughout, the slash operator is determinant-normalized, $k > 0$ is even, and $f(z) = \sum_{n\geq 1} a_n e(nz)$ has polynomially bounded coefficients. Write $D_\alpha(s)$ for its completed additive Mellin transform.

Lemma 26.1 (Transport to a truncated Mellin transform at infinity)**.** *Let $\gamma = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \mathrm{GL}_2^+(\mathbb{R})$, $c \neq 0$, and $\alpha = a/c$. Suppose that D_α continues meromorphically with no poles except, possibly, a simple pole at k. For every fixed $x \in \mathbb{R}$, the initially convergent integral*

$$J_{\gamma,x}(s) = \int_1^\infty (f|_k\gamma)(x + iy)y^{s-1}\, dy$$

continues meromorphically with no poles except, possibly, a simple pole at zero. No growth condition on the continued D_α is needed.

Proof. Put $\delta = \det(\gamma)/c^2 > 0$, $q = -i(x + d/c)$, and $t = 1/y$. For a nonzero constant C independent of t,

$$(f|_k\gamma)(x + i/t) = Ct^k(1 + qt)^{-k} f\left(\alpha + \frac{i\delta t}{1 + qt}\right).$$

Use a smooth cutoff supported near $t = 0$ and identically one there. The part away from zero is entire in s. The finite Taylor–Mellin identity to order M expresses the remaining integral as

$$C \sum_{m=0}^{M-1} \frac{q^m}{m!} \prod_{j=1}^{m} (j - s)\, \delta^{-(k-s+m)} D_\alpha(k - s + m) + R_M(s).$$

Here R_M is holomorphic on $\Re s < M - B$ for a fixed B depending only on the polynomial coefficient bound. Entire cutoff terms have been included in R_M. This is the same finite Taylor identity as the local Taylor–Fricke relation: Taylor-expand the change in the argument of f, integrate the derivatives by parts, and use the coefficient bound to bound the Taylor remainder. For any prescribed compact set in s, taking M large makes the remainder holomorphic on that set.

The mth displayed summand can have a pole only at $s = m$. For $m > 0$ the factor $m - s$ cancels that simple pole. The $m = 0$ term has at most a simple pole at zero. This proves the assertion locally on every compact set, and hence on the whole plane. □

Lemma 26.2 (Dilation recurrence detected by a one-sided Mellin transform)**.** *Let $H \in C^\infty(\mathbb{R})$ have at most exponential growth, and let $SH(t) = H(t + L)$, where $L > 0$. Suppose $P(S)H = 0$ for a*

nonzero polynomial P whose constant coefficient is nonzero. If

$$\int_0^\infty H(t)e^{st}\,dt$$

continues meromorphically with only a possible simple pole at zero, then H is constant. If $P(1)\neq 0$, this constant is zero.

Proof. The span of the integer translates of H is finite dimensional, and S acts invertibly on it. Decompose into generalized eigenspaces. For an eigenvalue $\rho\neq 0$, choose μ with $e^{\mu L}=\rho$. The corresponding component has the form

$$e^{\mu t}\sum_{j=0}^{m-1} t^j h_j(t), \qquad h_j(t+L)=h_j(t),$$

with smooth periodic h_j. This follows inductively from $(S-\rho)^m H=0$: after multiplication by $e^{-\mu t}$, solve the successive ordinary finite-difference equations, subtracting a polynomial in t times the periodic highest difference at each step. Expand each h_j into its smooth Fourier series. Its Fourier coefficients decrease faster than every power. The Laplace integral can therefore be taken termwise in an initial left half-plane and then continued normally away from the resulting discrete poles. The term $c_{j,n}t^j e^{(\mu+2\pi in/L)t}$ contributes

$$\frac{(-1)^{j+1}j!c_{j,n}}{(s+\mu+2\pi in/L)^{j+1}}.$$

Choose the logarithms μ in one half-open strip of imaginary width $2\pi/L$. Distinct eigenvalues then give disjoint pole sets. At each pole, the highest nonzero polynomial degree cannot cancel. The assumed pole restriction therefore kills every Fourier coefficient except the degree-zero term with exponent zero. Consequently H is constant; substituting into $P(S)H=0$ gives the last assertion. □

Conditional level-25 finish. Assume the established Ramanujan and Hecke transformation identities, and assume additionally that $D_{-1/5}$ and $D_{-3/25}$ have only possible simple poles at k. This paragraph does not assert that the latter analytic hypotheses have yet been proved in every parameter stratum.

Set

$$C_4=\begin{pmatrix}4 & 1\\ -25 & -6\end{pmatrix},\quad C_3=\begin{pmatrix}3 & 1\\ -25 & -8\end{pmatrix},\quad v=f|C_4-f,\quad u=f|C_3-f.$$

The denominator-four and denominator-six paired Ramanujan relations, using $C_6=C_4^{-1}$ and $D_6=D_4^{-1}$ projectively, give

$$v|M=v,\qquad M=\begin{pmatrix}1 & 1/2\\ -25/3 & -19/6\end{pmatrix}.$$

Take

$$\sigma=\begin{pmatrix}-1/5 & -3/10\\ 1 & 1\end{pmatrix}.$$

It has positive determinant and satisfies $\sigma^{-1}M\sigma=\operatorname{diag}(-3/2,-2/3)$. Both $\sigma\infty$ and $C_4\sigma\infty$ equal $-1/5$. The first lemma says that the truncated Mellin transform of $g=v|\sigma$ on the positive imaginary axis has only a possible simple pole at zero. On the other hand

$$g((9/4)z)=(9/4)^{-k/2}g(z).$$

Apply the second lemma to $H(t)=g(ie^t)$, with $L=\log(9/4)$ and $P(X)=X-(9/4)^{-k/2}$. As $k>0$, $P(1)\neq 0$, so $v=0$ by the identity theorem.

The exact Hecke-two identity now reduces to

$$u|(\lambda_2 + A + A^{-1}) = 0, \qquad A = \begin{pmatrix} 1 & -1/3 \\ 0 & 2 \end{pmatrix}.$$

The normalization of λ_2 here is the coefficient of f in the determinant-normalized Hecke relation. Put $H(t) = u(-1/3 + ie^t)$. Its truncated Mellin transform has only a possible simple pole at zero: apply the first lemma to C_3, whose image of infinity is $-3/25$, and subtract the entire truncated transform of the exponentially decreasing f. For $SH(t) = H(t + \log 2)$ the recurrence is

$$\big(2^{k/2}S^2 + \lambda_2 S + 2^{-k/2}\big)H = 0.$$

Thus H is constant, and the identity theorem makes u constant on the upper half-plane. Finally the paired denominator-three relation and the Fricke law give

$$u|K_3 = \eta u, \qquad K_3 = \begin{pmatrix} 1 & 26/75 \\ -3 & -1 \end{pmatrix}.$$

Since $k > 0$ and the lower-left entry is nonzero, no nonzero constant has this slash eigenrelation. Hence $u = 0$, with no generic-eigenvalue exception.

The paired relations also give the D_3, D_4 invariances. Together with T, $U = \left(\begin{smallmatrix} 1 & 0 \\ 25 & 1 \end{smallmatrix}\right)$, $E = \left(\begin{smallmatrix} 2 & -1 \\ -25 & 13 \end{smallmatrix}\right)$, and their Fricke conjugates these generate $\Gamma_0(25)$ (the separate exact group certificate checks index 30). Thus f is modular. The polynomial coefficient bound gives polynomial growth after every rational cusp scaling; periodicity at that cusp then excludes negative Fourier powers. This proves holomorphy at every cusp and $f \in M_k(\Gamma_0(25))$. No vanishing of the cusp constants is claimed or required.

Corollary 26.3 (Level 25 with zero Hecke eigenvalue at two)**.** *Under the original level-25 hypotheses, if $\lambda_2 = 0$, then $f \in M_k(\Gamma_0(25))$, for both Fricke signs and every permitted positive even weight and remaining eigenvalue.*

Proof. The independently rebuilt quotient and first-source certificates prove that every old150 twist has only a possible simple pole at $s = k$. In particular this holds at $-1/5$ and $-3/25$. The preceding local Mellin argument and exact index-30 generation therefore apply. No assertion of cuspidality is made. □

27. The second positive-sign level-25 polar chart

This closes one of the two projective charts when the Fricke sign is $\eta = 1$ and $a = \lambda_2 \neq 0$. The first chart is treated in Section 28. The conclusion at level 25 is membership in M_k.

Use the source-free 1050-grid quotient, with its68 gauge coordinates, and put $h = s - k/2$,

$$x = 2^h, \quad y = 3^h, \quad z = 5^h, \quad w = 7^h,$$
$$X = x^2,\ Y = y^2,\ Z = z^2,\ W = w^2, \quad A = ax,\ B = \lambda_3 y,\ C = \lambda_7 w.$$

After monomial leading elimination, the three free coordinates are $630, 672, 756$ (indices start at zero). The denominator-five vanishing conditions and exact first-Taylor zero-row constraints eliminate one coordinate by a unit pivot. The surviving projective coordinates are v_{672}, v_{756}. Here consider $(v_{672}, v_{756}) = (0, 1)$.

It is essential to include the nonzero residual lower-germ equations. There are400 such rows in three unknown lower coordinates. At $s - 1$, choose the three independent rows numbered $94, 278, 226$ in the recorded residual ordering, and let their matrix be L. For each residual row$r$, the identity

$$(\det L)r - (r \operatorname{adj} L)L = 0$$

eliminates the full lower meromorphic germ. Clear denominators in all four coefficients of this identity and divide their common polynomial gcd. The resulting polynomial identity remains valid

at zeros of $\det L$. Apply these same coefficients to the first Taylor sources. All400 resulting identities are included. This procedure neither divides by a factor vanishing at the candidate pole nor assumes a lower pole-order bound.

Every term of every row has the same parity after substituting $a = A/x$, $\lambda_3 = B/y$, $\lambda_7 = C/w$; the verifier checks this term by term. Removing a common nonzero Laurent monomial gives402 polynomial constraints in $u, C, B, A, Z, W, Y, X, t$, including $tXYZWA - 1$. In this second chart u is an unused variable. The exact rational calculation, with Magma's global modular algorithm disabled, gives the following ideal memberships:

$$(A+1)(A+2) = 0, \qquad (A+1)(Y+3) = 0, \qquad (A+1)(B+4) = 0.$$

On the branch $A = -2$, it gives

$$Y = -3, \qquad 2X^2 = 3.$$

On the branch $A = -1$, it gives

$$Y(X+1) = X^2, \qquad BX = X - 2, \qquad (X+4)F_8(X) = 0,$$

where

$$F_8(X) = X^8 + 121X^7 - 1256X^6 + 6903X^5 - 1161X^4 - 48312X^3 - 58008X^2 - 17520X + 1584.$$

These are necessary relations on both complete parameter branches, not a generic elimination that discards determinant-zero points. They are checked by `level25_plus_poles_chart2_certificate.py`, which prepares the exact characteristic-zero ideal-membership batch.

Let $\rho = \log 9/\log 4$. Since $8 < 9 < 16$, one has $3/2 < \rho < 2$, and the exponential definitions imply

$$|Y| = |X|^\rho.$$

The branch $A = -2$ is impossible: $|X|^2 = 3/2 > 1$ would imply $|Y| < 3/2$, whereas $|Y| = 3$.

On the branch $A = -1$, $X + 1 \neq 0$, since $X \neq 0$ and $Y(X+1) = X^2$. Thus $|Y| = |X|^2/|X+1|$. The root $X = -4$ is impossible because $|X+1|^2 > |X| > 1$. To handle F_8, use radius 10^{-5} disks with centers

$$-130.991096, \quad -0.956837, \quad 0.072143, \quad 4.209464,$$
$$-0.915249 \pm 0.101240i, \qquad 4.248412 \pm 5.570476i.$$

All displayed decimals denote exact rational numbers. The verifier `level25_plus_poles_chart2_roots.py` expands $F_8(c+v)$ and checks, with rational arithmetic,

$$|F_8(c)| + \sum_{j=2}^{8} \left| \frac{F_8^{(j)}(c)}{j!} \right| r^j < |F_8'(c)|r.$$

It uses rational upper and lower bounds for the complex moduli. Rouché's theorem gives one root per disk; disjointness and the degree show that all roots are covered. On every disk it also proves one of:

$$|X| > 1, \quad |X+1|^2 > |X|;$$
$$|X| < 1, \quad |X+1| > 1;$$
$$|X| < 1, \quad |X+1|^2 < |X|.$$

The first implies $|Y| < |X|^{3/2} < |X|^\rho$; the second implies $|Y| < |X|^2 < |X|^\rho$; the third implies $|Y| > |X|^{3/2} > |X|^\rho$. Every possibility contradicts the exponential modulus relation. Hence this projective polar chart is empty.

The first-Taylor source has a common factor $i(s-k)$. Accordingly the argument applies at a rightmost active pole away from k, and at k when its order is at least two. Allowed simple poles at k above the chosen point contribute holomorphically to lower Taylor equations, because the corresponding Taylor binomial coefficient vanishes. The treatment of pole order and of the68

discarded source-free directions is therefore the same as in the quotient theorem; this argument adds no order hypothesis.

28. The first positive-sign level-25 polar chart

Assume $\eta = 1$ and $a = \lambda_2 \neq 0$. Use the source-free 1050-grid quotient, the 68 gauge coordinates, and all 400 primitive lower-germ syzygies described in the second-chart argument. After unit elimination, the two free highest residues are v_{672}, v_{756}. This section considers the first projective chart $(v_{672}, v_{756}) = (1, u)$.

Put $X = x^2, Y = y^2, Z = z^2, W = w^2, A = ax, B = \lambda_3 y, C = \lambda_7 w$. The exact termwise Laurent-unit normalization gives 402 polynomial constraints, including $tXYZWA - 1$, with Z and all auxiliary variables retained. Denote their ideal by I_{full}. The exact rational Groebner calculation has 79 basis elements. From this basis select the 15 elements independent of C, Z, W, t, saved explicitly in `level25_plus_poles_chart1_rows.json`. The final verifier recomputes the full ideal and checks every selected element belongs to it.

In a new ring $\mathbb{Q}[t, u, B, A, Y, X]$, let I be the ideal generated by these 15 rows and $tAXY - 1$. The new saturation variable is independent of the previous one. Since A, X, Y are nonzero at every physical point, every point of the full chart projects to a point of this smaller ideal. This is a necessary-condition reduction; it makes no assertion that every point of the smaller ideal lifts to the full chart.

The ideal I is zero-dimensional of degree 70. The verifier checks the following exact ideal memberships:

$$(X - 4)P_{67}(X) \in I,$$
$$A + 2, \quad 213Y^3 + 4585Y^2 + 500Y + 12096 \ \in I + (X - 4),$$

and

$$Y - H_{66}(X) \in I + (P_{67}(X)).$$

The rational polynomials P_{67} and H_{66} are recorded by ascending coefficient lists in `level25_plus_poles_chart1_shape.json`. Their degrees are respectively 67 and 66. In particular, expressing Y introduces no denominator depending on X and loses no exceptional parameter values. All computations in `level25_plus_poles_chart1_certificate.py` use characteristic zero and disable Magma's global modular algorithm.

At every physical point, $X = 4^{s-k/2}$, $Y = 9^{s-k/2}$, hence

$$|Y| = |X|^{\rho}, \qquad \rho = \frac{\log 9}{\log 4}.$$

If $X = 4$, then $|Y| = 9$. The cubic equation above is impossible, because

$$4585 \cdot 9^2 = 371385 > 213 \cdot 9^3 + 500 \cdot 9 + 12096 = 171873.$$

The quadratic term strictly dominates the sum of the other terms on that circle. Thus this branch is excluded, including any putative weight-two exceptional case.

If $P_{67}(X) = 0$, use the separate exact root certificate `level25_root_chart1_modulus_certificate.json` and its verifier. It supplies 67 disjoint rational Rouché disks covering every root and, on each disk, proves that $Y = H_{66}(X)$ contradicts $|Y| = |X|^{\rho}$. The exponent is bounded by exact inequalities $1584/1000 < \rho < 1585/1000$. Thus the degree-67 branch is excluded too. The first projective chart is empty at every physical point.

Together with the second-chart argument, this eliminates every nonzero highest vector with zero denominator-five components in the positive Fricke, nonzero-λ_2 sector. The quotient theorem ensures that a highest vector at a rightmost active pole is nonzero on the original 150-grid. The independent denominator-five pole argument forces its denominator-five components to vanish. The two projective-chart contradictions then force the vector itself to vanish.

Here a pole is active unless it is a simple pole at $s = k$. The common first-Taylor factor is $i(s-k)$; at $s = k$, a pole of order at least two still produces a nonzero highest polar coefficient after multiplication by this factor. Allowed simple poles at k above a lower candidate point contribute holomorphically because the corresponding Taylor binomial coefficient vanishes. Primitive lower-germ syzygies eliminate the complete lower meromorphic germs before coefficients are extracted. No bound on lower Laurent orders is assumed.

Consequently, for $\eta = 1$ and $\lambda_2 \neq 0$, every quotient coordinate, and in particular every original 150-grid additive twist, is holomorphic away from $s = k$ and has at most a simple pole there. Combined with the separate $\lambda_2 = 0$ certificate, this establishes the pole statement for the positive Fricke sign for all Hecke eigenvalues. This section does not assert completion of the negative-sign first chart or of the whole level-25 converse; the intended final conclusion is $f \in M_k$.

28.1. **Exact exclusion of the degree-67 exponential locus.** Let $P, H \in \mathbb{Q}[X]$ be the coefficient arrays in `level25_plus_poles_chart1_shape.json`, of degrees 67 and 66. There is no $h \in \mathbb{C}$ satisfying

$$X = 4^h, \qquad Y = 9^h, \qquad P(X) = 0, \qquad Y = H(X).$$

The derivation of these polynomial relations from the original Taylor equations is a separate algebraic certificate. Here we prove the exclusion of their physical exponential locus.

Put $\rho = \log 9/\log 4$. Exact integer comparisons give

$$4^{1584} < 9^{1000} < 4^{1585}, \qquad 1584/1000 < \rho < 1585/1000.$$

The necessary modulus relation is $|Y|^2 = (|X|^2)^\rho$. The verifier `level25_root_chart1_modulus.py` checks 67 pairwise disjoint rational disks of radius $r = 10^{-250}$. Their centers are saved in the accompanying certificate. Numerical approximations were used only to select centers; the subsequent checks use rational arithmetic exclusively and can be rerun from the saved centers without numerical root finding.

For a polynomial $F = \sum_j a_j X^j$, put $M_F(R) = \sum_j |a_j| R^j$. At a center c, with $R = |\Re c| + |\Im c| + r$, the Taylor integral remainder gives

$$|P(c+v) - P(c) - P'(c)v| \le \tfrac{1}{2} r^2 M_{P''}(R) \quad (|v| \le r).$$

Rational outward bounds verify

$$|P(c)| + \tfrac{1}{2} r^2 M_{P''}(R) < r|P'(c)|.$$

Rouché's theorem therefore gives exactly one root, counted with multiplicity, in each disk. Since there are 67 disks, they exhaust all roots of P.

Polynomial evaluation uses rational Horner arithmetic rounded outward to denominator 10^{450}. If v approximates the current complex value with error e in the norm $|\Re z| + |\Im z|$, then multiplication by c, addition of the next exact coefficient, and rounding to w replace the error by an outward bound for

$$(|\Re c| + |\Im c|)e + |\Re(cv + a - w)| + |\Im(cv + a - w)|.$$

Thus no floating-point evaluation of a large rational coefficient is trusted. The same procedure bounds $P'(c)$ and $H(c)$; the variation of H across the disk is at most $r M_{H'}(R)$.

These estimates produce positive rational intervals $[u_-, u_+]$ for $|X|^2$ and $[v_-, v_+]$ for $|H(X)|^2$. Each u interval lies entirely above or below 1. If $u_- > 1$, the verifier proves one of

$$v_+^{1000} < u_-^{1584}, \qquad v_-^{1000} > u_+^{1585}.$$

If $u_+ < 1$, it proves one of

$$v_+^{1000} < u_-^{1585}, \qquad v_-^{1000} > u_+^{1584}.$$

Each comparison contradicts the necessary exponential modulus relation. All 67 disks pass. The certificate records each rational center, radius, modulus interval and comparison direction, together with hashes of the exact coefficient input and verifier.

29. The second negative-sign level-25 polar chart

Assume $\eta = -1$ and $a = \lambda_2 \neq 0$, and use the same source-free 1050-grid quotient and its 68 gauge coordinates as in the leading quotient calculation. The negative-sign leading reduction has free zero-based coordinates $420, 756, 882$. The denominator-five vanishing and first-Taylor equations leave two projective coordinates. Here we exclude the second chart, in which the first surviving coordinate is zero and the second is one.

Use all 439 residual lower-germ rows, with the primitive adjugate syzygies of `level25_plus_poles_residual_syzygies.m`. Clear all syzygy coefficient denominators and divide the gcd of all coefficients, then apply the identical coefficients to the first Taylor sources. Each polynomial lower-germ identity is checked before it is used. Thus no invertibility assumption on the selected determinant is introduced, and no bound on the lower Laurent orders is required.

The resulting second chart, saved by the negative-sign exporter as `level25_cusp_aux_minus_seed0_chart2.txt`, is processed by `level25_plus_poles_negative2_prepare.py`. Set

$$X = x^2, \quad Y = y^2, \quad Z = z^2, \quad W = w^2, \qquad A = ax, \quad B = \lambda_3 y, \quad C = \lambda_7 w.$$

The adapter checks the exponent parity of every term after the substitutions $a = A/x$, $\lambda_3 = B/y$, $\lambda_7 = C/w$. It removes only common Laurent monomials in variables known to be nonzero and preserves all auxiliary variables. There are 417 resulting constraints, including the saturation $tXYZWA - 1$.

Over $\mathbb{Q}$, with Magma's global modular algorithm disabled, these constraints generate the unit ideal. The exact verifier asserts both $1 \in I$ and that the computed Groebner basis is $[1]$. A standalone six-row subset is saved in `level25_plus_poles_negative2_certificate.m`; the adapter also checks that all six rows occur literally in the regenerated full list and that their ideal alone is the unit ideal. Consequently this chart has no point over $\mathbb{C}$, without any extra specialization or generic-parameter exception. The highest-Laurent interpretation, including the permitted simple pole at $s = k$, is inherited from the source-free quotient and primitive lower-germ argument.

This proves only the second negative chart. It makes no claim about the first negative chart, and the level-25 target is still M_k.

Independent audit of the two second charts at level 25. This audit concerns $\lambda_2 \neq 0$ and both Fricke signs, and only the second of the two surviving projective charts. It does not assert that either first chart is empty. The target at level 25 remains M_k.

The original 1050-grid exporter was run afresh in isolated scratch paths. In addition to its usual export, the independent run executes the original symbolic source-free-kernel assertions before adding gauges: all 68 basis directions vanish on the old 150-grid, satisfy every original leading row, and annihilate every positive Taylor source. This justifies the same gauge quotient at every parameter value, rather than merely at a generic numerical specialization. No coefficient in the exported leading or first-source matrices was changed.

The first-source coefficient of a matrix row is the leading multiplier times $CQ/(Cj + DQ)$; that of a translated Ramanujan row is its leading multiplier times Q/t. Leading elimination is replayed on the first source with every scalar shifted by $s \mapsto s - 1$. Thus the freshly saved residual matrices are the actual lower-germ and top-source coefficients of the original first-Taylor identities. Their exact coefficient tables are compared with the stable worker tables.

The selected 3×3 lower block is used solely to form its adjugate. Every primitive syzygy is checked as an exact rational-function identity after clearing all four coefficient denominators and dividing their common polynomial gcd. The resulting coefficients are polynomials and annihilate the lower

block identically, including at zeros of the chosen determinant. Applying them to the corresponding top-source rows therefore introduces no generic-invertibility hypothesis and no comparison of pole orders at different points. All 400 positive-sign and 439 negative-sign residual rows are included.

Subsequent elimination divides only by nonzero rational scalars and monomials in the exponential variables and λ_2. The latter is a unit in this sector. The substitutions $A = \lambda_2 x$, $B = \lambda_3 y$, $C = \lambda_7 w$, $X = x^2$, $Y = y^2$, $Z = z^2$, $W = w^2$ are checked term by term, including exponent parity. No factor involving B or C is removed. The saturation $tXYZWA - 1$ covers exactly the required nonzero variables.

For the positive sign the full 402-row system gives the stated two A branches by exact ideal membership. Both the $A = -2$ relations and the $A = -1$ degree-eight polynomial relations are rechecked with deterministic rational Gröbner bases. The independent execution also verifies all eight disjoint rational Rouché disks and their modulus inequalities. For the negative sign the full normalized system has 417 rows. The six displayed certificate rows are checked to occur literally in that freshly regenerated list; both their ideal and the complete ideal have basis [1]. All final algebra disables Magma's global-modular algorithm process-wide and in its explicit Gröbner-basis calls.

The analytic passage from these relations to exclusion of a rightmost active pole uses the existing quotient lemma: at $s = k$ it concerns orders at least two, and simple poles at k are allowed. The source-free and polynomial-syzygy checks supply the algebraic provenance needed by that lemma; they do not replace its Mellin/Taylor justification. Source hashes, intermediate output hashes, native batch completion records, and exact comparison results are recorded in `level25_secondcharts_independent_manifest.json`.

Independent extension to the first positive chart. The same freshly reconstructed positive-sign 1050-grid rows and all 400 primitive lower-germ syzygies were used for the first chart. Its full 402 normalized constraints were regenerated without dropping the Z-dependent rows. With deterministic rational Gröbner bases, the full ideal has basis size 79, and each of the 15 saved smaller certificate rows was checked for membership in that full ideal.

The ideal generated by these 15 necessary relations, together with $tAXY - 1$, has dimension zero and quotient dimension 70. Its exact membership checks give $(X - 4)P_{67}(X) = 0$. On the branch $X = 4$, they give $A = -2$ and

$$213Y^3 + 4585Y^2 + 500Y + 12096 = 0.$$

The physical relation requires $|Y| = 9$. The exact inequality $4585 \cdot 9^2 > 213 \cdot 9^3 + 500 \cdot 9 + 12096$ makes the quadratic term larger in modulus than the sum of all other terms on that circle, which rules out this branch.

On the other branch the checker verifies $Y - H_{66}(X)$ belongs to the ideal with P_{67} adjoined. This is an ideal-membership certificate, not an inference from numerical roots or a generic elimination. The exact coefficients of both polynomials are bound to the source manifest. The independent replay of the rational-disk verifier treats its stored centers solely as proposed rational inputs. It rechecks disjointness, all 67 Rouché inequalities, outward-rounded rational bounds on $|X|^2$ and $|H_{66}(X)|^2$, and $4^{1584} < 9^{1000} < 4^{1585}$. All 67 disks exclude $|Y| = |X|^{\log 9/\log 4}$, and the degree count excludes any omitted root.

Together with the independently audited second positive chart, this closes the positive-sign nonzero-λ_2 quotient pole argument, subject to the same analytic quotient lemma and permitted simple poles at $s = k$. This extension makes no assertion about the first negative chart.

Corollary 29.1 (Level 25 with positive Fricke sign)**.** *Under the original level-25 hypotheses, if $\eta = 1$, then $f \in M_k(\Gamma_0(25))$ for every permitted positive even weight and every Hecke eigenvalue, including $\lambda_2 = 0$.*

Proof. For $\lambda_2 = 0$, the earlier all-sign certificate applies. For $\lambda_2 \neq 0$, the two projective chart exclusions above prove that all old 150-grid twists have only a possible simple pole at $s = k$. In particular the local Mellin criterion at $-1/5$ and $-3/25$ applies. The resulting transformations and exact index-30 group certificate prove modularity, and polynomial cusp growth proves holomorphy at every cusp. The conclusion does not assert cuspidality. □

30. Two exceptional strata in the first negative level-25 chart

Work in the first projective chart of the negative Fricke-sign, nonzero-λ_2 pole sector. Retain the source-free 1050-grid quotient and all 439 primitive lower-germ syzygies. Write $X = x^2, Y = y^2, A = \lambda_2 x, B = \lambda_3 y$ as before. Every physical point satisfies $|Y| = |X|^\rho$, where $\rho = \log 9/\log 4$ lies strictly between $3/2$ and 2.

Select the 202 normalized equations independent of the prime-five and prime-seven parameters, and append $tAXY - 1$. These are necessary conditions for the full chart; discarding other equations does not remove any physical solution. The exact characteristic-zero verifier checks the following ideal memberships separately on $A = -1$ and $A = -2$. No division by $A + 1$ or $A + 2$ is used in these strata.

On $A = -1$ the necessary relations are

$$Y(X + 1) = X^2, \qquad F_4(X)F_8(X) = 0,$$

where

$$F_4(X) = 5X^4 + 5X^3 - 312X^2 - 228X + 48$$

and

$$F_8(X) = X^8 + 121X^7 - 1256X^6 + 6903X^5 - 1161X^4 - 48312X^3 - 58008X^2 - 17520X + 1584.$$

Since $X \neq 0$, the first relation implies $X + 1 \neq 0$ and $Y = X^2/(X + 1)$. The F_8 roots are excluded by the rational disk verifier `level25_plus_poles_chart2_roots.py`. The four F_4 roots are excluded by `level25_root_negative_quartic.py`. Both use the same modulus formula, with exact rational isolating regions and $3/2 < \rho < 2$. Thus the whole $A = -1$ stratum is impossible.

On $A = -2$ the necessary relations are

$$(X - 4)(2X^2 - 3) = 0, \qquad (X - 4)(Y + 3) = 0.$$

If $X \neq 4$, then $2X^2 = 3$ and $Y = -3$. Consequently $|X|^2 = 3/2 > 1$ and $|Y| = |X|^\rho < 3/2$, contradicting $|Y| = 3$. On the remaining branch $X = 4$, the verifier obtains

$$213Y^3 + 4585Y^2 + 500Y + 12096 = 0.$$

The physical modulus is $|Y| = 9$, but on that circle

$$|4585Y^2| = 371385 > 213 \cdot 9^3 + 500 \cdot 9 + 12096 = 171873.$$

Thus the quadratic term cannot be cancelled by the other three terms, and this branch is impossible too.

These arguments exclude both exceptional strata at every complex parameter value satisfying the exponential modulus relation. They do not settle the complementary stratum $A \neq -1, -2$, which requires its own exact certificate. All ideal computations disable Magma's global modular algorithm. Their pole-order interpretation is the same as the primitive lower-germ argument: the complete lower germs are eliminated before any Laurent coefficient is extracted, and allowed simple poles at $s = k$ are preserved.

An additional quartic locus at level 25. Suppose

$$5X^4 + 5X^3 - 312X^2 - 228X + 48 = 0, \qquad Y(X+1) = X^2.$$

There is no physical solution $X = 4^h, Y = 9^h$. Indeed, exact rational evaluation shows a sign change of the quartic on each interval

$$(-8.038483, -8.038482), \quad (-0.900349, -0.900348),$$
$$(0.170754, 0.170755), \quad (7.768076, 7.768077).$$

These decimals denote rational endpoints. Four disjoint sign-changing intervals contain four distinct real roots, exhausting the degree. On the first and fourth intervals, $|X| > 1$ and $|X+1|^2 > |X|$. On the second, $|X| < 1$ and $|X+1|^2 < |X|$. On the third, $|X| < 1$ and $|X+1| > 1$. All inequalities follow by rational endpoint bounds and are checked in `level25_root_negative_quartic.py`. Writing $3/2 < \rho = \log 9/\log 4 < 2$, these inequalities give, respectively,

$$|Y| < |X|^{3/2} < |X|^\rho, \qquad |Y| > |X|^{3/2} > |X|^\rho, \qquad |Y| < |X|^2 < |X|^\rho.$$

Each contradicts the physical modulus identity. This excludes only the stated algebraic locus; its implication from the Taylor system is supplied by the separate negative-sign stratum certificate.

31. The denominator-zero branch in the first negative level-25 chart

Retain the first negative Fricke-sign projective chart, and write $X = x^2, Y = y^2, A = \lambda_2 x$ as before. The strata $A = -1, -2$ have already been excluded separately. Thus assume $A(A+1)(A+2)XY \neq 0$. The two-row rational substitution used for the complementary generic branch has denominator

$$D_0 = A(X - Y) + X^3 - X^2 - X.$$

Here we exclude its zero locus without making that substitution.

Take the 202 original normalized equations independent of the prime-five and prime-seven variables. Adjoin D_0 and the saturation

$$tA(A+1)(A+2)XY - 1.$$

The ideal generated by these equations over $\mathbb{Q}$ is the unit ideal. The verifier `level25_plus_poles_negative1_dzero_certificate.py` uses the regenerated original rows, disables Magma's global modular algorithm, and asserts both $1 \in I$ and that its Groebner basis is $[1]$. The original saturation $tAXY - 1$ is replaced by the displayed one, with a fresh auxiliary value of t; all physical points in the stated stratum admit this value. No solution is lost within the stratum.

No compatibility equation for the numerator is assumed, and no factor of D_0 is divided out. Consequently this proves the entire $D_0 = 0$ branch empty, including every possible exceptional Hecke parameter. The complementary branch $D_0 \neq 0$ is treated by the rational-substitution certificate in Section 32.

32. Complete negative-sign pole control at level 25

We prove that every additive twist on the original 150-grid is holomorphic away from $s = k$ and has at most a simple pole there, for Fricke sign $\eta = -1$ and every choice of the Hecke eigenvalues. The conclusion at this level is membership in M_k; simple poles encoding cusp constants are permitted throughout.

Use the meromorphically continued, source-free 1050-grid quotient constructed in Section 25. It preserves every original 150-grid twist. Its 68 discarded directions are annihilated by the leading matrix and every positive Taylor source, identically in the spectral parameter and eigenvalues. The augmented old-grid-zero matrix, together with the quotient gauge, has full rank by Laurent-monomial pivots. Thus a nonzero highest Laurent vector of this quotient cannot vanish on the original grid. These assertions hold at every parameter specialization, rather than only on a generic open set.

Call a pole *active* unless it is simple and located at $s = k$. If any active pole exists, choose one furthest to the right in its integer-translation class. Such a choice exists because the family is holomorphic in a common right half-plane. A simple pole at k above the chosen point contributes holomorphically to every lower Taylor equation: for a shift $m \geq 1$, the coefficient $\binom{s+m-k}{m}$ vanishes at $s = k - m$ and cancels that simple pole. Consequently all the leading and Taylor principal-part arguments apply to this rightmost active pole.

Let v be its highest nonzero Laurent vector. The original-grid component of v is nonzero by the preceding full-rank statement. The negative denominator-five pole theorem in Section 23 forces every denominator-five component of v to vanish. This theorem applies also to order at least two at $s = k$, and it eliminates complete lower meromorphic germs before extracting a Laurent coefficient whenever their orders are not already bounded by an invertible system.

If $\lambda_2 = 0$, specialize the continued $Q = 1050$ quotient to this value and $\eta = -1$. Write $x = 2^{s-k/2}$, $y = 3^{s-k/2}$, $z = 5^{s-k/2}$ and $w = 7^{s-k/2}$. The old150-zero augmented matrix still has 982 monomial pivots and 68 free coordinates. Its kernel is the specialization of the previously certified kernel, whose identity submatrix on the gauge coordinates is unchanged. In particular, all positive Taylor sources still vanish on that kernel.

The leading matrix together with these gauge rows has 1046 nonzero rational Laurent-monomial pivots and four free coordinates, $420, 714, 840, 882$. Retain every zero row and residual row, and transport the first Taylor source through the same row operations at $s - 1$. The zero rows give identities of whole meromorphic germs, so no bound on an invisible lower-point pole is assumed. Add the five denominator-five zero-coordinate conditions already proved for the highest Laurent vector. Further monomial pivots eliminate coordinates 420 and 840, leaving 68 normalized polynomial relations in the two residue variables at 714 and 882. All divisions so far are by nonzero rational constants and monomials in x, y, z, w; there is no restriction on the Hecke eigenvalues at 3 or 7.

If the remaining vector is nonzero, scale it into one of the two projective charts $(v_{714}, v_{882}) = (1, \xi)$ or $(0, 1)$. In either chart, adjoining $\tau xyzw - 1$ to the 68 polynomial relations gives the unit ideal over $\mathbb{Q}$. Both explicit polynomial lists and the exact ideal assertions are given in `level25_cusp_aux_zero2_minus_certificate.m`; they are regenerated by the `level25_cusp_aux_prime7_firstjet.py`, `level25_cusp_aux_prime7_jetrank.py`, and `level25_cusp_aux_prime7_jetideals.py` scripts with arguments -1 and `zero2`. Thus both charts are impossible. At $s = k$ with pole order at least two, the coefficient of $(s-k)^{-(d-1)}$ in the exact zero-row identity $(s-k)\mathbf{F}(s)$ gives the same first-jet constraints; the rightmost-active-pole argument above handles every other case. This contradicts $v \neq 0$ and completes the $\lambda_2 = 0$ case.

Hence suppose $a = \lambda_2 \neq 0$. Set

$$h = s - k/2, \qquad x = 2^h,\ y = 3^h,\ z = 5^h,\ w = 7^h,$$
$$A = ax,\ B = \lambda_3 y,\ C = \lambda_7 w, \qquad X = x^2,\ Y = y^2,\ Z = z^2,\ W = w^2.$$

In particular $AXYZW \neq 0$ and

$$|Y| = |X|^\rho, \qquad \rho = \frac{\log 9}{\log 4}. \tag{25.N1}$$

Exact germ elimination. The negative leading reduction uses 1047 allowed monomial pivots and has free coordinates $420, 756, 882$, with indices starting at zero. There are 439 residual equations on the three lower free germs. Choose the three rows $184, 85, 117$ in the recorded residual ordering, forming a matrix L. For every residual row r, the identity

$$(\det L)r - (r \operatorname{adj} L)L = 0$$

eliminates the complete lower germ. Clear the denominators of all four coefficients, divide their common polynomial gcd, and apply the same resulting coefficients to the corresponding Taylor

sources. The verifier checks the complete polynomial row identity for each of the 439 rows. It never divides a meromorphic germ by $\det L$; the identities remain valid at every zero of that determinant.

Together with the zero-row Taylor constraints and denominator-five vanishing, these identities leave two residue coordinates. Their two projective charts exhaust every nonzero highest vector. Exact termwise parity checks permit replacing $x, y, z, w, a, \lambda_3, \lambda_7$ by X, Y, Z, W, A, B, C after removing only a common nonzero Laurent monomial from each row. Each chart has 417 normalized constraints, including the nonvanishing saturation. No prime-five or prime-seven variable is set to zero in this operation.

The second chart is the unit ideal, as proved in Section 29. In the first chart write its normalized residue coordinate as u. Selecting the 202 constraints independent of C, Z, W gives necessary equations in u, B, A, Y, X; dropping the other constraints loses no physical point. The exceptional strata $A = -1, -2$ are impossible by Section 30, using exact ideal membership and the physical relation (25.N1).

The remaining first-chart stratum. Assume $A \neq -1, -2$ and put

$$D = A(X - Y) + X^3 - X^2 - X, \qquad U = AX(X - 1) + XY + X - Y.$$

The separate exact $D = 0$ certificate excludes that stratum, without any assumption on U. We may therefore assume $D \neq 0$. The first and third recorded pure rows form an affine linear system in B, u after dividing the first row by the nonzero factor $A + 1$. Its determinant is $-AX^2YD$. Exact Cramer identities give

$$u = U/D, \qquad B = V/(XD),$$

where

$$\begin{aligned} V = {} & A^3XY - A^3X + A^2XY + 2A^2X - A^2Y \\ & + AX^3Y - AX^3 - 2AX^2Y + 3AX^2 - 2AX + AY \\ & - 2X^2Y - 2X^2 - XY^2 + 2XY + X + Y^2 - Y. \end{aligned}$$

These identities are checked directly in `level25_cusp_aux_minus_cramer_certificate.m`.

Substitute these expressions into the first 100 of the exact pure rows. Each selected row occurs literally in the independently regenerated 202-row list. Taking rational-function numerators and removing only factors among $A, Y, X, A + 1, A + 2, D$, all nonzero here, gives 14 distinct polynomial equations. Let I be their ideal in $\mathbb{Q}[A, Y, X]$, and write

$$S = AXY(A + 1)(A + 2)D.$$

The self-contained exact transfer certificate `level25_plus_poles_negative1_generic_audit_transfer.m` contains the 14 equations and the 15 positive equations explicitly. It verifies

$$S^2 \in I + (U).$$

Since $S \neq 0$, a point of this stratum satisfying I has $U \neq 0$. We may introduce a new residue coordinate $v = -D/U$. For each of the 15 explicit positive first-chart equations, substitute

$$u_+ = v = -D/U, \qquad B = V/(XD),$$

leaving A, Y, X unchanged. The numerator of every resulting rational function belongs to I directly, as checked by exact characteristic-zero normal forms. The denominators are products of the already nonzero X, D, U and rational constants. Thus our negative-chart point would give a point of the positive 15-equation system with $AXY \neq 0$.

The argument of Section 28 excludes every such algebraic point satisfying (25.N1): its exact eliminants reduce to $X = 4$ with the impossible cubic on $|Y| = 9$, or to the degree-67 polynomial and its exact rational-disk modulus certificate. Only the 15 equations, $AXY \neq 0$, and (25.N1) are used for that exclusion. We are not asserting a change of Fricke sign for a modular function or using an

unproved transformation law. This is a transfer between two explicit polynomial necessary-condition systems. It excludes the final first-chart stratum.
Pole-order conclusion. All projective charts and all values of λ_2 are now excluded for a nonzero highest vector with vanishing denominator-five entries. For $s_0 \neq k$, the common first-Taylor scalar $i(s_0 - k)$ is nonzero. At $s_0 = k$ and pole order at least two, multiplication by $i(s - k)$ still leaves a nonzero highest polar coefficient, one order lower. The complete lower-germ syzygies justify extraction of these coefficients without any bound on the pole orders at lower points. Allowed simple poles at k have already been accounted for above. Therefore no rightmost active pole exists. Every coordinate of the continued quotient, and in particular every original 150-grid additive twist, is holomorphic off $s = k$ and has at most a simple pole at $s = k$.

Combined with the positive-sign pole theorem, this gives the required pole statement for both Fricke signs and every Hecke eigenvalue. The separately verified dilation–Mellin and matrix-generation argument then gives level-25 modularity and holomorphy at all cusps, hence $f \in M_k$. No cuspidality conclusion is asserted for level 25.
Independent audit of the final negative first-chart transfer. Starting with this auditor's original 1050-grid reconstruction and all 439 primitive lower-germ syzygies, the negative first-chart dump was normalized afresh. The termwise exact rational normalization gives 202 pure equations and the saturation equation. The selected 100 equations were taken from this fresh rational input itself; the older finite-field diagnostic file was not used as evidence.

The two linear equations in u, B and their three Cramer determinants were rechecked exactly. On the stated stratum $AXY(A+1)(A+2)D \neq 0$, these give $u = U/D$ and $B = V/(XD)$. Substitution into all 100 rows, with a check of every denominator factor and every numerator factor removed, yields 14 distinct rational polynomials. Their literal coefficients agree with the frozen transfer certificate. The 15 positive obstruction polynomials in that certificate also agree literally, after variable renaming, with the previously independently audited positive first-chart rows.

The new deterministic rational calculation gives a 35-element Gröbner basis for the unsaturated three-variable ideal I and verifies $S^2 \in I + (U)$, where $S = AXY(A+1)(A+2)D$. At every physical point of the current stratum $S \neq 0$, so $U = 0$ is impossible. Substituting $u_+ = -D/U$ and $B = V/(XD)$ into each of the 15 positive obstruction rows gives a rational function whose denominator has only factors X, D, U; its numerator is checked to lie directly in I. No additional saturation, generic-eigenvalue premise, or nonzero assumption is hidden in this step.

Consequently the constructed algebraic point satisfies all 15 positive rows. Since $AXY \neq 0$, setting $t = (AXY)^{-1}$ extends it to the exact saturated positive system used in the independently checked shape proof. The transfer keeps X and Y fixed and hence keeps $|Y| = |X|^{\log 9/\log 4}$. The positive $X = 4$ cubic exclusion and the independently rechecked 67 rational root exclusions therefore apply. No positive-sign automorphic function is constructed or needed.

The exceptional strata $A = -1, -2$ and $D = 0$ are separate inputs, bound to the root agent's independent partial-audit manifest. Together these strata and the verified transfer cover the whole negative first chart. The preceding quotient lemma still allows simple poles at $s = k$; the level-25 conclusion is M_k, not cuspidality.

Theorem 32.1 (Complete converse at level 25)**.** *Under the hypotheses defining $\mathscr{E}(25, k)$ in* [3]*, with $k \geq 2$ even, the associated function belongs to $M_k(\Gamma_0(25))$. This holds for both Fricke signs and every complex unramified Hecke eigenvalue. No cuspidality assertion is made.*

Proof. The zero-Hecke-2 certificates and the positive and negative nonzero-Hecke-2 chart arguments prove, in every case, that every original 150-grid additive twist has at most a simple pole at $s = k$ and no other pole. Apply the established local Mellin criterion to the twists at $-1/5$ and $-3/25$. The dilation and Hecke relations then force the required individual transformations; the exact index-30

certificate proves invariance under $\Gamma_0(25)$. Polynomial cusp growth gives holomorphy at every cusp, while allowing nonzero cusp constants. □

33. Level 26: continuation from a shifted Taylor relation

Write $D_j(s) = F(s, j/78)$, $h = s - k/2$, and $x = 2^h$, $y = 3^h$, $z = 13^h$. Set $a = \lambda_3 = a_3 3^{1-k/2}$ and $(\eta, \sigma_2, \sigma_{13}) = (i^k \epsilon_f, \lambda_2, \lambda_{13})$. The three latter parameters are signs; a is arbitrary complex.

Proposition 33.1. *For every allowed choice of parameters all 78 functions D_j continue meromorphically to the whole plane.*

The leading relations alone have rank 77. Thus an invertible leading system, as used above, is unavailable. A relation between the first Taylor coefficients supplies the missing equation.

Lemma 33.2 (Continuation using a shifted first Taylor relation)**.** *Suppose an r-component Mellin vector V satisfies, for arbitrarily large M,*

$$A(s)V(s) = \sum_{m=1}^{M-1} i^m \binom{s+m-k}{m} C_m(s)V(s+m) + H_M(s),$$

where H_M is holomorphic on $\Re s > B - M$ and all matrix coefficients are meromorphic. Let $L(s-1)A(s-1) = 0$ identically. If

$$\begin{pmatrix} A(s) \\ L(s-1)C_1(s-1) \end{pmatrix}$$

has column rank r over the field of meromorphic functions, then V continues meromorphically to $\mathbb{C}$.

Proof. Apply $L(s-1)$ to the equations at $s-1$. The coefficient of $V(s)$ is $i(s-k)L(s-1)C_1(s-1)$. Every other unknown is $V(s+j)$ with $j \geq 1$. The remainder is meromorphic on $\Re s > B - M + 1$. Append this equation to the equations at s, and choose r independent rows. The resulting square matrix is invertible as a meromorphic matrix; the extra factor $s - k$ is not identically zero. If V is known on $\Re s > b$, choose M with $B - M + 1 < b - 1$. The right side is then meromorphic on $\Re s > b - 1$ and determines V there. Start with absolute convergence and iterate. The identities agree on overlaps by uniqueness of meromorphic continuation. □

Here is the exact finite certificate at level 26. The known slash transformations and eigenvalues are

$$(T, 1), \quad (U_{26}, 1), \quad \left(\begin{pmatrix} 2 & -1 \\ -26 & 14 \end{pmatrix}, -\sigma_2 \right), \quad (W_{26}, \eta).$$

The determinant-two transformation follows from the original and c_2-twisted Fricke laws, exactly as at level 22. Form all words of length at most three in these matrices and their inverses, retaining their eigenvalues. Include their leading transformation rows whenever both rational points lie on the 78-grid; include a single entire coordinate row when the image is infinity. Also include all Ramanujan and Taylor–Fricke rows with $q \mid 78$ and signed $t \mid 78^2/\operatorname{lcm}(26, q^2)$, and the exact Hecke rows at $2, 3, 13$. The quadratic Hecke term occurs only at 3. These definitions are implemented in `level26_jet_builder.py`.

The exact arithmetic in `level26_first_jet.py` is over $\mathbb{Q}(x, y, z, a)$. For each of the eight sign triples it checks:

(1) The leading matrix has 77 pivots and no remaining nonzero rows. A normalized kernel vector v is constructed by back substitution and checked against every original row.
(2) Every pivot numerator and denominator, considered as a polynomial in x, y, z with coefficients in $\mathbb{Q}[a]$, has at least one coefficient that is a nonzero rational constant. Consequently no pivot vanishes identically for any fixed complex a.

(3) The same elimination operations, with (x, y, z) replaced by $(x/2, y/3, z/13)$, are applied to $C_1(s-1)v(s)$. Among the rows whose leading part is identically zero, one has nonzero resulting scalar $b(s)$. Both the numerator and denominator of this scalar satisfy the same constant-coefficient test. This is the row labelled $H_{3,18}$ after elimination.

The last row is a genuine left syzygy: its original row label does not mean that an exact Hecke identity itself has a nonzero Taylor source. The nonzero source comes from the transformation rows subtracted during elimination.

For any fixed complex a, distinct monomials in x, y, z become exponentials with distinct real exponents, by unique factorization of powers of $2, 3, 13$. A polynomial with the constant coefficient specified above therefore cannot vanish identically on the exponential curve. This proves that both the 77 pivots and $b(s)$ are nonzero meromorphic functions for every a, including $a = 0$. Since the leading kernel is one-dimensional and $b(s) \neq 0$, the augmented matrix in the lemma has rank 78. The lemma proves the proposition. The eight exact results and their source hashes are recorded in `level26_first_jet.json` and its manifest.

This argument proves continuation only. Dividing by nonconstant exponential polynomials is legitimate for meromorphic continuation, but cannot be used to discard their zeros in a pole-cancellation argument. No bounded pole strip or modularity is asserted here.

Level 26: a denominator-156 quotient. Put $Q = 156$, $h = s - k/2$, $x = 2^h$, $y = 3^h$, $z = 13^h$, and $a = a_3 3^{1-k/2}$. Retain the three signs and the row conventions of Section 33. Use precisely the existing row builder, with its grid size changed from 78 to 156.

Let $A(s)$ denote the leading matrix and $C_m(s)$ its positive Taylor coefficient matrices before multiplication by the scalar Taylor factors. Exact elimination using only nonzero constant multiples of monomials in x, y, z gives 148 pivots. For the sign choice $(1, 1, 1)$ the free coordinates are

$$60, 95, 101, 108, 111, 123, 129, 139.$$

All eight sign cases have the same free coordinates in the supplied elimination order. Let $w_j(s)$ be the reconstructed kernel candidate obtained by setting free coordinate j to one and the other free coordinates to zero. For the six odd free coordinates, the exact checks prove

$$A(s)w_j(s) = 0, \qquad w_j(s)_{2r} = 0, \qquad C_m(s)w_j(s+m) = 0 \quad (m \geq 1).$$

The last assertion is an all-orders support calculation, not a check of finitely many Taylor orders: for every individual matrix row with lower left entry $C \neq 0$, its target coordinate of w_j vanishes; for every Ramanujan–Fricke row, the entire target sum vanishes. These equalities remain true after any shift of s. All entries of the w_j have monomial denominators in x, y, z. Some depend on a/y and $1/y^2$; they must not be described as constant vectors.

Define the holomorphic family of projections

$$P(s)v = v - \sum_{j \text{ odd free}} v_j w_j(s).$$

The free-coordinate normalization gives $P(s)^2 = P(s)$, and its image is the fixed coordinate subspace on which the six listed odd coordinates vanish. Thus $A(s) = A(s)P(s)$ and $C_m(s) = C_m(s)P(s+m)$. Every even coordinate is unchanged by P. Consequently the exact Taylor–Fricke continuation system descends to $\bar{D}(s) = P(s)(F(s, r/156))_{0 \leq r < 156}$, a vector in a fixed 150-dimensional space. It is legitimate to continue this projected vector without asserting continuation of the six discarded functions.

The remaining leading rows involve only coordinates 60 and 108. Two of these rows have a determinant which is a nonzero exponential polynomial for every fixed complex a, including $a = 0$. This is checked by finding a monomial in x, y, z whose coefficient is a nonzero rational constant independent of a; the distinct exponential frequencies are independent because 2, 3 and 13 are

multiplicatively independent. The supplied certificate records such a determinant separately for each sign choice. Therefore the usual finite Taylor continuation induction applies directly to this 150-dimensional quotient: invert the leading matrix meromorphically, use the already continued shifted arguments and the holomorphic Taylor remainder, and extend successively to every left half-plane. In particular it recovers meromorphic continuation of every denominator-78 twist.

Pole constraints and their present limitation. For signs $(1,1,1)$ the remaining leading and zero-row first-Taylor constraints reduce, by monomial units only, to three rows on the two even free coefficients. Their exact entries are recorded in the accompanying first-consistency output. The original four normalized rows satisfy $r_1 = r_2 + (z-1)r_3$, with r_4 the first-Taylor row. Hence r_2, r_3, r_4 suffice. The gcd of their two-by-two minors is a monomial unit, rather than the single degree-six obstruction found on $Q = 78$; nevertheless this gcd calculation does not imply that the minors have no common zero. The saturated ideal still has a nonempty algebraic locus. No global pole exclusion follows from the calculation alone.

At a rightmost pole $s_0 \neq k$ of the quotient, the leading pole coefficient satisfies these three rows: the first-Taylor scalar factor $s_0 - k$ may then be canceled. At $s_0 = k$ that cancellation is invalid. Higher-jet exploratory calculations divide the nth shifted principal coefficient by $\prod_{r=0}^{n-1}(s_0 - k - r)$, so their conclusions do not cover zeros of that product. In particular neither weight two nor poles at $s = k$ have been disposed of by this refinement alone. The enlarged calculation and modularity argument below complete all sign cases.

34. Level 26: auxiliary prime 5 and an enlarged closed system

We first establish continuation. The higher-Taylor germ argument in Theorem 35.1, together with its exact sign transport, removes poles for both bad-13 signs. Put $h = s - k/2$, $(x,y,z,w) = (2^h, 3^h, 13^h, 5^h)$, $(a,b) = (\lambda_3, \lambda_5)$, and let $(\eta, \sigma_2, \sigma_{13})$ be the three signs.

Uniform continuation. Use words of length at most three in the original known transformations T, U_{26}, E_2, W_{26} and their inverses, all closed Ramanujan–Fricke rows, and Hecke rows at $2, 3, 5, 13$. The prime-5 eigenvalue b is independent of a and may be any complex number, including zero. On the 390-grid a complete-support-closed subsystem has 195 coordinates; it contains all 78-grid and all 5-grid coordinates. On the 780-grid there is a closed subsystem of 320 coordinates, containing the doubled 195-coordinate subsystem, every 78-grid coordinate, and all coordinates of denominators $4, 5, 12, 20$. The two support sets are saved in `level26_prime5_support.json` and `level26_prime5_wide_support.json`.

For every sign choice the leading matrices have rank 195 and 320, respectively. Every elimination pivot passes the uniform exponential coefficient audit: a monomial in the four exponential variables has a nonzero rational coefficient independent of both a and b. Therefore no eigenvalue specialization makes a pivot identically zero. The usual closed Taylor strip induction continues both finite vectors meromorphically to the entire plane. This makes no claim about coordinates outside the specified support sets.

Positive sign at 13: use the full germ argument. The leading and first-source rank calculation recorded in `level26_prime5_positive_rank.json` does not by itself exclude poles. After elimination, the lower germs can have higher pole order than the top germs; vanishing parameter coefficients can then contribute to the top Laurent coefficient. Thus one cannot infer that the first-source rows annihilate that coefficient. The full germ argument below avoids this residue-order issue.

The replacement is Theorem 35.1 below. Its polynomial syzygies eliminate the entire lower germs before any Laurent coefficient is taken. The exact symmetry $(\eta, \sigma_{13}, z) \mapsto (-\eta, -\sigma_{13}, -z)$ transports the negative-sign certificate to every positive-sign case. No restriction on either auxiliary eigenvalue is introduced.

The weight point. For every sign choice, specialize the 320-system leading matrix at $(x,y,z,w) = (2,3,13,5)$, leaving a, b arbitrary. In the two projective charts for its remaining two coordinates, the

resulting polynomial ideals over $\mathbb{Q}$ both have Groebner basis $\{1\}$. This uses only leading equations, with no division by the vanishing first Taylor scalar. Consequently no simple pole at $s = k = 2$ can be the only active obstruction. The new polynomial resultant and exact Abel-tail argument in Section 35 proves full leading rank at $s = k$ for every positive even weight, all signs, and arbitrary a, b. This leading-rank result is an input to the full germ argument below; it does not alone establish entireness. That argument treats both signs by eliminating entire lower germs before taking Laurent coefficients.

35. Level 26: the weight-point obstruction for every even weight

This calculation proves that the leading matrix of the closed 320-coordinate 780-system has full rank at $s = k$ for every positive even weight k, all eight sign choices, and arbitrary complex λ_3, λ_5. It does not by itself exclude other active poles for either value of σ_{13}.

Write $k = 2m$, so $m \geq 1$, and put

$$(x, Y, z, W) = (2^m, 9^m, 13^m, 25^m), \qquad A = \lambda_3 3^m, \quad B = \lambda_5 5^m.$$

In particular all four exponential parameters are positive and nonzero, whereas A, B remain arbitrary complex numbers. The Laurent-monomial elimination already recorded in `level26_prime5_wide_certificate.json` leaves two free coordinates. Divide the nonzero vector into its two projective charts.

A polynomial elimination without specialization exceptions. In the chart where the first free coordinate is 1 and the second is t, the normalized leading rows give five polynomials $p_0, \ldots, p_4 \in \mathbb{Z}[A, t, B, x, Y, z, W]$. Their normalization divides only nonzero exponential monomials and nonzero rational constants. In every sign case, p_0, p_1 are independent of B and linear in t, and

$$c_B := \partial_B p_4 \in \{x^2, -x^2\}.$$

Set $b_0 = p_4 - p_0 - c_B B$. A common zero of the leading rows has $B = -b_0/c_B$. Consequently the following three polynomial equations are necessary:

$$p_0 = p_1 = q = 0, \qquad q = x^2 p_3\big|_{B=-b_0/c_B}.$$

The displayed q is a polynomial: the only introduced denominator is $\pm x^2$, and p_3 has degree at most one in B. Form

$$r = x^{-u} \operatorname{Res}_t(p_0, p_1), \qquad v = x^{-v_0} \operatorname{Res}_t(p_0, q), \qquad D = \operatorname{Res}_A(r, v),$$

where u, v_0 remove their greatest common powers of x. All three expressions are polynomials. Since $x \neq 0$, any common zero of the leading rows gives $r = v = 0$, hence $D = 0$. This implication is valid at every specialization: polynomial resultants vanish when their specialized polynomials have a common root, including specializations at which their degrees drop. No function-field Groebner-basis specialization is being invoked.

In the other chart, where the free coordinates are $(0, 1)$, the coefficients $u_0 = [t]p_0$ and $u_1 = [t]p_1$ give necessary equations. They are independent of B, and their A-resultant D_0 is again a necessary zero.

Exact nonvanishing on every physical weight. For either resultant, expand

$$D(2^m, 9^m, 13^m, 25^m) = \sum_{j=1}^{J} c_j b_j^m, \qquad b_1 > b_2 > \cdots > b_J > 0,$$

with integer coefficients and distinct positive integer bases. Distinct monomials have distinct bases by unique prime factorization. The certificate lists all coefficients and verifies the following integers

exactly: for a sign $e \in \{1, -1\}$ and a threshold M,

$$S_i = \sum_{j \le i} e c_j b_j^M \ge 0 \quad (1 \le i \le J), \qquad S_J > 0.$$

For every real $r \ge 0$, summation by parts gives

$$\sum_j e c_j b_j^{M+r} = \sum_{i<J} S_i (b_i^r - b_{i+1}^r) + S_J b_J^r > 0.$$

This proves nonvanishing for every integer $m \ge M$. The finitely many smaller positive integers are evaluated exactly. With signs ordered as $(\eta, \sigma_2, \sigma_{13})$, the thresholds for D are

signs	$---$	$--+$	$-+-$	$-++$	$+--$	$+-+$	$++-$	$+++$
M	7	7	1	1	3	3	6	6

For D_0, $M = 1$ in all eight cases. The only zero finite evaluations of D are $m = 1$ in the cases $---, --+, +--, +-+$. At that single weight, the original leading equations have ideal $\{1\}$ in both projective charts, as verified by the exact rational Groebner-basis calculation already recorded in the wide certificate. Thus both charts are impossible in every case, proving full leading rank at $s = k$.

Reproduction and consequence. The coefficient arrays, first resultants, sign thresholds, finite integer evaluations, and weight-two checks are recorded in `level26_prime5_weightk_certificate.json`. For each three-letter sign code, first run `level26_prime5_weightk_eliminate.pyCODEmagma` to export the normalized leading polynomials, then run `level26_prime5_weightk_resultant.pyCODEprepare`. Run the resulting bounded Magma source in `/tmp`; finally run `level26_prime5_weightk_resultant.pyCODE` to verify the integer Abel certificates. Each final resultant takes less than a second in the recorded runs. The script divides only powers of x and its output explicitly lists any exceptional finite weights.

The same leading-rank calculation applies whenever $h = s - k/2$ is a positive integer: its coefficients depend only on h and the eigenvalues, not separately on s, k. In particular it excludes rightmost poles at both $s = k$ and $s = k + 1$ for every sign. This supplies the normalization exceptions in the full germ argument that follows.

35.1. **Level 26: exclusion of poles for both bad-13 signs.** Let $k > 0$ be even. Use the established meromorphic closed subsystem on $U/780$, where $|U| = 320$ and U contains every lift of the original 78-grid. All three signs are arbitrary. We first treat ramified sign -1 at 13, then transfer the full germ certificate to sign $+1$. No restriction is imposed on either auxiliary Hecke eigenvalue $a = \lambda_3$ or $b = \lambda_5$.

Theorem 35.1. *Every additive Mellin transform in this closed* 320*-coordinate system is entire, for all eight choices of* $(\eta, \sigma_2, \sigma_{13})$*. In particular, every transform on the original* 78*-grid is entire.*

Proof. First suppose $\sigma_{13} = -1$. Suppose there is a pole, and choose one, s_0, rightmost in its class modulo integer translation. Right-half-plane holomorphy guarantees such a choice. All strictly positive integer shifts of every coordinate are then holomorphic at s_0.

The independent positive-integer-h leading-rank certificate, with $h = s - k/2$, gives rank 320 whenever h is a positive integer, uniformly in both auxiliary eigenvalues and all signs. In particular $s_0 = k$ and $s_0 = k + 1$ are impossible. We may consequently work with the following normalized meromorphic germs, for $n = 0, 1, 2$:

$$V_n(S) = \frac{D(S - n)}{i^n \prod_{j=0}^{n-1} (S - k - j)}.$$

All denominators are nonzero in a sufficiently small neighborhood of $S = s_0$.

The finite Taylor identities, with sufficiently many terms, give exact relations of germs modulo holomorphic functions. All terms at $D(S+m)$, $m>0$, are holomorphic. After the displayed normalization, the coefficient of a source at the mth preceding layer is its shifted leading scalar times $c^m/m!$, with $c=C(Cr+D)/\det(g)$ for a known matrix $g=\left(\begin{smallmatrix}A & B\\ C & D\end{smallmatrix}\right)$, and $c=\ell_q t/780$ for a translated Ramanujan–Fricke row. This is precisely the finite system generated by `level26_negative_poles_generate.py`.

At each of the three layers, 318 leading pivots are nonzero Laurent monomials in the exponential variables. Their elimination expresses the whole layer in terms of two free germs, earlier layers, and holomorphic functions. Retaining every residual and zero row produces 185 linear equations on the six free germs

$$(v_{0,1},v_{0,2},v_{1,1},v_{1,2},v_{2,1},v_{2,2}).$$

The top free coordinates are $300,540$, except in the sign case $(-,+,-)$, where they are $480,540$. Since a pole exists, the highest Laurent vector of the two top free germs is nonzero.

Put

$$x=2^{S-k/2},\quad Y=3^{2(S-k/2)},\quad z=13^{S-k/2},\quad W=5^{2(S-k/2)},\quad A=a3^{S-k/2},\quad B=b5^{S-k/2}.$$

Multiplication by nonzero exponential monomials makes every row polynomial in x,Y,z,W,A,B. Removing rational scalar repetitions leaves 90 distinct rows. This step does not divide by a, b, or any possibly vanishing parameter polynomial.

Here is the crucial elimination of the four lower germs. Choose four rows whose lower 4×4 block is L, with nonzero polynomial determinant d, and let E be their top 4×2 block. For any other row $(r_{\mathrm{top}},r_{\mathrm{low}})$, set

$$c=r_{\mathrm{low}}\operatorname{adj}(L),\qquad g=\gcd(d,c_1,c_2,c_3,c_4),\qquad q=d/g,\quad e=c/g.$$

All entries of q,e are polynomials. The certificate explicitly checks the polynomial identity $qr_{\mathrm{low}}=eL$. Consequently

$$(qr_{\mathrm{top}}-eE)(v_{0,1},v_{0,2})^{\mathsf{t}}$$

is holomorphic. This is valid even where d or g vanishes: one takes a polynomial combination of the original germ equations, and never divides a holomorphic error term by g. The four lower germs have been eliminated without any bound on their pole orders.

Take the highest Laurent coefficient of the two top germs. In the chart $(v_{0,1},v_{0,2})=(0,1)$, the polynomial equations, together with $txYzW-1=0$, generate the unit ideal in every sign case. Thus normalize the nonzero first coordinate to one and put $u=v_{0,2}/v_{0,1}$. In this chart the primitive equations and the same saturation relation give a zero-dimensional ideal. Exact elimination onto (x,Y) gives

$$p_{\eta,\sigma_2}(x)=0,\qquad Y=q_{\eta,\sigma_2}(x).$$

The backend checks that both projected polynomials belong to the original chart ideal. Their exact rational coefficients are saved in the four `level26_negative_poles_*_projection.json` files. Write p,q for the $(-,-,-)$ pair: p is monic of degree 21 and q has degree 20.

The exact sign-comparison certificate proves the following identities and congruences:

$(\eta,\sigma_2,\sigma_{13})$	$p_{\eta,\sigma_2}(x)$	$q_{\eta,\sigma_2}(x)\pmod{p(\pm x)}$
$(-,-,-)$	$p(x)$	$q(x)$
$(-,+,-)$	$-(x^2+5x+8)p(-x)$	$q(-x)$
$(+,-,-)$	$(x^2-5x+8)p(x)$	$q(x)$
$(+,+,-)$	$-p(-x)$	$q(-x)$.

In either quadratic factor, the corresponding Y polynomial is congruent to 9. The quadratic roots are nonreal conjugates with product 8, so $|x|^2=8$. Physically this forces $\Re(S-k/2)=3/2$, whereas $Y=9$ forces that real part to be 1. These branches are impossible.

For the degree-21 branch, the independent rational-disk certificate `level26_root_modulus_audit.py` encloses all 21 roots of p in disjoint rational disks. Rational Rouché inequalities certify one root in each disk; the degree count certifies that none is omitted. Rational bounds for $|x|^2$ and $|q(x)|^2$, together with

$$2^{3169} < 9^{1000} < 2^{3170},$$

prove on every disk that

$$|q(x)| \neq |x|^{\log 9/\log 2}.$$

This contradicts the necessary physical identity $|Y| = |x|^{\log 9/\log 2}$. Reflection $x \mapsto -x$ preserves the modulus and therefore excludes the reflected branches too. All projective possibilities for the nonzero top Laurent vector are now excluded. Hence no pole exists.

Transport to the positive bad-13 sign. Let ι be the automorphism of the Laurent coefficient field which fixes x, y, w, a, b and sends z to $-z$. The complete closed Taylor systems, including their leading rows and all their source terms, are transported by

$$(\eta, \sigma_2, \sigma_{13}; z) \longmapsto (-\eta, \sigma_2, -\sigma_{13}; -z).$$

Here the map is an identity of systems over a formal coefficient field; it is not a claim that -13^h is the physical exponential at the same value of h.

For a known transformation g write $\chi(g) = (-1)^{v_{13}(\det g)}$. This is invariant under scalar rescaling of a rational matrix and is multiplicative. The generators T, U_{26}, E_2, W_{26} have respective values $1, 1, 1, -1$. Consequently changing the Fricke sign changes the known-map eigenvalue $\mu(g)$ to $\chi(g)\mu(g)$, for every word used in the builder. The leading scalar of its grid row is

$$\mu(g)\,\mathrm{mon}\left(\frac{(Cj+DQ)^2}{\det(g)Q^2}\right).$$

Its monomial has z-exponent congruent to $v_{13}(\det g)$ modulo two. Thus applying ι changes this scalar by exactly $\chi(g)$, as required. Rows whose image is infinity have no such scalar and are unchanged.

For a translated Ramanujan–Fricke row put $\ell_q = \mathrm{lcm}(26, q^2)$. Up to its fixed overall sign, its scalar is

$$\eta(-\sigma_2)^{[2|q]}(-\sigma_{13})^{[13|q]}\,\mathrm{mon}(\ell_q t^2/Q^2).$$

The z-exponent is odd when $13 \nmid q$ and even when $13 \mid q$. This agrees with the change of the displayed signs under the displayed sign transport. Untranslated Ramanujan rows are unchanged. In the Hecke row at 13 the only sign-bearing coefficient is $-\sigma_{13}/z$, so it transforms correctly too. The Hecke rows at $2, 3, 5$ are fixed. All destinations and the complete-support condition depend only on the rational matrices and grid, so the same 320-coordinate support and the same source destinations are preserved.

Every source coefficient at order m is its shifted leading scalar times $c^m/m!$, where

$$c = \frac{C(Cj+DQ)}{\det(g)Q} \quad \text{or} \quad c = \frac{\ell_q t}{Q}.$$

These slopes are rational and independent of the signs. Moreover ι commutes with the shift $(x, y, z, w) \mapsto (x/2^n, y/3^n, z/13^n, w/5^n)$. This proves the symmetry at every Taylor order, before any elimination. The checker `level26_positive_repair_check.py` independently compares every closed raw row and every source of orders one and two for all four sign pairs, retaining row multiplicities and exact rational function coefficients.

Now take a hypothetical rightmost pole of a positive bad-13 system. The existing all-sign positive-integer-h leading-rank certificate excludes $s_0 = k, k+1$. Normalize its three layers by

$$V_n(S) = D(S-n)\Big/\left(i^n \prod_{j=0}^{n-1}(S-k-j)\right), \qquad n = 0, 1, 2.$$

The denominators are units near s_0. Apply the negative-sign certificate with signs $(-\eta, \sigma_2, -1)$ and with its formal z replaced by $-13^{S-k/2}$. All monomial pivots remain units. Every primitive lower-column syzygy remains a polynomial identity after this substitution. Its combination of the full germ equations is holomorphic, even at zeros of the selected determinant or of the common gcd removed from the complete syzygy. Thus the lower germs are removed without comparing their pole orders to those at the top.

Both projective-chart certificates transfer unchanged. In the first chart their projected necessary equations involve only x, Y; in the second chart their ideal is the unit ideal. The saturation factor $xYzW$ stays nonzero under the substitution. Since $x = 2^h$ and $Y = 9^h$ retain their physical values, the existing exact modulus exclusion applies without alteration. The correspondence is

positive case	negative certificate
$(-,-,+)$	$(+,-,-)$
$(-,+,+)$	$(+,+,-)$
$(+,-,+)$	$(-,-,-)$
$(+,+,+)$	$(-,+,-)$.

Hence the top Laurent vector is zero, a contradiction. The closed 320-coordinate vector is entire for all eight signs. No new Gröbner-basis or root-enclosure calculation is required: the repair transports the existing polynomial germ certificates, rather than using the incorrect first-source residue inference. □

Reproducibility and scope. The symbolic row generator, exact Laurent normalization, primitive polynomial syzygies, chart ideals, and projection extraction are respectively the files with suffixes `generate.py`, `eliminate.py`, `backend.m`, and `extract.py` under the prefix `level26_negative_poles_`. The file `level26_negative_poles_signs.py` checks every cross-sign factorization and congruence and binds them to the exact root-disk certificate. Magma runs use a bounded process alarm, one thread, and a 500-MB memory limit; the final backend disables the Monte-Carlo global-modular algorithm for all internal computations. This proof uses no unproved Ramanujan estimate, no generic-eigenvalue assumption, and no comparison between pole orders at different points.

35.2. **An exact modulus exclusion for a finite algebraic locus.** Suppose a residue calculation forces $P(x) = 0$ and $Y = Q(x)$, where $P, Q \in \mathbb{Q}[X]$. In the level-26 application $x = 2^h$ and $Y = 9^h$, so the necessary relation is

$$|Y|^2 = (|x|^2)^\rho, \qquad \rho = \frac{\log 9}{\log 2}.$$

The exact integer inequalities $2^{3169} < 9^{1000} < 2^{3170}$ give $3169/1000 < \rho < 3170/1000$.

The verifier `level26_root_modulus_audit.py` chooses rational complex centers from numerical approximations, but proves every subsequent assertion using rational arithmetic. Write $P(c+u) = \sum_j p_j u^j$ at each center c. For a rational radius r it checks

$$|\Re p_0| + |\Im p_0| + \sum_{j\ge 2}(|\Re p_j| + |\Im p_j|)r^j < \max(|\Re p_1|, |\Im p_1|)r.$$

Rouché's theorem then puts exactly one root, counted with multiplicity, in the disk $|x - c| < r$. The pairwise disjoint disks number $\deg P$, so they exhaust all roots. Numerical root finding therefore serves only to select proposed centers; it is not evidence of completeness.

Writing $Q(c+u) = \sum_j q_j u^j$, the rational number $e = \sum_{j\ge 1}(|\Re q_j| + |\Im q_j|)r^j$ bounds $|Q(x) - Q(c)|$. The inequalities

$$|c|^2 - 2(|\Re c| + |\Im c|)r - r^2 \le |x|^2 \le |c|^2 + 2(|\Re c| + |\Im c|)r + r^2$$

and their analogues at $Q(c)$ with radius e give rational intervals $[u_-, u_+]$ and $[v_-, v_+]$ for the two squared moduli. Endpoints are rounded outward using exact integer division. Every interval is positive and lies strictly on one side of $u = 1$.

If $u_- > 1$, the physical relation is impossible whenever either $v_+^{1000} < u_-^{3169}$ or $v_-^{1000} > u_+^{3170}$. If $u_+ < 1$, it is impossible whenever either $v_+^{1000} < u_-^{3170}$ or $v_-^{1000} > u_+^{3169}$. These are exact rational comparisons. The saved certificate for the sign triple $(-1, -1, -1)$ verifies this exclusion on all 21 root disks of radius 10^{-40}, with the source polynomial hash recorded. This excludes that finite algebraic locus; its derivation from the original Taylor equations is a separate certificate obligation. Independent source-to-projection audit at level 26. The four cases $(\eta, \sigma_2, \sigma_{13}) = (-,-,-), (-,+,-), (+,-,-), (+,+,-)$ were independently regenerated from the original 780-grid builder. The independent runner executes the original source, redirecting scratch outputs to a disjoint directory prefix; it does not start with the exported polynomial ideals. Source and output hashes are recorded at each generation, normalization, and extraction step.

The normalization substitutes $Y = y^2$, $W = w^2$, $A = ay$, $B = bw$, checks the resulting integral exponents, and removes only invertible exponential monomials and nonzero rational row scalars. The auxiliary eigenvalues remain arbitrary. In particular, no division by A or B or by any nonconstant parameter polynomial occurs in this normalization.

For the lower-germ elimination, the four chosen pivot rows merely select a polynomial minor. Both the adjugate identity and the primitive polynomial syzygy identity are checked exactly. The resulting identities remain valid at zeros of the minor or of the common factors removed from the syzygy coefficients: the primitive coefficients themselves are polynomials and satisfy the original lower-column identity. Thus this calculation neither assumes a generic point nor compares pole orders at different translates.

All final Magma runs set `SetGBGlobalModular(false)` before constructing the chart ideals. Both explicit Gröbner-basis calls also set `GlobalModular:=false`; the process-wide setting covers internal elimination and order-change computations. The second chart has basis [1] in every case. In the first chart the basis sizes are $40, 42, 42, 40$, respectively. The exported univariate polynomial degrees are $21, 23, 23, 21$, and the corresponding Y polynomial degrees are $20, 22, 22, 20$. Each exported polynomial is additionally checked for membership in the original saturated chart ideal. Fresh extracted rational coefficient strings agree exactly with the saved four projection files; no floating approximation enters this comparison.

The machine-readable audit is `level26_negative_independent_manifest.json`. It records the actual source hashes, intermediate hashes, completion evidence and projection comparisons. This audit supplies the algebraic implication from the normalized germ equations to the finite (x, Y) locus. The analytic germ normalization, positive-integer leading-rank exclusion, and rational-disk exclusion of the physical locus are separate inputs to the complete pole proof, not consequences of a polynomial comparison alone.

35.3. **Level 26: modularity from the simple-pole restriction.** Assume the original level-26 hypotheses and, additionally, that $D_{2/13}(s)$ and $D_{\pm 1/3}(s)$ continue meromorphically with no poles except possibly simple poles at $s = k$. Here D_r is the completed additive Mellin transform of f and $k > 0$ is even. We prove $f \in S_k(\Gamma_0(26))$. In particular the pole theorem on the continued denominator-780 subsystem is more than sufficient. This subsection does not itself establish that pole theorem.

Write $\delta = -\sigma_2$, and use the determinant-normalized slash operator. The original scalar laws are $f|T = f$, $f|E = \delta f$, $f|W = \eta f$, where

$$T = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad E = \begin{pmatrix} 2 & -1 \\ -26 & 14 \end{pmatrix}, \quad W = \begin{pmatrix} 0 & -1 \\ 26 & 0 \end{pmatrix}.$$

Matrices are identified up to nonzero scalar, as the weight is even. Set

$$P = \begin{pmatrix} 4 & -1 \\ 26 & -6 \end{pmatrix}, \quad Q = \begin{pmatrix} 4 & 1 \\ -26 & -6 \end{pmatrix}, \quad h = f|P - \delta f, \quad j = f|Q - \delta f.$$

The Ramanujan-4 and Ramanujan-6 functional equations have conductors 208 and 468, respectively, and both have sign $\delta\epsilon_f$. Their center matrices are respectively (P, Q) and (P^{-1}, Q^{-1}). Mellin inversion gives

$$h|t_{1/4} + j|t_{-1/4} = 0, \qquad h|P^{-1}t_{1/6} + j|Q^{-1}t_{-1/6} = 0, \qquad t_u = \begin{pmatrix} 1 & u \\ 0 & 1 \end{pmatrix}.$$

In the second equation the common factor $-\delta$ from the inverse matrix defects has been removed. Thus $j = -h|t_{1/2}$ and

$$h|M = h, \qquad M = t_{1/2}Q^{-1}t_{-1/3}P = \begin{pmatrix} 20 & -3 \\ 52 & -6 \end{pmatrix}. \tag{11}$$

Also $Q = W^{-1}P^{-1}W^{-1}$, so the first paired equation gives

$$h|K = \delta\eta h, \qquad K = P^{-1}W^{-1}t_{-1/2} = \begin{pmatrix} 26 & -7 \\ 104 & -26 \end{pmatrix}. \tag{12}$$

The matrices K and $P^2 = (5, -1; 26, -5)$ are involutions, while $P^4 = 1$. The latter identity gives the automatic relation

$$h|(1 + \delta P + P^2 + \delta P^3) = 0.$$

Since $P^2MP^{-2} = M^{-1}$, the holomorphic function $w = h + h|P^2$ satisfies

$$w|M = w, \qquad w|P = -\delta w.$$

Consequently $w|MP = -\delta w$. But $(\operatorname{tr}(MP))^2/\det(MP) = 49/18$ is nonintegral and strictly between 0 and 4. Thus MP is an infinite-order elliptic matrix. The positive-weight holomorphic elliptic lemma gives $w = 0$, so

$$h|P^2 = -h. \tag{13}$$

Now put

$$Z = KP^2 = \begin{pmatrix} 52 & -9 \\ 156 & -26 \end{pmatrix}.$$

It commutes with M, and (12)–(13) give $h|Z = -\delta\eta h$, hence $h|Z^2 = h$. Both M and Z are hyperbolic, with common fixed points

$$r_\pm = \frac{13 \pm \sqrt{13}}{52}.$$

Their expanding eigenvalues after determinant-one normalization are

$$\alpha = \frac{7 + \sqrt{13}}{6}, \qquad \beta = \frac{3 + \sqrt{13}}{2}.$$

Their logarithms are rationally independent. Indeed β is an algebraic unit, satisfying $X^2 - 3X - 1 = 0$, whereas α is not an algebraic integer: its conjugate sum is $7/3$. If a nonzero integral power of α were an integral power of β, the right side would be an algebraic unit, and integrality of a root of a monic polynomial would force α to be integral, a contradiction. The same argument applies to β^2.

It follows that the subgroup generated by M and Z^2 is dense in the connected one-parameter hyperbolic subgroup with these two fixed points. Continuity of the slash action and (11) therefore make h invariant under that whole subgroup. Conjugating the subgroup to positive dilations shows that

$$h(z) = c\big((z - r_+)(z - r_-)\big)^{-k/2} \tag{14}$$

for a constant c. To see this directly after conjugation, the invariance becomes $g(tz) = t^{-k/2}g(z)$ for every $t > 0$; differentiating gives $zg'(z) = -(k/2)g(z)$. The integer $k/2$ makes the displayed model single-valued and holomorphic in the upper half-plane.

We now use the pole hypothesis, at the single rational point $P\infty = 2/13$. The truncated Mellin transport lemma in Section 26.1 says that

$$\int_1^\infty (f|P)(iy)y^{s-1}\,dy$$

continues with no poles except possibly a simple pole at zero, provided $D_{2/13}$ has only a possible simple pole at k. The same assertion holds for the truncated Mellin transform of h, since $f(iy)$ decays exponentially. On the other hand, if $c \neq 0$ in (14), then $h(iy) = ci^{-k}y^{-k} + O(y^{-k-1})$, so its truncated Mellin transform has a nonzero simple pole at $s = k$. As $k > 0$, this is impossible. Therefore $h = 0$ and $f|P = \delta f$; the paired relation also gives $f|Q = \delta f$.

Recovering the denominator-3 transformation. Put $A_q = D_{1/q} - D_{-1/q}$, $x = 2^{s-k/2}$, $y = 3^{s-k/2}$, and $a = a_3 3^{1-k/2}$. The known matrix $ETW = (13, -1; -156, 13)$ gives the entire, rapidly decreasing odd denominator-12 transform and its functional equation

$$A_{12}(s) = \sigma_2\epsilon_f 1872^{k/2-s}A_{12}(k-s).$$

The newly proved P, Q laws similarly give

$$A_4(s) = \sigma_2\epsilon_f 208^{k/2-s}A_4(k-s),$$

with entireness and rapid vertical decay, directly from the two Mellin endpoints. Exact Hecke identities at 2 and 3 give

$$2A_{12} = (\sigma_2 x^{-1} + x^{-2})A_3 + (1 + a/y + y^{-2})A_4.$$

The coefficient $C_3 = \sigma_2 x^{-1} + x^{-2}$ is nonzero at $s = k$, since $x = 2^{k/2} > 1$. Thus the assumed only possible simple pole of A_3 at k is removable. The coefficient ratios under reflection are $C_3/C_3^* = \sigma_2 x^{-3}$ and $(1 + a/y + y^{-2})/(1 + ay + y^2) = y^{-2}$. Substitution therefore proves

$$A_3(s) = \epsilon_f 234^{k/2-s}A_3(k-s).$$

Rapid decay follows from the displayed quotient for A_3, using the maximum principle in uniform small disks around the zeros of C_3. The Ramanujan-3 equation has the same conductor and sign, so Mellin inversion gives

$$f|G = f, \qquad G = \begin{pmatrix} 3 & -1 \\ -26 & 9 \end{pmatrix}.$$

Generation and cusp decay. The scalar kernel of T, E, W, P, G has 14 explicit integral Schreier generators, formed with representatives $1, E, W, EW$. The accompanying exact exporter verifies their determinants and lower-left divisibility by 26, and verifies their words in $\mathrm{PSL}_2(\mathbb{Z})$. Bounded Magma enumeration gives index 42, equal to the index of $\Gamma_0(26)$. Hence f is modular. The Atkin–Lehner-2 matrix E and Fricke matrix W act transitively on the four squarefree-level cusps; their eigenlaws carry the exponential decay at infinity to every cusp. Thus $f \in S_k(\Gamma_0(26))$.

The matrix and quadratic-field identities are checked in `level26_modularity_agent_complete.py`; the group enumeration is exported by `level26_modularity_agent_generation.py`. All uses of the pole restriction have been stated explicitly, and no unramified Hecke eigenvalue has been divided out.

Corollary 35.2. *Under the original level-26 hypotheses, $f \in S_k(\Gamma_0(26))$ for every permitted positive even weight, every choice of signs, and every Hecke eigenvalue.*

Proof. The enlarged pole calculation makes all 320 continued twists entire, including $D_{2/13}$ and $D_{\pm 1/3}$. The preceding modularity argument applies. Fresh regeneration of all eight enlarged systems, their continuation pivots and weight-polynomial inputs is recorded in `level26_root_positive_audit.json`; exact rational resultant and integer-tail checks independently verify the weight-point certificate. For the four negative bad-13 sign cases, the independent source-to-projection audit and rational-disk certificate above verify the remaining pole exclusion. □

36. The converse theorem at level 28

Theorem 36.1. *Under the hypotheses defining $\mathscr{E}(28,k)$ in* [3]*, with $k \geq 2$ even, the associated function f belongs to $S_k(\Gamma_0(28))$. The assertion holds for every allowed root number and ramified sign and every complex unramified Hecke eigenvalue, including zero.*

Proof. We use the determinant-normalized weight-k slash operator and put

$$a_n = f_n n^{(k-1)/2}, \qquad f(z) = \sum_{n\geq 1} a_n e^{2\pi i n z}, \qquad F(s,r) = (2\pi)^{-s}\Gamma(s)\sum_{n\geq 1} a_n e^{2\pi i n r} n^{-s}.$$

The Mellin variable is unnormalized, with center $k/2$. Write $h = s - k/2$, $x = 2^h$, $y = 3^h$, $z = 7^h$, $a = \lambda_3 = \sqrt{3} f_3$, $\sigma = \sqrt{7} f_7 \in \{\pm 1\}$, and $\eta = i^k \epsilon_f \in \{\pm 1\}$. Convergence of the original Dirichlet series somewhere gives polynomial growth of its coefficients, which suffices for all the finite Taylor–Mellin expansions below.

Known transformations and the exact finite systems. Let $T = t_1$, $t_r = \left(\begin{smallmatrix} 1 & r \\ 0 & 1 \end{smallmatrix}\right)$, $U_m = \left(\begin{smallmatrix} 1 & 0 \\ m & 1 \end{smallmatrix}\right)$, and $W_L = \left(\begin{smallmatrix} 0 & -1 \\ L & 0 \end{smallmatrix}\right)$. The Euler factor at 2 is trivial, so $a_{2n} = 0$ and $f|t_{1/2} = -f$. Mellin inversion of the original functional equation gives $f|W_{28} = \eta f$. Consequently $f|U_{28} = f$ and

$$f|E = f, \qquad E = \begin{pmatrix} 2 & -1 \\ -28 & 16 \end{pmatrix}, \qquad E = 2W_{28}^{-1} t_{1/2} W_{28} t_{-1/2}.$$

The Euler–Ramanujan lemma in the manuscript supplies the entire Ramanujan twists and their Fricke laws with conductor $\ell(q) = \operatorname{lcm}(28, q^2)$ and slash multiplier $\eta_q = \eta(-\sigma)^{[7|q]}$.

Here are the finite rows used in the proof. Set $Q = 84$, and denote by e_j the coordinate with index $j \bmod Q$. Form every word of length at most three in $T, U_{28}, E, W_{28}, t_{1/2}$ and their inverses, retaining its actual scalar multiplier $\mu \in \{\pm 1\}$. For $g = \left(\begin{smallmatrix} A & B \\ C & D \end{smallmatrix}\right)$ and each grid point $r = j/Q$ whose image is again on the grid, include

$$e_j - \mu \rho^h e_\beta, \qquad \rho = \frac{(Cr+D)^2}{\det g}, \qquad \beta = Qgr \pmod Q. \tag{28.8}$$

If $Cr + D = 0$, include the entire-coordinate row e_j instead. For every $q \mid Q$ with $\ell(q) \mid Q^2$, include

$$\sum_{(v,q)=1} e_{vQ/q}, \qquad \sum_{(v,q)=1} e_{vQ/q+t} - \eta_q \left(\frac{\ell(q) t^2}{Q^2}\right)^h \sum_{(v,q)=1} e_{vQ/q+u}, \tag{28.9}$$

where $d = Q^2/\ell(q)$, t runs through the positive and negative divisors of d, and $u = -d/t$. Finally include the exact Hecke rows

$$\sum_{v=0}^{p-1} e_{j+vQ/p} - \lambda_p p^{-h} e_{pj} + \chi_0(p) p^{-2h} e_{p^2 j}, \quad p = 2,3,7, \quad 0 \leq j < Q/p, \tag{28.10}$$

with $(\lambda_2, \lambda_3, \lambda_7) = (0, a, \sigma)$. All indices are reduced modulo Q.

These rows are built by `level28_jet_builder.py` with its grid changed from 42 to 84. For arbitrarily large M, Taylor expansion gives

$$A_0(s)\mathbf{F}(s) + \sum_{m=1}^{M-1} A_m(s)\mathbf{F}(s+m) = \mathbf{H}_M(s), \tag{28.11}$$

where A_0 consists of the rows above, the A_m are entire, and $\mathbf{H}_M$ is holomorphic on $\Re s > B - M$ for a fixed coefficient-growth bound B. In a transformation row (28.8), the positive Taylor term has coefficient

$$\mu\rho^h(i\kappa)^m\binom{s+m-k}{m}, \qquad \kappa = \frac{C(Cr+D)}{\det g}.$$

For a Fricke-sum row, replace $\mu\rho^h$ by the coefficient in (28.9), take $\kappa = \ell(q)t/Q$, and use the corresponding target sum. The Hecke and unshifted entire-sum rows have no higher terms. These are finite Taylor identities with holomorphic remainders; no infinite Taylor-series interchange is needed.

The constant odd mode and meromorphic continuation. Over $\mathbb{Q}(x, y, z, a)$, elimination using only nonzero rational Laurent-monomial pivots gives 82 pivots and free coordinates 24,53. The free-53 kernel direction is the constant vector

$$w_j = \begin{cases} -1, & j = 11, 17, 25, 31, \\ 1, & j = 53, 59, 67, 73, \\ 0, & \text{otherwise.} \end{cases} \tag{28.12}$$

The exact verifier `level28_cusp_probe_support.py` checks $A_0 w = 0$ and the following stronger assertions in all four sign cases. In each transformation row, either $C = 0$ or $w_\beta = 0$. In each Fricke-sum row, $\sum_{(v,q)=1} w_{vQ/q+u} = 0$. Hence $A_m w = 0$ for *every* $m \geq 1$, since the same coordinate or sum multiplies every power of the Taylor slope. This proves all-order closure exactly.

In the initial half-plane define

$$\widetilde{F}_j(s) = F(s, j/84) - w_j F(s, 53/84).$$

This definition assumes no continuation of the subtracted function. Equation (28.11) holds unchanged for $\widetilde{\mathbf{F}}$; its 53-coordinate is zero, and every even coordinate is unchanged. The remaining 83-coordinate leading system has generic full rank for every fixed complex a. Indeed, after the 82 monomial pivots one residual contains a monomial in x, y, z with nonzero coefficient independent of a, in both numerator and denominator. For example, in the case $\eta = \sigma = -1$ it is

$$\frac{2y^4 + 2ay^3 + y^2z - 3y^2 + ayz - ay + z^2 - z}{2x^2y^2}.$$

The coefficient of z^2 is 1. The verifier checks the same criterion in all four cases. Unique factorization at $2, 3, 7$ shows that the selected minor is not identically zero on the exponential curve for any specialization of a.

Invert this minor meromorphically in (28.11). Initially every coordinate is holomorphic for $\Re s > B$; if continuation has reached $\Re s > B - n$, taking $M = n + 1$ continues it one further strip, since every positive shift is already known there. The identity theorem identifies the continuations on overlaps. Thus every quotient coordinate is meromorphic on $\mathbb{C}$. In particular every original 42-grid twist continues, because $\widetilde{F}_{2j}(s) = F(s, j/42)$. This also independently supplies the original-grid continuation proved by the smaller first-jet certificate `level28_first_jet.py`.

Restriction and removal of poles. Choose a rightmost pole s_0 in an integer-translation class of the quotient family. Such a choice exists because all coordinates are holomorphic in a common right half-plane. Its highest nonzero Laurent vector satisfies $A_0(s_0)v = 0$. The monomial pivots and the zero 53-coordinate force its 24-coordinate to be nonzero. This avoids any assumption about a possibly higher auxiliary odd-grid pole.

If $s_0 \neq k$, the first Taylor syzygies have nonzero scalar $s_0 - k$. Combining their zero-row sources with the leading residuals gives

$$y^2 = q, \qquad qa + (q+1)y = 0, \qquad z = -\eta r, \quad q = 15 - 6\eta\sigma, \quad r = 12 - 5\eta\sigma. \tag{28.13}$$

The verifier `level28_root_constraints.py` computes and asserts this exact saturated rational ideal in every sign case; only nonzero monomials are inverted. If $\eta\sigma = -1$, then $|y|^2 = 21$, $|z| = 17$, which is impossible since $7^{10} < 9^9$ and $21^9 < 17^{10}$ imply $21^{\log 7/\log 9} < 17$. If $\eta\sigma = 1$, then $\Re h = 1$, $y^2 = 9$, $z = \pm 7$. The phases force $\Im h = 0$: otherwise $\log 3/\log 7$ would be rational. Therefore the only candidate is

$$h = 1, \qquad \eta = \sigma = -1, \qquad a = -10/3. \tag{28.14}$$

At (28.14), retain all lower-point Laurent vectors. Including the gauge row $e_{53} = 0$, the falling-factorial-normalized systems are

$$A_0(s_0 - n)v_n = \sum_{m=1}^{n} \frac{C_m(s_0 - n)}{m!} v_{n-m}, \tag{28.15}$$

where C_m contains the mth slope power from (28.11). The exact verifier `level28_root_exceptional_jet.py` checks full column rank 84 at the first two shifts, solves both systems, and finds at the third shift the nonzero residual $68600/81$ in the Ramanujan row $q = 7$. For $s_0 \neq k$, $s_0 - k = 1 - k/2$ is a negative integer, so all normalization factors are nonzero. This excludes (28.14) away from k.

At $s_0 = k$, put $m = k/2 \geq 1$, so $y = 3^m, z = 7^m$. The leading ideal alone contains

$$(y^2 - 1)\begin{cases} (z+\eta)^2 - 4y^2, & \eta\sigma = 1, \\ (z+\eta)^2 - 12y^2 - 4, & \eta\sigma = -1. \end{cases} \tag{28.16}$$

These exact ideal memberships are checked in `level28_cusp_probe_weight.py`. For every $m \geq 2$, $7^m - 1 > 4 \cdot 3^m$: it holds at $m = 2$, and multiplying by 7 proves the next step. Hence $(z+\eta)^2 \geq (z-1)^2 > 16y^2 > 12y^2 + 4$, excluding (28.16). For $m = 1$, the squares $(7+\eta)^2$ are 36 or 64, so only $\eta = \sigma = -1$ remains. The leading ideal then also contains $(y^2 - 1)(ay + y^2 + 1)$, forcing $a = -10/3$.

Suppose the pole at $s_0 = k = 2$ has order at least two. Set $e = s - 2$ and $P_n(e) = \prod_{j=0}^{n-1}(e - j)$. Full rank at the lower points bounds their pole orders by one less than the top. Normalize the relevant leading coefficients by the nonzero $i^n P_n'(0)$. The identity

$$\binom{e - n + m}{m} P_{n-m}(e) = P_n(e)/m!$$

gives precisely (28.15), so its third-shift inconsistency excludes every higher-order pole as well. A simple pole at k contributes holomorphically to all lower Taylor equations, because the binomial factor has a zero. Repeating the rightmost-pole argument below that harmless contribution excludes all other poles. We have proved: the quotient is entire except possibly for a simple pole at 2 in the single case $k = 2$, $\eta = \sigma = -1$, $a = -10/3$.

We now remove that last pole. Direct multiplication gives

$$ETW_{28} = -\frac{1}{3} t_{-1/12} W_{1008} t_{-1/12}. \tag{28.17}$$

The translates at $\pm 1/12$ are therefore exchanged by Fricke and decay exponentially at both Mellin endpoints. Their difference $B_-(s) = F(s,1/12) - F(s,-1/12)$ is entire, rapidly decreasing on fixed vertical strips, and satisfies

$$B_-(s) = -\epsilon_f 1008^{-h} B_-(k-s).$$

Since $F(s,5/12) = -F(s,-1/12)$ and $F(s,3/4) = -F(s,1/4)$, the exact Hecke-3 row at $1/4$ gives

$$(a + y + y^{-1})F(s,1/4) = yB_-(s). \tag{28.18}$$

Its denominator is symmetric under $s \mapsto k-s$, so

$$F(s,1/4) = -\epsilon_f 112^{-h} F(k-s,1/4) \tag{28.19}$$

meromorphically. This coordinate is unchanged in the quotient, since $w_{21} = 0$. In weight two it is holomorphic at 0, hence (28.19) excludes its possible pole at 2. The exceptional leading kernel normalized at coordinate 24 has coordinate 21 equal to $7/6 \neq 0$. Thus its entire residue vector vanishes, and every quotient coordinate, including every original 42-grid twist, is entire. The exact center identity, actual Hecke row, and residue $7/6$ are checked in `level28_cusp_probe_poles_chi4.py`.

A genuine new transformation and modularity. Equation (28.18) gives rapid vertical decay of the now entire quarter twist away from small fixed disks about the periodic zeros of $a+y+y^{-1}$. The maximum-modulus principle gives the same bounds inside those disks. Mellin inversion in (28.19) consequently gives $(f|t_{1/4})|W_{112} = -\eta(f|t_{1/4})$, and hence

$$f|G_4 = -f, \qquad G_4 = \begin{pmatrix} 4 & -1 \\ -28 & 8 \end{pmatrix}. \tag{28.20}$$

Indeed, $t_{1/4}W_{112}t_{-1/4}W_{28}^{-1}$ is projectively G_4^{-1}. This argument works for all weights and eigenvalues, not merely the exceptional parameters used to remove the last pole.

Put $\delta = \mathrm{diag}(1,2)$ and $g = f|\delta$. The transformations $t_{1/2}, U_{14}, G_4$ act on f with multiplier -1. After conjugation by δ they become, up to scalar matrices,

$$T, \qquad U = \begin{pmatrix} 1 & 0 \\ 7 & 1 \end{pmatrix}, \qquad E_7 = \begin{pmatrix} 2 & -1 \\ -7 & 4 \end{pmatrix}.$$

These generate $\Gamma_0(7)$ projectively. For an elementary certificate, take the left-coset transversal I, ST^j, $0 \le j \le 6$, where $S = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$. The bottom row modulo 7 identifies the eight cosets. The nonidentity Schreier generators for right multiplication by S, T are

$$T, \quad T^{-1}U, \quad T^{-1}E_7^{-1}, \quad E_7T, \quad E_7U, \quad U^{-1}E_7^{-1}, \quad U^{-1}T, \quad U^{-1}.$$

All 16 matrix identities, including the omitted identity generators, are checked by `level28_cusp_probe_generation.py`. Independently, the companion Magma file enumerates index 8 in $\langle S, T \mid S^2 = (ST)^3 = 1\rangle$, using the checked words $U = ST^{-7}S$ and $E_7 = ST^3ST^{-2}ST^{-1}$.

Each of T, U, E_7 reduces modulo 2 to a transposition in $\mathrm{SL}_2(\mathbb{F}_2) \simeq S_3$. Thus its multiplier -1 is the sign of that permutation, and the multipliers of all words agree with an actual group character. It follows that $\Gamma_0(7) \cap \Gamma(2)$ fixes g. Conjugation by δ identifies this subgroup exactly with $\Gamma_0(28)$: its matrices have even upper-right entry and lower-left entry divisible by 14, and $\delta \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \delta^{-1} = \left(\begin{smallmatrix} a & b/2 \\ 2c & d \end{smallmatrix}\right)$. The converse inclusion follows from the same formula. Hence $f|\Gamma_0(28) = f$.

Cuspidality. There is one cusp of each denominator $d \mid 28$, since $\gcd(d, 28/d) \le 2$. Equivalently, the elementary primitive-pair cusp reduction takes every cusp to some $1/d$, $d \mid 28$; the only possible unit numerators modulo the common divisor are already 1. The scalar transformations $G_4, W_{28}, H = t_{1/2}$ send infinity to representatives of all six classes:

$$\infty, \quad 0 = W_{28}\infty, \quad \tfrac{1}{2} = HW_{28}\infty, \quad -\tfrac{1}{7} = G_4\infty, \quad \tfrac{5}{14} = HG_4\infty, \quad \tfrac{1}{4} = W_{28}G_4\infty.$$

For explicit conversion to the standard representatives, $\left(\begin{smallmatrix}13 & 2\\ 84 & 13\end{smallmatrix}\right)$ maps $-1/7$ to $1/7$ and $\left(\begin{smallmatrix}3 & -1\\ 28 & -9\end{smallmatrix}\right)$ maps $5/14$ to $1/14$; both belong to $\Gamma_0(28)$. Also $U_{28}\infty = 1/28$ and $T0 = 1$. The exponential decay of f at infinity is carried by these scalar transformations to vanishing at every cusp. Therefore $f \in S_k(\Gamma_0(28))$. □

Exact certificates. The finite calculations in this proof are independently reproducible from the following source files; none assumes modularity or excludes an eigenvalue by division.

- `level28_jet_builder.py`, `level28_pole_reduction.py`, and `level28_root_probe.py`: exact leading and Taylor rows.
- `level28_cusp_probe_support.py`: all four sign cases, all-orders annihilation of the constant odd mode, and uniform quotient rank 83.
- `level28_root_constraints.py`: all four exact first-jet ideals; `level28_root_exceptional_jet.py`: the shifted third-order obstruction.
- `level28_cusp_probe_weight.py` and `level28_root_weight_probe.py`: exact leading-ideal weight exclusions.
- `level28_cusp_probe_poles_chi4.py`: final pole cancellation identities, nonzero residue, and conjugation matrices.
- `level28_cusp_probe_generation.py` and its `.m` companion: explicit Schreier generation and independent Magma index 8.

37. A complete converse proof at level 30

This proof uses the original and Ramanujan-twist Fricke identities, two infinite-order elliptic arguments, and an exact group-generation certificate. The previously proved cancellation of poles on the 30-grid is not needed for this completion. Write $\sigma_p = \sqrt{p} f_p \in \{\pm 1\}$ for $p = 2, 3, 5$ and $\eta = (-1)^{k/2}\epsilon_f$. All slash operators are determinant-normalized; matrices that differ by a nonzero scalar have the same action because k is even. The original and modulus-two identities already give

$$f|T = f, \qquad f|W = \eta f, \qquad f|A = -\sigma_2 f, \quad T = \begin{pmatrix}1 & 1\\ 0 & 1\end{pmatrix}, \quad W = \begin{pmatrix}0 & -1\\ 30 & 0\end{pmatrix}, \quad A = \begin{pmatrix}16 & 1\\ 30 & 2\end{pmatrix}.$$

For example $W^{-1}t_{1/2}W_{60}t_{-1/2} = 2A^{-1}$, so the modulus-two eigenvalue is $-\sigma_2$.

37.1. **The modulus-six transformation.** The conductor of the Ramanujan-6 twist is 180, and its Fricke multiplier relative to that of f is $\delta_6 = (-\sigma_2)(-\sigma_3) = \sigma_2\sigma_3$. For $a = \pm 1$,

$$W^{-1}t_{a/6}W_{180}t_{-a/6} = \begin{pmatrix}6 & -a\\ -30a & 6\end{pmatrix}.$$

Put $B = \left(\begin{smallmatrix}6 & 1\\ 30 & 6\end{smallmatrix}\right)$. The two displayed matrices are B and $6B^{-1}$. Consequently, after interchanging the two terms if necessary, the Ramanujan identity is

$$f|(B^{-1} - \delta_6)t_{1/6} + f|(B - \delta_6)t_{-1/6} = 0.$$

For $u = f|(B^{-1} - \delta_6)$ we have $f|(B - \delta_6) = -\delta_6 u|B$, hence

$$u = \delta_6 u|Bt_{-1/3}.$$

The matrix $Bt_{-1/3} = \left(\begin{smallmatrix}6 & -1\\ 30 & -4\end{smallmatrix}\right)$ has determinant 6 and squared trace divided by determinant equal to $2/3$. It is an infinite-order elliptic matrix. A positive-weight holomorphic function cannot have eigenvalue 1 or -1 under such a matrix: conjugation to the disk makes a nonzero Taylor coefficient require an irrational rotation raised to a positive integer power to be 1 or -1. Thus $u = 0$, giving

$$f|B = \sigma_2\sigma_3 f.$$

37.2. **The modulus-four transformation.** The conductor of the Ramanujan-4 twist is 240, and the relative Fricke multiplier is $\delta_4 = -\sigma_2$. The exact matrices

$$\tfrac{1}{2}W^{-1}t_{1/4}W_{240}t_{-1/4} = C = \begin{pmatrix} 4 & -1 \\ -30 & 8 \end{pmatrix}, \qquad \tfrac{1}{2}W^{-1}t_{-1/4}W_{240}t_{1/4} = C' = \begin{pmatrix} 4 & 1 \\ 30 & 8 \end{pmatrix}$$

give

$$f|(C - \delta_4)t_{1/4} + f|(C' - \delta_4)t_{-1/4} = 0. \tag{15}$$

Set $L = WB^{-1}$ and $R = WB$. Direct multiplication shows, projectively,

$$C' = LC^{-1}R, \qquad f|L = f|R = \rho f, \qquad \rho = \eta\sigma_2\sigma_3.$$

Let $v = f|(C - \delta_4)$. Since $\rho^2 = \delta_4^2 = 1$,

$$f|(C' - \delta_4) = \rho f|(C^{-1} - \delta_4)R = -\rho\delta_4 v|C^{-1}R.$$

Substitution in (15) yields

$$v = \rho\delta_4 v|M, \qquad M = C^{-1}Rt_{-1/2} \sim \begin{pmatrix} 10 & -2 \\ 30 & -5 \end{pmatrix}.$$

Here $\det M = 10$, $\operatorname{tr} M = 5$ for the displayed representative, and $(\operatorname{tr} M)^2/\det M = 5/2$. This is again elliptic of infinite order. Indeed, finite projective order would make $(\operatorname{tr} M)^2/\det M = 2 + \zeta + \zeta^{-1}$ an algebraic integer; a rational algebraic integer is an integer, whereas $5/2$ is not. The same positive-weight argument gives $v = 0$. We have proved

$$f|C = -\sigma_2 f.$$

37.3. **Exact generation and cuspidality.** Consider $H = \langle T, A, B, W, C\rangle$ projectively. The determinant square classes of A, B, W are the three independent classes $2, 6, 30$; the class of C equals that of A. Their scalar eigenvalues likewise satisfy

$$\chi(T) = 1, \quad \chi(A) = \chi(C) = -\sigma_2, \quad \chi(B) = \sigma_2\sigma_3, \quad \chi(W) = \eta.$$

Thus these eigenvalues factor through the determinant square-class group. Every element of its kernel fixes f.

Take the eight representatives $A^aB^bW^c$, with $a, b, c \in \{0, 1\}$, for these square classes. The Schreier construction gives 26 nonidentity kernel generators. After scalar normalization every generator has integer entries, determinant one, and lower-left entry divisible by 30. The independent exact verification in `level30_modulus4_generation.py` expresses them as words in $\mathrm{PSL}_2(\mathbb{Z}) = \langle S, T \mid S^2 = (ST)^3 = 1\rangle$. Todd–Coxeter enumeration terminates with index 72, which equals $[\mathrm{PSL}_2(\mathbb{Z}) : \overline{\Gamma_0(30)}]$. Containment and equality of indices therefore prove that the kernel contains $\Gamma_0(30)$, and hence f is modular of weight k on $\Gamma_0(30)$ with trivial character. No additional root-number identity has been assumed in this argument.

Finally A, B, W are Atkin–Lehner normalizers of exact divisors $2, 6, 30$. Their eight products above take infinity to all eight cusps of $\Gamma_0(30)$. The same verification computes the gcd with 30 of their reduced cusp denominators and obtains, exactly once each,

$$1, 2, 3, 5, 6, 10, 15, 30.$$

For squarefree level these labels distinguish all cusps (the standard cusp classification has $\varphi(\gcd(d, 30/d)) = 1$ for each $d \mid 30$). Each product acts on f by a scalar and f vanishes at infinity, so f vanishes at every cusp. This proves

$$\boxed{f \in S_k(\Gamma_0(30)).}$$

38. An exact pole-elimination certificate at level 30

This section proves entireness of the additive Mellin transforms on the 1/30 grid. It does not yet prove the missing individual transformation laws needed for modularity.

Put $a_n = f_n n^{(k-1)/2}$ and

$$F(s,r) = (2\pi)^{-s}\Gamma(s)\sum_{n\geq 1} a_n e(nr) n^{-s}, \qquad \lambda_p = \sqrt{p} f_p \in \{1,-1\} \quad (p=2,3,5).$$

Let $\eta = (-1)^{k/2}\epsilon_f$, $h = s - k/2$, and

$$x = \lambda_2 2^{-h}, \qquad y = \lambda_3 3^{-h}, \qquad z = \lambda_5 5^{-h}, \qquad E = \eta\lambda_2\lambda_3\lambda_5.$$

All three variables are nonzero. Write $V_j(s) = F(s, j/30)$, with subscripts read modulo 30. For $q \mid 30$ define

$$U_q = \{30a/q : 0 \leq a < q,\ (a,q) = 1\}, \qquad D_q = 30/q.$$

The exact Hecke equations are

$$H_{p,j}V := \sum_{a=0}^{p-1} V_{j+30a/p} - X_p V_{pj} = 0, \quad 0 \leq j < 30/p, \quad (X_2, X_3, X_5) = (x, y, z).$$

The rows $R_q V = \sum_{a\in U_q} V_a$ are entire Ramanujan twists. For $t \in \mathbb{Z}\setminus\{0\}$ with $|t| \mid D_q$, set

$$u = -D_q/t, \quad c_{q,t} = E(-1)^{\omega(q)} \prod_{p\mid D_q} X_p^{1-2\mathbf{1}_{p\mid t}}, \qquad T_{q,t}V = \sum_{a\in U_q} (V_{a+t} - c_{q,t} V_{a+u}).$$

Here the coefficient is exactly $\eta_q (30q(t/30)^2)^{s-k/2}$, since $\eta_q = \eta\prod_{p\mid q}(-\lambda_p)$. The Taylor–Fricke identity gives, for any positive integer M,

$$T_{q,t}V(s) - c_{q,t}(s)\sum_{m=1}^{M-1} (iqt)^m \binom{s+m-k}{m} \sum_{a\in U_q} V_{a+u}(s+m) \quad \text{holomorphic for } \Re s > B - M. \tag{16}$$

Here B can be fixed using the polynomial coefficient bound, and the identity is initially derived in the common domain of definition by Mellin transform. The $M = 1$ case starts continuation by one strip, and successive values of M continue farther left.

We record exact finite linear algebra certificates. They are identities in $\mathbb{Q}[x^{\pm1}, y^{\pm1}, z^{\pm1}]$, not numerical rank tests. Use columns $0, \ldots, 29$ in that order. Let the following list of 26 rows be denoted by $\mathcal{B}$:

$$\begin{aligned}
&R_1, T_{1,1}, T_{1,-1}, T_{1,2}, T_{1,-2}, T_{1,3}, T_{1,-3}, T_{1,5}, T_{1,-5},\\
&R_2, T_{2,1}, T_{2,-1}, T_{2,3}, T_{2,-3},\\
&R_3, T_{3,1}, T_{3,-1}, T_{3,2}, T_{3,-2},\\
&R_5, T_{5,1}, T_{5,-1}, T_{5,2}, T_{5,-2}, T_{6,1}, T_{6,-1}.
\end{aligned}$$

Appending $H_{2,2}, H_{2,3}, H_{2,4}, H_{2,7}$ gives a square matrix A_E with

$$\det A_E = \frac{2E}{x}\Delta_E, \qquad \Delta_E = xyz + xy - 2Exz - Eyz - Ez. \tag{17}$$

The Laurent polynomial on the right is nonzero. After substituting the exponentials defining x, y, z, it remains nonzero as an analytic function of s: distinct Laurent monomials in $2^{-h}, 3^{-h}, 5^{-h}$ have distinct frequencies by unique prime factorization. Thus (16), the entire $R_q V$, and the exact $H_{p,j}V$ continue every V_j meromorphically to $\mathbb{C}$ by induction over successive leftward strips.

A second certificate controls all specializations, including possible zeros of other convenient pivots. Delete the last row $T_{6,-1}$ of $\mathcal{B}$, append $H_{2,2}, H_{2,3}, H_{2,4}, H_{2,7}$, and delete column 27. This 29×29 determinant is exactly

$$-xz. \tag{18}$$

Consequently every nonzero common kernel vector of all the rows has nonzero coordinate 27; such a kernel has dimension at most one.

For completeness the kernel on $\Delta_E = 0$ can be given without exceptional denominators. Set

$$C_E = xy - 2Ex - Ey - E, \qquad z = -xy/C_E.$$

The equation $\Delta_E = 0$ and $xyz \neq 0$ imply $C_E \neq 0$. Normalize $v_{27} = 1$. The vector has $v_{30-j} = -v_j$, $v_0 = v_1 = v_2 = v_{14} = v_{15} = v_{16} = v_{28} = v_{29} = 0$, and

j	3	4	5	6	7	8
v_j	-1	$\frac{xy-x-1}{x^2}$	$\frac{-E(x+1)C_E}{x^2}$	$-\frac{x+1}{x^2}$	$\frac{xy+x+y}{x^2}$	$\frac{xy+x+y}{x^2}$

j	9	10	11	12	13	17
v_j	$-\frac{x^2+x+1}{x^2}$	$-\frac{EC_E}{x^2}$	$-\frac{x^2y+x^2+x+1}{x^2}$	$\frac{1}{x}$	$-\frac{xy-x-1}{x}$	$\frac{xy-x-1}{x}$

(The last entry repeats the value determined by antisymmetry.) Substitution verifies every row equation. Equation (18) therefore proves this formula is the unique normalized kernel at every point of the nonzero torus on $\Delta_E = 0$.

The decisive input is a row syzygy involving R_{10}. Write $R_{10} = \sum_{b \in \mathcal{B}} \alpha_b(x, y, z) b$. Its nonzero coefficients are

b	α_b
R_1	$xz(y-1)/y$
$T_{1,3}, T_{1,-3}$	$1/2$
$T_{1,5}, T_{1,-5}$	$z/(2y)$
R_2	$-(y-z)/y$
$T_{2,1}, T_{2,-1}$	$-Ex/(2y)$
$T_{2,3}, T_{2,-3}$	$-Ex/2$
R_3	$Ex(z-1)/(2y)$
$T_{3,2}, T_{3,-2}$	$-Ex/(2y)$
R_5	$Ex(y+1)/(2y)$
$T_{5,2}, T_{5,-2}$	$Ex/(2y)$
$T_{6,1}, T_{6,-1}$	$-Ex/(2y)$.

Every coefficient is holomorphic after the exponential substitution, since $y \neq 0$. In particular this relation has no hidden exceptional locus. Combining the Taylor identities using this syzygy cancels all terms involving $V(s)$ identically, before taking residues. For the kernel v above one computes

$$\sum_{(q,t) \in \mathcal{B}} \alpha_{T_{q,t}}(2x, 3y, 5z) c_{q,t}(2x, 3y, 5z) qt \sum_{a \in U_q} v_{a - D_q/t} = \frac{20E(3x+1)}{x}. \tag{19}$$

This is an exact rational-function identity on $\Delta_E = 0$.

Proposition 38.1. *Under the hypotheses defining $\mathscr{E}(30, k)$, every additive Mellin transform $F(s, j/30)$ is entire.*

Proof. Suppose a pole exists. Choose a pole s_0 that is rightmost within its integer-translation class $s_0 + \mathbb{Z}$. Such a choice exists because the original Dirichlet series are holomorphic in a common right half-plane. No assertion about a globally rightmost pole is required. Let $v \neq 0$ be the vector of coefficients of the highest-order Laurent terms of the V_j at s_0. All positive integral shifts are

holomorphic there. Equation (16) and the exact rows give $A_E(s_0)v = 0$, so $\Delta_E(s_0) = 0$. By (18), $v_{27} \neq 0$.

Apply the exact row syzygy at $s = s_0 - 1$, choosing M sufficiently large that its remainder is holomorphic there. All terms with $m \geq 2$ are holomorphic by the choice of s_0. The top Laurent coefficient of the $m = 1$ term, using (19), is

$$i(s_0 - k)\,\frac{20E(3x(s_0)+1)}{x(s_0)}v_{27}.$$

It must vanish. Thus $s_0 = k$ or $x(s_0) = -1/3$. Notice that possible poles of larger order at $s_0 - 1$ do not affect this argument: the entire vector $V(s)$ was canceled identically by the syzygy.

If $s_0 = k$, put $n = k/2 \geq 1$. Dividing Δ_E by xyz gives

$$1 + \lambda_5 5^n - \eta\lambda_5(2\lambda_2 3^n + \lambda_3 2^n + 6^n).$$

For $n \geq 3$, this cannot vanish, since $6^n > 5^n + 2 \cdot 3^n + 2^n + 1$. This holds at $n = 3$ and persists after increasing n, by dividing all terms by 6^n. For $n = 1, 2$, the 16 sign choices give absolute value at least 2.

If $x(s_0) = -1/3$, let $\alpha = \Re(s_0 - k/2) = \log_2 3 > 3/2$. The equation $\Delta_E = 0$ gives $z = y/(1+2y)$ when $E = -1$, and $z = -y/(1+4y)$ when $E = 1$. In either case,

$$5^{-\alpha} = |z| \geq \frac{|y|}{1+4|y|} = \frac{1}{3^\alpha + 4}.$$

But $5^\alpha > 3^\alpha + 4$: the difference $5^t - 3^t$ is increasing for $t > 0$, and $5^{3/2} - 3^{3/2} > 4$. This is a contradiction in both cases. Hence no pole exists. □

Relation to the complete proof. This independent argument establishes entireness on the denominator-30 grid. Its matrix and syzygy identities are verified by exact-arithmetic scripts. Entireness alone does not supply individual twisted functional equations; the preceding elliptic and group-generation argument supplies a complete converse without needing that additional inference.

39. Levels 33 and 34: a modulus-six elliptic argument

The same construction used at levels 30 and 35 works for $N = 33, 34$. In fact, for $N \in \{30, 33, 34, 35\}$ put

$$d = 36 - N = \gcd(N, 36), \quad \ell = \operatorname{lcm}(N, 36) = 36N/d, \quad G = \begin{pmatrix} 6 & -1 \\ -N & 6 \end{pmatrix}.$$

Write $\eta = (-1)^{k/2}\epsilon_f$, and let δ be the ratio of the Ramanujan-6 and original Fricke multipliers. Explicitly,

N	30	33	34	35
δ	$\lambda_2\lambda_3$	$-\lambda_3$	$-\lambda_2$	1

$$\lambda_p = \sqrt{p}\, f_p.$$

All these multipliers belong to $\{\pm 1\}$. For $a = \pm 1$,

$$\frac{d}{6}W_N^{-1}t_{a/6}W_\ell t_{-a/6} = \begin{pmatrix} 6 & -a \\ -Na & 6 \end{pmatrix}.$$

The two matrices are G and dG^{-1}. The two Fricke laws thus give

$$f|(G - \delta)t_{1/6} + f|(G^{-1} - \delta)t_{-1/6} = 0.$$

For $u = f|(G - \delta)$, this becomes $u = \delta u|G^{-1}t_{-1/3}$. A representative of the latter projective matrix is

$$M = \begin{pmatrix} 6 & -1 \\ N & 6 - N/3 \end{pmatrix}, \qquad \frac{(\operatorname{tr} M)^2}{\det M} = \frac{d}{9}.$$

For the four stated levels this is a nonintegral rational number strictly between zero and four. Hence M is elliptic of infinite projective order. The positive-weight holomorphic elliptic lemma proves $u = 0$, and therefore

$$f|G = \delta f. \tag{6.1}$$

All comparisons and trace ratios are checked exactly in `modulus6_elliptic.py`.

The centers $1/6$ and $-1/6$ are consequently genuine Fricke centers. Indeed

$$(dG^{-1})W_N = \frac{d}{6} t_{1/6} W_\ell t_{-1/6},$$

and the analogous identity holds at the negative center. Thus $F(s, \pm 1/6)$ are entire, rapidly decreasing in fixed vertical strips, and satisfy individual functional equations of conductor ℓ and sign $\epsilon_f \delta$. Put

$$h = s - k/2, \quad x = 2^h, \quad D(s) = F(s, 1/6) - F(s, -1/6), \quad C(s) = F(s, 1/3) - F(s, -1/3).$$

At level 33 the good-prime Hecke relation gives

$$C(s) = \frac{xD(s)}{x + x^{-1} + \lambda_2}, \qquad C(s) = \epsilon_f \delta \, 99^{-h} C(k - s). \tag{33.1}$$

At level 34 the prime 2 is simply ramified, so its quadratic Euler term is absent. Writing $\sigma = \lambda_2 \in \{\pm 1\}$ gives instead

$$C(s) = \frac{xD(s)}{x + \sigma}, \qquad C(s) = -\epsilon_f \, 306^{-h} C(k - s). \tag{34.1}$$

For the second equation use $\delta = -\sigma$ and $(x^{-1} + \sigma)/(x + \sigma) = \sigma/x$. Dividing C by $i\sqrt{3}$ gives the completed quadratic twist modulo 3. At level 33 its only possible poles satisfy $x + x^{-1} = -\lambda_2$; at level 34 they are simple and satisfy $x = -\sigma$, hence lie on $\Re s = k/2$. These formulas do not yet exclude the poles. The Ramanujan-3 sum is entire, so they also give meromorphic continuation of both individual additive twists modulo 3.

39.1. **Uniform auxiliary continuation at level 33.** Include (6.1), with multiplier $-\lambda_3$, among the known transformations when constructing the $Q = 66$ system. Use words of length at most three in T, U_{33}, W_{33}, the known modulus-two matrix, and G, together with inverses. Include the closed Ramanujan–Fricke rows and Hecke rows at $2, 3, 11$. The exact builder is `level33_jet_builder.py`. For every choice of $(\eta, \lambda_3, \lambda_{11})$, the verifier `level33_uniform_continuation.py` finds 66 pivots. Each pivot numerator and denominator has an exponential monomial with a nonzero rational coefficient independent of λ_2. Distinct prime factorizations give distinct real exponential rates. Thus the pivots are not identically zero for any complex λ_2, including zero. The matrix is meromorphically invertible on every specialization; the closed Taylor recurrence proves continuation of all 66 additive transforms by strip induction. All eight cases and exact pivots are recorded in `level33_uniform_continuation.json`.

39.2. **A closed continuation subsystem at level 34.** At level 34, use the analogous known transformations, including (6.1) with multiplier $-\lambda_2$, on $Q = 102$. Let

$$U = (\mathbb{Z}/102\mathbb{Z}) \setminus \{11, 23, 28, 31, 37, 40, 62, 65, 71, 74, 79, 91\}.$$

Retain only rows whose complete Taylor support lies in U, not just rows whose leading support does so after cancellation. This is a 90-coordinate closed system. The exact verifier `level34_closed_continuation.py` finds 90 pivots, with the same parameter-independent exponential-coefficient audit, in each of the eight sign cases. Hence every $F(s, j/102)$ for $j \in U$ is meromorphic for every complex λ_3. The set includes all additive twists of denominators 3 and 34, and in particular the Fricke partners $\mp 3/34$ of $\pm 1/3$. The exact rows, pivots and source hashes are recorded in

`level34_closed_continuation.json`. This certificate alone does not continue the other twelve level-34 coordinates. The larger system below continues all original 102-grid coordinates and removes the quadratic-twist poles. The subsequent Ramanujan-5 argument completes level 34; the level-33 section completes that level separately.

40. Level 34: removal of the quadratic-twist poles using prime 5

This section proves that the completed quadratic twist modulo three is entire for every positive even weight, every ramified sign triple, and arbitrary complex Hecke eigenvalues at 3 and 5. The following section then supplies the remaining transformation and group-generation argument.

Use the classical Mellin variable and write

$$h = s - k/2, \quad (x, y, z, w) = (2^h, 3^h, 17^h, 5^h), \qquad (a, b) = (\lambda_3, \lambda_5), \quad (\eta, \sigma, \tau) \in \{\pm 1\}^3.$$

The already established modulus-six relation supplies the scalar eigenlaw for $G = \left(\begin{smallmatrix} 6 & -1 \\ -34 & 6 \end{smallmatrix}\right)$, with eigenvalue $-\sigma$. Together with the original Fricke and Ramanujan identities it gives

$$C(s) = F(s, 1/3) - F(s, -1/3) = \frac{xD(s)}{x + \sigma}, \qquad D \text{ entire},$$

and the quadratic-twist functional equation. Hence every possible pole of C is simple and lies at $x = -\sigma$, on $\Re s = k/2$.

A larger closed system. Put $Q = 510$. The file `level34_finish_prime5_support.json` specifies a set U of 414 residues; it contains every multiple of 5, and thus every original 102-grid coordinate. Generate words of length at most three in $T, U_{34}, E_2, W_{34}, G$ and their inverses, with eigenvalues $1, 1, -\sigma, \eta, -\sigma$. Include their known-map Taylor rows, the Ramanujan rows and Taylor–Fricke rows for all $q \mid 510$ with $\operatorname{lcm}(34, q^2) \mid 510^2$, and the exact Hecke rows at $2, 3, 5, 17$. Retain a Taylor row only if its *complete* source and target support belongs to U. These rules yield 2,838 rows.

Over $\mathbb{Q}(x, y, z, w, a, b)$ the leading system has rank 414. For each of the eight sign triples, every pivot numerator and denominator has an exponential monomial whose coefficient is a nonzero rational constant, independent of a, b, and whose exponent is not shared by any other parameter monomial. Consequently these pivots are not identically zero after any specialization of a, b: unique prime factorization makes the distinct exponential monomials linearly independent as functions of h. Closed Taylor–Fricke strip induction therefore continues all 414 coordinates meromorphically to the plane, uniformly in the sense that no eigenvalue specialization is excluded. This is a statement about U, not the entire 510-grid.

The first Taylor constraints. Perform a second elimination allowing only nonzero rational Laurent monomials in x, y, z, w as pivots. It has rank 411 and leaves three free coordinates. Let L denote its remaining leading rows. Apply the same row operations at $s - 1$ to the first Taylor sources; its identically zero leading rows give fourteen nonzero source rows, denoted J. Both L and J, and the expression for C on the three free coordinates, are exported exactly in the certificate.

For clarity, a known matrix $g = \left(\begin{smallmatrix} A & B \\ C_0 & D_0 \end{smallmatrix}\right)$ with eigenvalue μ gives leading coefficient

$$\mu\rho^h, \qquad \rho = \frac{(C_0 j + D_0 Q)^2}{\det(g) Q^2},$$

and normalized first source coefficient $\mu\rho^{h-1} C_0 (C_0 j + D_0 Q)/(\det(g) Q)$, at its image coordinate. For a Ramanujan row of level ℓ and translation t/Q, the corresponding coefficients are $\eta_q(\ell t^2/Q^2)^h$ and $\eta_q(\ell t^2/Q^2)^{h-1} \ell t/Q$ on the complete target sum. The omitted common scalar in J is $i(s_0 - k)$ when the upper pole is at s_0. Exact Hecke and entire Ramanujan rows have zero source. These formulas describe the checked first-jet calculation completely.

Rightmost active poles, including the weight-two endpoint. Suppose C has a pole at $s_c = k/2 + it$. Among the points $s_c + n$, $n \geq 0$, at which the continued vector has a pole, ignore a point $s = k$ if its vector pole is at most simple. Call all other poles active. There is a rightmost active point $s_0 = s_c + n$: s_c itself is active and all coordinates are holomorphic in a common right half-plane. Take the nonzero highest Laurent coefficient vector v at s_0, and express it in the three monomial-elimination coordinates.

A simple vector pole at $s = k$ contributes holomorphically to every strictly lower Taylor equation. Indeed the multiplier of $V(s+m)$ contains $\binom{s+m-k}{m}$, which has a zero at $s = k - m$ for $m > 0$. Thus every positive shift above s_0 is harmless, including any ignored simple weight-k pole. The leading equations give $Lv = 0$. If $s_0 \neq k$, the first Taylor zero rows at $s_0 - 1$ give $Jv = 0$ after division by $i(s_0 - k) \neq 0$. If $s_0 = k$, activity means its order r is at least two. In these same zero rows take the Laurent coefficient of order $r - 1$, rather than r: the first Taylor scalar has a simple zero, and this again gives $Jv = 0$. All lower-variable terms have been eliminated identically as meromorphic functions, not merely at a single point. This argument does not exclude or divide by weight two.
Positive integer shifts. Suppose $n \geq 1$ and put

$$u = -\sigma x = 2^n \geq 2, \qquad Y = y^2, \qquad A = ay.$$

At such a point C is holomorphic, so its Laurent coefficient is zero. For every sign triple, one of the three columns of J is identically zero, and the same column of C is zero. A nonzero Laurent-monomial pivot in J eliminates one of the other two coordinates. If the remaining coordinate were zero, a leading pivot equal to a nonzero Laurent monomial times $x - \sigma$ would also kill the zero column, contradicting $v \neq 0$. Normalize this remaining coordinate to one.

The C coefficient is a nonzero Laurent monomial times

$$H = (22u^2 - 4u + 8)(A + Y) + 39u^2 - 28u + 116.$$

Thus $H = 0$. After solving this for A, another row of J is a nonzero Laurent monomial, divided by $(11u^2 - 2u + 4)^2$, times $N(u) - D(u)Y$, where

$$D(u) = 748u^6 - 1126u^5 + 540u^4 - 1168u^3 + 176u^2 - 288u,$$
$$N(u) = 1122u^7 - 6054u^6 + 27603u^5 - 44208u^4 + 17676u^3 - 14688u^2 + 1632u - 1344.$$

All coefficients, in descending order, of the following shifted polynomials are strictly positive:

$$\begin{aligned} D(v+2): &\quad 748, 7850, 34160, 77792, 95568, 57216, 11264; \\ (9uD - 2N)|_{u=v+2}: &\quad 4488, 64806, 350334, 949524, 1404744, 1155888, 553536, 169344; \\ (9uD + 2N)|_{u=v+2}: &\quad 8976, 103422, 547146, 1680492, 3115992, 3314448, 1708992, 236160. \end{aligned}$$

Therefore $D(u) > 0$ and $|N(u)/D(u)| < 9u/2$ for every real $u \geq 2$. But $Jv = 0$ forces $Y = N(u)/D(u)$, whereas $|Y| = 9^n \geq (9/2)2^n = 9u/2$. This contradiction excludes every positive integer shift, including an active pole at $s = k$.
The critical line. It remains that $n = 0$, so $x = -\sigma$. At this specialization a nonzero Laurent-monomial first-source pivot removes one free coordinate and a nonzero Laurent-monomial leading pivot removes a second. If the final coordinate vanishes all three vanish, so normalize it to one. Evaluate every row of L and J on the resulting vector, put $a = A/y$, and clear denominators. Each numerator has only even powers of y, so replace y^2 by Y.

The resulting finite polynomial system in $\mathbb{Q}[b, w, z, A, Y]$ has reduced Groebner basis $\{1\}$ for every sign triple. The exact polynomials, pivot rows, and source hashes are saved with the certificate; `level34_finish_prime5_critical.py` recomputes all eliminations and the basis over $\mathbb{Q}$. This proves inconsistency over $\mathbb{C}$ without any restrictions on a, b and without needing $|y| = |z| = |w| = 1$. All divisions used before clearing denominators were by Laurent monomials, nonzero at the physical exponential values. Hence $v = 0$, the final contradiction.

Thus C is entire. The Ramanujan sum $F(s,1/3)+F(s,-1/3)$ was already entire; both individual additive twists modulo three are therefore entire as well. The established functional equations remain valid by analytic continuation.

For inverse Mellin arguments, the quotient $C = xD/(x+\sigma)$ also preserves rapid decay in closed vertical strips. Outside fixed small disks around the simple, equally spaced zeros of $x+\sigma$, the denominator is bounded away from zero on each strip. Inside those disks, apply the maximum principle to the now entire quotient using the boundary estimate. Thus the known rapid strip estimates for D give those for C, and the individual twists inherit the corresponding estimates from their sum and difference.

41. Completion of the level-34 converse theorem for all eigenvalues

The entire quadratic twist modulo three, established in Section 40, supplies the following known scalar laws. Put $\delta = -\sigma_2$ and write

$$T = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad E = \begin{pmatrix} 2 & -1 \\ -34 & 18 \end{pmatrix}, \quad B = \begin{pmatrix} 6 & -1 \\ -34 & 6 \end{pmatrix}, \quad W = \begin{pmatrix} 0 & -1 \\ 34 & 0 \end{pmatrix}, \quad G = \begin{pmatrix} 11 & -1 \\ 34 & -3 \end{pmatrix}.$$

Then $f|T = f|G = f$, $f|E = f|B = \delta f$, and $f|W = \eta f$. For the last law, combine the now entire Ramanujan-3 sum and quadratic-3 difference: their opposite functional-equation signs give the swapped individual laws. The matrix $W^{-1}t(-1/3)W_{306}t(-1/3) = (3,-1;34,-11)$ is G^{-1} projectively. All slash operators are determinant-normalized in positive even weight; the matrix identities below are projective.

Consider

$$P = \begin{pmatrix} 4 & -1 \\ 34 & -8 \end{pmatrix}, \qquad h = f|P - \delta f.$$

The good-prime Ramanujan sum at 5 has an entire completed Mellin transform and Fricke conductor $25\cdot 34 = 850$, with the original Fricke eigenvalue η on the Fourier-function side. Indeed, for $x = 5^{s-k/2}$ its Mellin multiplier has the form $-1 + cx^{-1} - x^{-2}$, whose reflection ratio is x^{-2}; here c is the normalized Hecke scalar. No condition on c is needed. Its finite sum of translates therefore satisfies a genuine Fricke identity by Mellin inversion.

The four relevant determinant-one matrices are

$$G_1 = \begin{pmatrix} 5 & -1 \\ -34 & 7 \end{pmatrix}, \quad G_{-1} = \begin{pmatrix} 5 & 1 \\ 34 & 7 \end{pmatrix}, \quad G_2 = \begin{pmatrix} 5 & 2 \\ -68 & -27 \end{pmatrix}, \quad G_{-2} = \begin{pmatrix} 5 & -2 \\ 68 & -27 \end{pmatrix}.$$

The identities

$$W^{-1}t_{a/5}W_{850}t_{b/5} = G_a, \qquad (a,b) = (1,-1),(-1,1),(2,2),(-2,-2)$$

factor the Ramanujan functional equation into

$$\begin{aligned} 0 = {} & (f|G_1 - f)|t_{1/5} + (f|G_{-1} - f)|t_{-1/5} \\ & + (f|G_2 - f)|t_{-2/5} + (f|G_{-2} - f)|t_{2/5}. \end{aligned}$$

Two terms vanish individually, by the short explicit words

$$G_2 = ETWG^{-1}WET, \qquad G_{-2} = W^{-1}G_2^{-1}W^{-1}.$$

Each has scalar character one, for every $\delta,\eta \in \{1,-1\}$. The other two matrices satisfy $G_1 = BP$ and $G_{-1} = PB^{-1}$, so the remaining equation is

$$\delta h|t_{1/5} + h|B^{-1}t_{-1/5} = 0.$$

Equivalently,

$$h|K = -\delta h, \qquad K = B^{-1}t_{-2/5} = \begin{pmatrix} 30 & -7 \\ 170 & -38 \end{pmatrix}.$$

This matrix has determinant 50 and trace -8, hence $(\operatorname{tr} K)^2/\det K = 32/25$. It is an elliptic matrix of infinite projective order: finite order would make this rational number an algebraic integer. Consequently K^2 is also infinite elliptic, and $h|K^2 = h$. The positive-weight holomorphic elliptic lemma gives $h = 0$. Thus $f|P = \delta f$, with no restriction on the eigenvalue at 3 or any other good prime.

Generation and cuspidality. Use the four determinant-squareclass representatives $1, E, W, EW$, whose characters are $1, \delta, \eta, \delta\eta$. For each representative r and each scalar generator in $\{T, E, B, W, G, P\}$, form $rG_0r'^{-1}$ with r' of the product squareclass. These matrices fix f, are integral of determinant one, and have lower-left entry divisible by 34. Exact Todd–Coxeter enumeration gives their subgroup index 54 in $\mathrm{PSL}_2(\mathbf{Z})$, equal to the index of $\Gamma_0(34)$. Hence f is modular for $\Gamma_0(34)$.

Finally the images of infinity under $1, E, W, EW$ are, respectively, $\infty, -1/17, 0, -1/18$. Their reduced denominators have gcd with 34 equal to $34, 17, 1, 2$, so they represent all four cusps of this squarefree level. Every corresponding slash transform is a scalar multiple of f and decays at infinity. Comparison with a determinant-one cusp scaling uses only an upper triangular matrix of positive dilation, which preserves vanishing of the constant term. Thus f vanishes at every cusp, and

$$f \in S_k(\Gamma_0(34)).$$

The exact checker `level34_aux_defect_complete.py` verifies all four center identities, the scalar words, the elliptic invariant, the complete index-54 enumeration, and the four cusp denominators.

42. Level 33: the quadratic twist, pole cancellation, and modularity

Assume the hypotheses of the original converse problem at level 33. Put $h = s - k/2$, $x = 2^h$, $y = 3^h$, $z = 11^h$, and $a = \lambda_2$. Write $u = \lambda_3$, $v = \lambda_{11}$ and let η be the slash-normalized Fricke sign. Thus $\eta, u, v \in \{\pm 1\}$; no restriction is imposed on the complex number a. Use the proved modulus-six law and uniform continuation of all $F(s, j/66)$ from the preceding section. In particular

$$C(s) = F(s, 1/3) - F(s, -1/3) = \frac{xD(s)}{x + x^{-1} + a}, \qquad C(s) = -\epsilon_f u\, 99^{-h} C(k - s), \tag{33.2}$$

where $D(s) = F(s, 1/6) - F(s, -1/6)$ is entire.

Proposition 42.1. *The function C is entire, for every sign triple and every complex a.*

Proof. If C has a pole, its functional equation permits us to choose one s_* with $\Re(s_* - k/2) \geq 0$. Choose a rightmost pole $s_0 = s_* + n$, $n \in \mathbb{Z}_{\geq 0}$, of the full 66-component grid in this integer translation class. This exists because every component is holomorphic in a common right half-plane. Set $r = 2^n$, $X = x(s_0)^2$ and $A = ax(s_0)$. The pole denominator in (33.2) implies

$$A = -X/r - r, \qquad |X| \geq r^2, \qquad |y(s_0)| \geq 3^n. \tag{33.3}$$

Use the highest nonzero Laurent coefficient of the grid at s_0; no assumption is made that its C component is nonzero.

Here are exact algebraic facts about the leading and first Taylor systems. Elimination divides only by nonzero rational Laurent monomials in x, y, z, gives 64 pivots, and leaves two free coordinates, 18 and 42. The remaining leading rows and the first-Taylor zero-row sources form a two-column matrix. If $s_0 \neq k$, its rank is at most one. Its minor ideal contains

$$(A + X)(A + X + 1)(2y - 3u). \tag{33.4}$$

After specializing each factor to zero, the reduced ideals are as follows:

$$\begin{aligned} A = -X: &\quad (X^2 - uy - 1), \\ y = 3u/2: &\quad (A + X,\ X^2 - 5/2), \\ A = -X - 1: &\quad (X - X_0(Y),\ z - \eta u(Z_0(Y) - 1) + v,\ (Y + 3)P(Y)), \qquad Y = -uy. \end{aligned}$$

Here

$$P(Y) = 704Y^7 + 8128Y^6 + 3072Y^5 + 19152Y^4 - 14256Y^3 + 35640Y^2 - 36693Y + 16767,$$

and the exact rational polynomials X_0, Z_0 are recorded in `level33_first_jet.py`. That verifier recomputes and asserts all three ideals for all eight sign choices; no eigenvalue is divided out. The separate verifier `level33_septic_roots.py` isolates all seven roots of P by exact rational intervals/rectangles. Rational interval evaluation proves at each root either $|Y| < 1$ or $|X_0(Y)| < |Y|$. Neither is compatible with $|X| = 4^t$, $|Y| = 3^t$, $t \geq 0$.

If $n \geq 1$, then $A + X + 1 = (r-1)(X/r - 1) \neq 0$, and $2y = 3u$ is impossible. If $A + X = 0$, (33.3) gives $X = r^2/(r-1)$, so $r = 2$ and $X = 4$. The first specialized ideal would then give $y = 15u$, contradicting $|y| = 3$. Thus $n \geq 1$ is impossible when $s_0 \neq k$.

If $n = 0$, then $A = -X - 1$. The septic roots have just been excluded. The remaining root $Y = -3$ gives

$$X = 4, \qquad y = 3u, \qquad z = -10\eta u - v.$$

Its real part is $t = 1$, so $|z| = 11$. Moreover the phases of X and y imply that a nonzero imaginary part would make $\log 2/\log 3$ rational. Hence s_0 is real. Positivity of y, z leaves just

$$(\eta, u, v) = (-1, 1, -1), \qquad h(s_0) = 1, \qquad a = -5/2. \tag{33.5}$$

The full higher Taylor systems at this point have rank 66 at each of the three lower integer shifts. The first two inhomogeneous systems are consistent, but the third has residual 300806 in the Hecke row $(2, 24)$, with the normalization fixed in `level33_exceptional_jet.py`. All source terms and intermediate unknown vectors are included. If $s_0 \neq k$, then $k \geq 4$ and $s_0 - k = 1 - k/2$ is a negative integer. The usual falling factorial normalizations are all nonzero, so this inconsistency excludes (33.5).

We treat $s_0 = k$ separately. The leading minor system alone contains

$$(A + X)(A + X + 1)(Xy - uX + y + u). \tag{33.6}$$

Here $x = 2^m, y = 3^m, z = 11^m$ with $m = k/2 \geq 1$, and $0 \leq n \leq m$. The last factor is positive for either value of u. For $n \geq 1$, the preceding inequalities exclude both other factors except $n = m = 1$, $a = -2$; direct exact evaluation gives leading rank 66 also there. For $n = 0$, substitute $A = -X - 1$ in the leading minors. For each sign triple, a selected nonzero minor is an exponential polynomial in m with largest base 144. Its leading term strictly dominates the sum of absolute values of the others for $m \geq 5$; the values $m = 2, 3, 4$ are checked exactly and nonzero. At $m = 1$ the leading rank is 66 except in (33.5), where it is 65. These assertions, including the explicit selected minors and dominance inequalities, are recomputed by `level33_weight_chain.py`.

Consequently the only possible top pole at k has $k = 2$ and (33.5). It cannot have order at least two. Indeed, at the coefficient one order below the highest Laurent term the first Taylor factor $s - k$ has a simple zero; the zero-row identities again give the first-jet system. The full-rank lower systems bound their pole orders by one less than the top order. Dividing their leading terms by $i^j P_j'(0)$, where $P_j(e) = \prod_{b=0}^{j-1}(e - b)$, gives the same normalized recurrence and the same third-order inconsistency, since

$$\binom{e - j + l}{l} P_{j-l}(e) = P_j(e)/l!.$$

Thus this possible pole is simple.

Finally a simple pole at $s = 2$ contributes holomorphically to the Taylor equations at $s = 1$ and $s = 0$: each relevant binomial coefficient has a zero at that point. Rank 66 at the first lower point forces the grid to be holomorphic at 1, and rank 66 at the second forces holomorphy at 0. But a pole of C at 2 would, by (33.2), force a pole of C at 0. This contradiction removes the last case and proves entireness. □

Theorem 42.2. *At level* 33*, the original converse hypotheses imply* $f \in S_k(\Gamma_0(33))$.

Proof. The Ramanujan-3 sum $F(s,1/3)+F(s,-1/3)$ is entire and has the same self-functional-equation sign and conductor as C. Hence each individual additive transform is entire and has that functional equation. The quotient in (33.2) is rapidly decreasing in fixed vertical strips: away from fixed small disks about the periodic zeros of its denominator this follows from D, and inside those disks it follows from holomorphy and the maximum principle. Mellin inversion therefore gives

$$f|C_+ = -uf, \quad f|C_- = -uf, \qquad C_+ = \begin{pmatrix} 3 & -1 \\ -33 & 12 \end{pmatrix}, \quad C_- = \begin{pmatrix} 3 & 1 \\ 33 & 12 \end{pmatrix}.$$

For example these matrices are the projective products $W_{33}^{-1} t_{\pm 1/3} W_{99} t_{\mp 1/3}$. We already know invariance under T and $E = \left(\begin{smallmatrix} 2 & -1 \\ -33 & 17 \end{smallmatrix}\right)$, and the laws $f|G = -uf$, $f|W_{33} = \eta f$ for $G = \left(\begin{smallmatrix} 6 & -1 \\ -33 & 6 \end{smallmatrix}\right)$.

The determinant square classes of C_+ and W_{33}, namely 3 and 33, are independent. Use the four representatives $C_+^a W_{33}^b$, $a,b \in \{0,1\}$. Every associated Schreier generator of determinant square class one has multiplier one. The exact verifier `level33_generation.py` produces 18 such nonidentity generators, checks that their primitive representatives are integral of determinant one with lower-left entry divisible by 33, checks their words in the standard generators of $\mathrm{PSL}_2(\mathbb{Z})$, and obtains index 48 by a complete coset enumeration. Since $[\mathrm{PSL}_2(\mathbb{Z}) : \overline{\Gamma_0(33)}] = 48$, these generators generate $\Gamma_0(33)$ projectively. Even weight deals with the central sign, proving modularity.

Finally C_+ is an exact-divisor Atkin–Lehner matrix for 3 and W_{33} is the Fricke matrix. The four representatives send infinity to cusps with denominator labels $1, 3, 11, 33$, checked in the same verifier. These are all cusp classes at the squarefree level 33. Their eigenlaws transfer the given vanishing at infinity to every cusp. Thus f is cuspidal. □

43. Level 35: a closed subsystem and meromorphic character twists

This subsection proves continuation only; it does not assert a twisted functional equation or modularity. Write $h = s - k/2$, $\eta = (-1)^{k/2}\epsilon_f$, $\lambda_p = \sqrt{p}\, f_p$, and $V_j(s) = F(s, j/210)$. The ramified parameters λ_5, λ_7 belong to $\{\pm 1\}$; λ_2, λ_3 are unrestricted complex numbers.

Let $U \subset \mathbb{Z}/210\mathbb{Z}$ consist of the following 92 residues:

$$\begin{aligned} U = \{&0,1,2,3,4,5,6,7,9,10,12,14,15,18,20,21,25,28,30,34,35,36,\\ &40,42,45,49,50,55,56,60,63,65,70,75,77,80,84,85,90,91,95,96,\\ &98,100,102,104,105,106,108,110,112,114,115,119,120,125,126,130,\\ &133,135,140,145,147,150,154,155,160,161,165,168,170,174,175,176,\\ &180,182,185,189,190,192,195,196,198,200,201,203,204,205,206,207,208,209\}. \end{aligned}$$

Then all V_j, $j \in U$, extend meromorphically to $\mathbb{C}$. In particular the completed twists by every Dirichlet character modulo 3, 5, or 7 are meromorphic: each is a finite linear combination of these V_j (with the usual Gauss-sum factor for primitive characters).

Here is a reproducible exact certificate. Generate words of length at most three in the known transformations

$$T = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \qquad U_{35} = \begin{pmatrix} 1 & 0 \\ 35 & 1 \end{pmatrix}, \qquad A_2 = \begin{pmatrix} 2 & -1 \\ -35 & 18 \end{pmatrix}, \qquad W_{35} = \begin{pmatrix} 0 & -1 \\ 35 & 0 \end{pmatrix},$$

and their inverses. Their eigenvalues are respectively $1, 1, 1, \eta$. Include their Taylor relations whenever both rational arguments belong to $U/210$. Include all Ramanujan–Fricke rows with $q \mid 210$ and $t \mid 210^2/\operatorname{lcm}(35, q^2)$, retaining a row only when *every* argument in both translated Ramanujan sums belongs to $U/210$. Finally include the exact Hecke rows at $2, 3, 5, 7$ whose nonzero coordinates belong to U. This restriction applies to the full Taylor support, not merely to the support remaining after cancellation in its leading term. Thus all higher Taylor corrections involve only this same subsystem.

The script `level35_closed_continuation.py` performs sparse elimination over

$$\mathbb{Q}(x,y,z,w,a,b), \qquad (x,y,z,w)=(2^h,3^h,5^h,7^h), \quad (a,b)=(\lambda_2,\lambda_3).$$

It handles the eight choices of $(\eta,\lambda_5,\lambda_7)$ and the four strata given by $a=0$ or $a\neq 0$, and $b=0$ or $b\neq 0$. Every pivot numerator and denominator is audited as follows: after grouping by the powers of x,y,z,w, at least one coefficient is a single nonzero monomial in the parameters allowed to be nonzero on that stratum. Prime factorization makes the corresponding exponentials in h distinct, so the pivot is not identically zero for any permitted specialization. All 32 cases give 92 pivots. Their labels and values are saved in `level35_closed_continuation.json`.

Consequently the leading coefficient matrix admits a meromorphic left inverse for every allowed pair of unramified eigenvalues. Beginning in the half-plane of absolute convergence, the Taylor–Fricke remainder estimate and this inverse extend the finite vector one strip at a time. Closure of the full Taylor support ensures that no uncontinued additive twist enters this induction. This proves the stated meromorphic continuation.

43.1. **Level 35: the modulus-six elliptic relation.** Use $t_r=\left(\begin{smallmatrix}1 & r\\ 0 & 1\end{smallmatrix}\right)$ and the determinant-normalized slash operator. For q coprime to 35, the matrix identity underlying the Ramanujan-twist functional equation is

$$q^{-1}W_{35}^{-1}t_{a/q}W_{35q^2}t_{-b/q}=\begin{pmatrix} q & -b\\ -35a & (35ab+1)/q\end{pmatrix}, \qquad 35ab\equiv -1 \pmod q.$$

The factor q^{-1} means scalar multiplication of the matrix; positive scalars do not change the slash action. For $q=6$ the units are $a=\pm 1$, and the congruence gives $b=a$. Since $35=6^2-1$, the two resulting matrices are

$$G=\begin{pmatrix} 6 & -1\\ -35 & 6\end{pmatrix}, \qquad G^{-1}=\begin{pmatrix} 6 & 1\\ 35 & 6\end{pmatrix}.$$

The Ramanujan-6 and original Fricke identities therefore imply exactly

$$f|(G-1)t_{1/6}+f|(G^{-1}-1)t_{-1/6}=0.$$

Set $g=f|(G-1)$. Since $f|(G^{-1}-1)=-g|G^{-1}$, this says

$$g|M=g, \qquad M=G^{-1}t_{-1/3}=\begin{pmatrix} 6 & -1\\ 35 & -17/3\end{pmatrix}.$$

Its determinant is 1 and its trace is $1/3$. Thus M is elliptic of infinite order: a rational trace of a finite-order determinant-one matrix would be an integer. A holomorphic function of positive integral weight fixed by such a matrix vanishes. Indeed, conjugating the fixed point to the disk origin makes the Taylor coefficient of degree n transform by an irrational rotation with nonzero exponent $k+2n$. Consequently $f|G=f$, and also $f|G^{-1}=f$.

43.2. **The quadratic twist modulo three.** The new matrix supplies actual Fricke centers:

$$G^{-1}W_{35}=\begin{pmatrix} 35 & -6\\ 210 & -35\end{pmatrix}, \qquad G^{-1}W_{35}: z\longmapsto \frac{1}{6}-\frac{1}{1260(z-1/6)}.$$

The matrix GW_{35} supplies the center $-1/6$ in the same way. Therefore $f(z\pm 1/6)$ each satisfies the Fricke identity at conductor 1260 with slash eigenvalue $\eta=(-1)^{k/2}\epsilon_f$. Their Fourier expansions decay exponentially at infinity; the Fricke identities give the same decay at zero after the weight factor. Their Mellin transforms $F(s,\pm 1/6)$ are entire and rapidly decreasing in every fixed vertical strip, and individually satisfy

$$F(s,\pm 1/6)=\epsilon_f 1260^{k/2-s}F(k-s,\pm 1/6).$$

Put $h = s - k/2$, $x = 2^h$, $\lambda = \sqrt{2}f_2$, and

$$D(s) = F(s, 1/6) - F(s, -1/6), \qquad C(s) = F(s, 1/3) - F(s, -1/3).$$

The Hecke-2 identity at the arguments $\pm 1/3$ gives

$$D(s) - C(s) = \lambda x^{-1} C(s) + x^{-2} C(s), \qquad C(s) = \frac{xD(s)}{x + x^{-1} + \lambda}.$$

The numerator $xD(s)$ has the functional equation of conductor $1260/4 = 315$ and sign ϵ_f; the denominator is invariant under $s \mapsto k - s$. Since the Gauss sum of the primitive character χ_3 is $i\sqrt{3}$, we obtain

$$F(s, \chi_3) = \frac{xD(s)}{i\sqrt{3}(x + x^{-1} + \lambda)}, \qquad F(s, \chi_3) = \epsilon_f 315^{k/2-s} F(k - s, \chi_3)$$

as meromorphic identities. Its only possible poles satisfy $2^h + 2^{-h} = -\lambda$. They have order at most two, lie in a bounded vertical strip, and occur on at most two vertical arithmetic progressions. This subsection does not remove those poles.

If these poles are removed, rapid vertical decay follows from the entire numerator, periodic denominator, and the maximum-modulus principle near its zeros. Combining this twisted functional equation with the Ramanujan-3 equation then gives individual Fricke equations for $F(s, 1/3)$ and $F(s, -1/3)$. Mellin inversion consequently gives

$$f \left| \begin{pmatrix} 3 & -1 \\ -35 & 12 \end{pmatrix} = f, \qquad f \right| \begin{pmatrix} 3 & 1 \\ 35 & 12 \end{pmatrix} = f.$$

The following pole argument supplies this missing condition.

43.3. **Uniform continuation of the complete 210-grid.** The new invariance under G strengthens the closed-subsystem result: *every* $F(s, j/210)$ extends meromorphically to the plane, for all allowed signs and eigenvalues. To verify this, repeat the closed-subsystem construction with $U = \mathbb{Z}/210\mathbb{Z}$, adding G and G^{-1} to the known generators before forming words of length at most three. The exact script `level35_full_continuation.py` obtains 210 pivots in each of the 32 sign and zero-parameter cases. Its pivot audit is precisely the nonvanishing exponential-coefficient audit explained above, so no exceptional complex value of either nonzero Hecke parameter is lost. The full grid is closed under every retained Taylor correction. The same strip induction consequently proves the assertion.

The exact pivot lists and source hash are saved in `level35_full_continuation.json`; all 32 cases were run successfully. Together with the preceding modulus-three quotient, this gives both uniform additive continuation and a meromorphic quadratic twisted functional equation. The following sections remove the poles and prove the remaining modular transformation laws.

43.4. **Exclusion of shifted poles away from the weight point at level 35.** This section removes the shifted poles needed for the complete proof below. Write $h = s - k/2$, $x = 2^h$, $y = 3^h$, $z = 5^h$, $w = 7^h$, and $a = \sqrt{2}f_2$, $b = \sqrt{3}f_3$. The calculation below is written for $(\eta, \sigma_5, \sigma_7) = (-1, -1, -1)$ and, by the exact sign covariance proved below, applies to all four sign triples with $\eta\sigma_5\sigma_7 = -1$. The positive sign class is treated at the end of this section. The unshifted case is treated separately; the exceptional point $s = k$ is treated after the unshifted cases.

Choose a pole s_χ of the quadratic twist modulo 3 with $\Re(s_\chi - k/2) \geq 0$, using its functional equation. Choose a rightmost pole $s_0 = s_\chi + n$, $n \geq 0$, of the complete 210-coordinate additive-twist vector in this integer class. Such a rightmost pole exists because the Dirichlet series are holomorphic in a fixed right half-plane. The candidate pole equation for the quadratic twist gives, at s_0,

$$r = 2^n, \qquad a = -x/r - r/x, \qquad |x| \geq r.$$

The first Taylor layer, eliminated by Laurent-monomial pivots, gives relations on the principal parts at s_0. The common first-Taylor factor is $i(s_0 - k)$, so its cancellation requires $s_0 \neq k$. The

exact construction is in `level35_firstjet_projection.py`. After 204 top-coordinate pivots the remaining coordinates are $24, 42, 84, 128, 158, 192$; denote their leading principal-part coefficients by v_j.

Put $P = x^2 + ax + 1$. Two one-coordinate equations give $Pv_{42} = Pv_{192} = 0$. On the shifted branch $n \geq 1$,

$$P = (r-1)(x^2/r - 1) \neq 0.$$

After these coordinates vanish, two further equations give

$$(x^4 - 2x^2 + xa - y^2)v_{84} = 0, \qquad (x^4 + xa - yb - 1)v_{84} = 0.$$

If $v_{84} = 0$, a third equation gives $Hv_{24} = 0$, where

$$H = 2x^3a + x^2a^2 + x^2 - 3 = -\frac{2r-1}{r^2}x^4 + (3-2r)x^2 + r^2 - 3.$$

This is nonzero for $|x| \geq r \geq 2$: with $u = |x|^2/r^2 \geq 1$, the leading term minus the absolute values of the remaining terms is at least

$$r^2\big((2r-1)u^2 - (2r-3)u - 1\big) + 3 > 0.$$

Finally two equations on v_{128}, v_{158} have determinant, up to a nonzero Laurent monomial,

$$(x-a)\big(a^2x^2 + ax^3 + ax + x^4 - x^2 + 1\big).$$

The second factor becomes $(r^2 - r + 1)(x^4/r^2 - x^2/r + 1)$, whose roots have modulus $\sqrt{r}$; the first factor vanishes only at modulus $r/\sqrt{r+1}$. Thus $v_{84} = 0$ forces all coordinates to vanish, a contradiction.

For $n \geq 2$ the first equation on v_{84} cannot vanish. Indeed, with $\alpha = \Re h \geq n \geq 2$, its vanishing would imply

$$16^\alpha \leq (2 + 1/r)4^\alpha + r + 9^\alpha \leq \tfrac{9}{4}4^\alpha + 2^\alpha + 9^\alpha < 16^\alpha.$$

The last inequality holds at $\alpha = 2$ and every normalized term strictly decreases thereafter. This excludes $n \geq 2$.

For $n = 1$, normalize $v_{84} = 1$. The two displayed equations give

$$a = -x/2 - 2/x, \qquad y^2 = x^4 - \tfrac{5}{2}x^2 - 2, \qquad by = x^4 - \tfrac{1}{2}x^2 - 3.$$

Solve for v_{24} using row 104 and then for v_{128}, v_{158} using rows 67 and 148 of the reduced matrix; their denominators are nonzero when $|x| \geq 2$ by the preceding estimates. Three remaining equations, affine in z, w, have determinant

$$\frac{1}{56}u^2(2u-3)^2F_7(u), \qquad u = x^2,$$

up to a nonzero rational scalar, where

$$F_7(u) = 14u^7 + 49u^6 - 391u^5 + 133u^4 + 1098u^3 - 224u^2 - 1448u - 176.$$

Consequently $F_7(u) = 0$ and $|u| \geq 4$.

Here is an exact root-location check, avoiding numerical root estimates. Compare F_7 on $|u| = 4$ with

$$q(u) = 14(u + 7301/1000)(u + 125/1000)(u - 2629/1000) \\ \cdot (u^2 + 2268u/1000 + 1493/1000)(u^2 - 3565u/1000 + 3497/1000).$$

Its five factors have lower bounds $3, 3, 1, 121/16, 4$ respectively. Thus $|q| > 7623/2$, whereas the coefficient bound gives exactly

$$|F_7 - q| \leq 249682520631353/4000000000000 < 7623/2.$$

Rouché's theorem shows that F_7 has exactly six roots inside the circle. Since $F_7(-8) = -3723120$ and $F_7(-7) = 748439$, its unique remaining root lies in $(-8, -7)$. But then

$$y^2 = u^2 - \tfrac{5}{2}u - 2 > 129/2, \qquad |y^2| = |u|^{\log_4 9} < 8^{\log_4 9} = 27,$$

a contradiction. Hence the shifted cases $n \geq 1$ are excluded under the stated sign and $s_0 \neq k$ qualifications.

Exact reduction of sign cases. The simultaneous substitutions

$$(\eta, \sigma_5, z) \longmapsto (-\eta, -\sigma_5, -z), \qquad (\eta, \sigma_7, w) \longmapsto (-\eta, -\sigma_7, -w)$$

preserve every leading equation and every Taylor coefficient, not just its vanishing locus. For a known matrix word, its eigenvalue changes by the parity of the number of Fricke factors, and this is the parity of both the 5- and 7-adic valuations of its determinant. The exponential factor has the opposite valuation, because its numerator and remaining denominator are squares. The two changes cancel. For a Ramanujan–Fricke row, the parity of the 5-adic valuation of $\operatorname{lcm}(35, q^2)$ is 1 when $5 \nmid q$ and 0 otherwise; this cancels the change of η and, when present, σ_5. The same argument applies at 7. The bad-prime Hecke row depends on σ_5/z or σ_7/w, so it is also unchanged. Taylor shifts multiply prime powers by positive rational constants and hence commute with both sign substitutions.

The executable verifier `level35_sign_symmetry.py` checks all 1,848 leading rows and their complete first Taylor rows for each of the eight sign triples. Thus there are two sign classes, indexed by $\eta\sigma_5\sigma_7$. In the shifted-pole argument above, the nonvanishing estimates and the final polynomial F_7 only involve x, y, so the proof extends unchanged to the whole negative class. For the positive class, the coefficients used to force $v_{42} = v_{192} = 0$, the two equations on v_{84}, the coefficient H of v_{24} in row 3, and the final 2×2 determinant are identical. Thus the argument excluding $n \geq 2$ applies to both classes.

For $n = 1$ in the positive class, use row 3 to solve for v_{24} and again rows 67 and 148 for v_{128}, v_{158}. The remaining rows 0, 8, 9, 12 are affine in z, w. The greatest common divisor of the determinants formed from rows $(0, 8, 9)$ and $(0, 8, 12)$ is, up to a nonzero rational scalar,

$$(u+2)^2(3u-2)^2(u^2+2u+2)F_+(u),$$

where

$$F_+(u) = 14u^7 + 21u^6 - 273u^5 + 91u^4 + 1098u^3 - 648u^2 - 1576u + 48.$$

All roots of the displayed extraneous factors have modulus less than 4. The certificate `level35_positive_gap1.py` checks the substitutions and the two determinants directly against the original six-column reduced rows, with $v_{42} = v_{192} = 0$ and $v_{84} = 1$.

Seven disjoint rational disks, checked using the linear Taylor term and Rouché's theorem by `level35_positive_gap1_roots.py`, exhaust the seven roots of F_+. Six lie inside $|u| < 4$. On the seventh, $4 < |u| < 8$ and $|u^2 - \frac{5}{2}u - 2| > 27$. This contradicts $|y^2| = |u|^{\log_4 9} < 27$. Therefore $n = 1$ is excluded in both sign classes, and hence for all eight sign triples, provided $s_0 \neq k$.

43.5. **Level 35: the unshifted leading residue.** We first prove the homogeneous leading-residue assertion, then control lower Laurent coefficients to deduce pole cancellation. Assume $s_0 \neq k$, put $h = s_0 - k/2$, and suppose $\alpha = \Re h \geq 0$ and $a = -x - x^{-1}$, where $x = 2^h, y = 3^h, z = 5^h, w = 7^h$. For the negative-product sign class $\eta\sigma_5\sigma_7 = -1$, every solution of the full leading and first-Taylor homogeneous system has zero quadratic-modulo-three residue. Sign covariance transports the algebraic identities to the remaining negative-product sign triples; then z, w may acquire fixed signs, while their absolute values remain $5^\alpha, 7^\alpha$.

The argument first treats the base triple and then checks the only additional central point introduced by this sign substitution.

Write $X = x^2, Y = y^2$, and set

$$A = X^2 + Y - 1, \quad C = X^2 - 3X - Y - 1, \quad D = X^2 - X + Y + 3,$$
$$P_1 = by + X + Y + 1, \quad P_2 = X^2 - X - by - 2.$$

Thus $P_1 + P_2 = A$. The remaining six residue coordinates after the 204 monomial pivots are $v_{24}, v_{42}, v_{84}, v_{128}, v_{158}, v_{192}$. The actual quadratic residue is $-2zw\, v_{192}/y^2$; this follows by seven monomial row operations on the original leading equations, and is also reproduced by the full first-Taylor verifier.

Rows 150 and 159 give $P_1 v_{42} = 0$ and $P_2 v_{84} + A v_{42} = 0$. When $P_1 = 0$ and $A \neq 0$, these imply $v_{84} = -v_{42}$, and row 45 then gives $(z-1)v_{192} = 0$. When both P_1, P_2 are nonzero, $v_{42} = v_{84} = 0$ and the same conclusion follows. Therefore, for $z \neq 1$, a nonzero v_{192} requires $P_2 = 0$.

First suppose $A \neq 0$. Then $v_{42} = 0$, and rows 45 and 53 give

$$C v_{84} = w(z-1)v_{192}, \qquad w = Xz - X - 1.$$

In particular C, v_{84} are nonzero. Normalize $v_{84} = 1$. If $D \neq 0$, row 104 determines v_{24}. Rows 16, 3 and 164 then give three polynomial equations in X, Y, z. These selected rows have no v_{128} or v_{158} terms; the executable verifier explicitly checks this, so neither coordinate is silently discarded. All row numbers are zero-based indices of the saved full first-Taylor matrix. Saturating the resulting ideal by the already nonzero product $(z-1)ACD$ and by X^3+1 gives the necessary relations $P_{17}(X) = 0$ and $Y = Q_{16}(X)$, where the exact rational polynomials are saved in `level35_unshifted_root_disks.json`. The branch $X^3 + 1 = 0$ is separately excluded: the same three equations together with $X^3 + 1$, saturated only by $(z-1)ACD$, generate the unit ideal. Both ideal-membership assertions are checked over $\mathbb{Q}$.

The checker `level35_unshifted_roots.py` uses exact rational complex arithmetic to isolate all 17 roots of P_{17}. For each specified rational center c and radius $r = 10^{-18}$, it verifies

$$|P'_{17}(c)|r > |P_{17}(c)| + \sum_{j=2}^{17} \frac{|P_{17}^{(j)}(c)|}{j!} r^j.$$

Rouché's theorem gives one root in each disk, and the disks are pairwise disjoint. Taylor bounds for Q_{16} prove that each disk has either $|X| < 1$, or $|Y| < 1$, or $|Y| > |X|^2$ throughout it. These alternatives all contradict the physical conditions $|X| \geq 1$ and $|Y| = |X|^{\log_4 9}$, since $0 < \log_4 9 < 2$. The saved centers are exact rational data; the verifier neither finds numerical roots nor relies on floating-point comparisons.

For $D = 0$, one has $Y = -X^2 + X - 3$ and $C = 2(X-1)^2$. Normalize $v_{192} = 1$. The coefficient of v_{24} in row 0 is

$$\frac{(X^2+2)(X^2-X+2)}{X}.$$

If it is nonzero, solve row 0; rows 13, 14, 19, 21 and 163 yield $(z-1)(X^2+2)$ in their polynomial ideal, contradicting this pivot. If $X^2 - X + 2 = 0$, then $|X| = \sqrt{2}$, so $|Y| = \sqrt{3}$, whereas $Y = -1$. If $X^2 + 2 = 0$, row 104, without division by this second pivot, gives

$$z^2 - 2(X+1)z - 11X - 1 = 0.$$

Here $|X+1| = \sqrt{3}$, $|z| = 5^{1/4} < 3/2$, and $|11X + 1| = 9\sqrt{3}$. The triangle inequality would give $9\sqrt{3} \leq |z|^2 + 2\sqrt{3}|z| < 33/4 < 9$, impossible.

Now suppose $A = 0$. If $P_2 \neq 0$ the previous zero-coordinate argument applies, so assume $P_2 = 0$. Then $Y = 1 - X^2$, $by = X^2 - X - 2$, and $C = (2X+1)(X-2)$. Normalize $v_{192} = 1$. Rows 45 and 53 imply

$$v_{42} + v_{84} = \frac{w(z-1)}{C}, \qquad w = Xz - X - 1.$$

The denominator C is nonzero. Row 104 has v_{24} coefficient $X-4$, also nonzero on the physical locus: $X=4$ would require $|Y|=9$ but $Y=-15$. After eliminating v_{24}, the coefficient of v_{42} in row 0 is

$$-\frac{X(X^2+1)(Xz-X-z)}{Xz-X-1}.$$

If this is nonzero, solve row 0. Rows 9, 26, 16 and 163 yield the ideal element $(z-1)(X-4)$, a contradiction. If $X^2+1=0$, then $\alpha=0$ and $Y=2$, impossible. If $Xz-X-z=0$, the relations give $z=X/(X-1)$ and $w=1/(X-1)$, so $(5/7)^\alpha=|z/w|=|X|=4^\alpha$ and hence $\alpha=0$. Thus $|X|=|X-1|=1$, which implies $|1-X^2|=\sqrt{3}$, contradicting $|Y|=1$.

It remains to justify $z\neq 1$ for the base sign triple. If $z=1$, then $\alpha=0$. The alternative $A=0$ would force $X^{12}=1$; combined with $z=1$ and the irrationality of $\log 2/\log 5$, this forces $h=0$, where $A=1$. Therefore $A\neq 0$. On $P_1=0$, rows 159 and 53 give $(w+1)v_{192}=0$. Otherwise $v_{42}=0$; row 45 gives $Cv_{84}=0$. On the unit circle $|X^2-3X-1|=|X-X^{-1}-3|\geq 3>|Y|$, so $C\neq 0$ and $v_{84}=0$. Row 53 again gives $(w+1)v_{192}=0$. But $5^h=1$ and $7^h=-1$ are incompatible by the irrationality of $\log 5/\log 7$.

For the other negative-product triples, the sole change is the possibility $h=0, z=1, w=-1$ after the sign substitutions. At this point $x=y=1, a=-2$, and the exact combination of specialized rows

$$2(R_{104}-R_{19}+R_{45})+5R_{150}=-8v_{192}$$

holds for every b. This excludes the remaining signed central case. The emitter checks this identity coefficient by coefficient over $\mathbb{Q}(b)$. Thus the leading-residue conclusion applies to all four negative-product sign triples.

The polynomial ideal memberships are reproduced by `level35_unshifted_residue.py`, which emits a small Magma verifier. Its selected input rows are in `level35_unshifted_rows.json`; the emitter verifies them against a freshly generated full first-Taylor matrix and checks the source hashes. All exact membership assertions and all rational root-exclusion checks pass.

Control of lower Laurent coefficients. The selected rows have no v_{128}, v_{158} coefficients. Before specializing at the pole, they are meromorphic identities modulo holomorphic germs for the four remaining coordinate functions $V_{24}, V_{42}, V_{84}, V_{192}$. This follows from exact elimination of the lower shifted variables and division by the common nonzero first-Taylor factor $i(s_0-k)$; all higher integer translates are holomorphic by the choice of s_0.

The explicit quotient for $C(s)$ bounds its pole order by two, so the same holds for V_{192} because the exact target multiplier $-2zw/y^2$ is a unit. Row 150 is a nonzero unit times $K(s)V_{42}(s)$ modulo holomorphic functions, where $K=ax-y^2-by$. At a zero of K on $ax+x^2+1=0$,

$$K'=(X+1)\log(3/2)-Y\log 3.$$

It cannot vanish when $\alpha\geq 0$: otherwise $|Y|\leq c(|X|+1)\leq 2c|X|<|X|$, with $c=\log(3/2)/\log 3<1/2$, whereas $|Y|\geq|X|$. Consequently V_{42} has pole order at most one.

The coefficient of V_{84} in row 159 is the unit $1/y^2$ times $L=x^4+ax-by-1$. At a zero of L on the same candidate divisor,

$$L'=\log(16/3)X^2+\log(3/2)X+\log(9/2).$$

Writing these three positive coefficients as a_0, b_0, c_0, one has $b_0<c_0<a_0$. The discriminant is negative, and the two roots have modulus $\sqrt{c_0/a_0}<1$. Therefore $L'\neq 0$ on $|X|\geq 1$. Row 159 shows that V_{84} has pole order at most two, and at most one unless $K(s_0)=L(s_0)=0$, equivalently $P_1=P_2=A=0$.

Finally, the coefficients of V_{24} in rows 0 and 163 satisfy the exact identity

$$XR_{0,24}+YR_{163,24}=-4 \qquad \text{on } ax+x^2+1=0.$$

At least one is a unit at the point. Thus the pole order of V_{24} is bounded by that of the other three coordinates. The entire closed four-coordinate vector consequently has pole order at most two. Apply the leading-residue result above to its order-two coefficient: the order-two coefficient of V_{192} vanishes. If V_{84} still had an order-two term, then $A = 0$, while V_{42} has order at most one. Row 45 forces C times this term to vanish. Here $C = (2X + 1)(X - 2) \neq 0$ on the physical locus: $X = -1/2$ violates $|X| \geq 1$, while $X = 2, Y = -3$ would force a rational relation between $\log 2$ and $\log 3$. Hence the order-two term of V_{84} also vanishes, and the coefficient identity bounds V_{24} by order one. Apply the leading-residue result once again, now to the order-one coefficient. It forces the remaining possible pole of V_{192}, and thus of C, to vanish.

This proves removal of an unshifted quadratic pole for every negative-product sign triple when $s_0 \neq k$. Combined with the shifted-pole exclusion, the negative-product class is reduced to the exceptional rightmost point $s_0 = k$. The positive-product sign class is treated separately by Section 43.6. The exceptional rightmost point $s_0 = k$ is treated below.

43.6. **Level 35: the positive-product unshifted case.** Assume the same unshifted pole setup as in the preceding subsection, with $s_0 \neq k$ and $\Re(s_0 - k/2) \geq 0$. The other sign class, $\eta\sigma_5\sigma_7 = +1$, reduces by the verified sign covariance to $(\eta, \sigma_5, \sigma_7) = (1, -1, -1)$. Its full first-Taylor projection has 204 monomial pivots, 326 residual rows, and quadratic target $2zwV_{192}/y^2$. The selected rows again involve only $V_{24}, V_{42}, V_{84}, V_{192}$.

Use the notation X, Y, A, C, P_1, P_2 from the negative-product case. Rows 150 and 159 are identical in the two sign classes. Thus the same preliminary argument reduces a nonzero leading quadratic residue, when $z \neq 1$, to $P_2 = 0$. Row 45 now reads

$$Cv_{84} + (\text{coefficient})v_{42} + w(z - 1)v_{192} = 0.$$

When $A \neq 0$, one has $v_{42} = 0$; normalizing $v_{192} = 1$ gives

$$v_{84} = -\frac{w(z - 1)}{C}, \qquad w = -Xz + X - 1.$$

The second identity follows from row 53. The coefficient identity $XR_{0,24} + YR_{163,24} = -4$ supplies a uniformly nonzero pivot for v_{24}. No eigenvalue or additional expression is divided out.

After these substitutions, rows 3, 13, 14, 16 and 18 yield a polynomial ideal. Saturation by

$$(z - 1)ACXYZ(-Xz + X - 1),$$

whose factors are nonzero on this branch, yields

$$(X + 1)P_{15}(X) = 0, \qquad (X + 1)(Y - Q_{14}(X)) = 0, \qquad (X + 1)(z - R_{14}(X)) = 0.$$

All three rational polynomials are explicitly recorded in `level35_positive_root_disks.json`. On the branch $X = -1$ the same ideal gives $z = 3$, contrary to $|z| = 1$ when $|X| = 1$. For the 15 roots of P_{15}, the exact rational disk verifier `level35_positive_roots.py` uses the same Rouché inequalities as the negative-product case. Each disk has one of

$$|X| < 1, \qquad |Y| < 1, \qquad |Y| > |X|^2, \qquad |z| < 1$$

throughout it. Each alternative contradicts the physical moduli $|X| = 4^\alpha$, $|Y| = 9^\alpha$, $|z| = 5^\alpha$, $\alpha \geq 0$. All comparisons use exact rational bounds, including evaluation of Q_{14} and R_{14} on the isolating disks.

For $A = 0$, still with $v_{192} = 1$, write

$$v_{84} = -\frac{w(z - 1)}{C} - v_{42}, \qquad w = -Xz + X - 1, \qquad C = (2X + 1)(X - 2).$$

Eliminate v_{24} with the same coefficient identity, retaining v_{42} as an unknown. Rows 3, 16, 164 and 13 generate the unit ideal after saturation by

$$Xz(-Xz+X-1)(2X+1)(X-2)(1-X^2)(z-1).$$

All factors are nonzero physically: the only additional checks are $C \neq 0$ as established in the preceding subsection, and $1-X^2 = Y \neq 0$. This computation never divides by a coefficient of the remaining unknown v_{42}.

The treatment of $z=1$ is the same until the possible signed central point $h=0, z=1, w=-1$. Here the positive-product rows satisfy, for all b, the exact identity

$$2(R_3+R_{14}+R_{19}-R_{45})+3R_{150} = -8v_{192}.$$

This rules out the remaining central leading residue. The emitter `level35_positive_residue.py` verifies this identity, verifies that rows 150 and 159 are unchanged, compares every selected row with the freshly generated full projection, and asserts that no selected row involves v_{128} or v_{158}. Its emitted Magma verifier checks both saturated ideal certificates exactly.

Finally the pole-order argument from the negative-product case applies unchanged: the simple-zero bounds for K and L, the bound of two for the quadratic quotient, and the nonzero V_{24} coefficient combination bound the closed four-coordinate vector by order two. The leading-residue result removes its quadratic order-two coefficient; row 45 removes any possible residual order-two V_{84} coefficient; and a second application removes the quadratic order-one coefficient. Thus the unshifted quadratic pole is excluded for *all eight sign triples* when $s_0 \neq k$. Positive-product shifted chains are treated in the shifted-pole section. The next subsection treats the exceptional point $s_0 = k$.

43.7. **The exceptional rightmost point at level 35.** The apparent exception $s_0 = k$ is removed using only the leading relations. In particular, no division by a Taylor factor is used here. Put $m = k/2 \geq 1$. If a rightmost grid pole $s_0 = k$ is above a quadratic pole s_χ with $\Re s_\chi \geq k/2$, then

$$n = k - s_\chi \in \{0,1,\dots,m\}, \quad x = 2^m, \quad y = 3^m, \quad a = -2^{m-n} - 2^{n-m}.$$

By the exact sign covariance, it suffices to use the two canonical sign classes and allow $z = \pm 5^m$, $w = \pm 7^m$.

The leading-only elimination in `level35_leading_projection.py` uses 204 Laurent-monomial pivots and leaves the coordinates $24, 42, 126, 128, 140, 158$. The selected-row identities below are verified by `level35_weight_point.py` against saved original reduced rows. We show that their kernel is zero. This contradicts the nonzero highest Laurent vector of any grid pole, regardless of its order or its quadratic component.

For $n \geq 1$, rows 46 and 125 give $Pv_{42} = Pv_{140} = 0$, with $P = x^2 + ax + 1 \neq 0$ as in the shifted-pole argument. Row 41 then gives

$$Ev_{126} = 0, \qquad E = x^4 - 2x^2 + ax - y^2.$$

For $n \geq 2$ the previous norm estimate gives $E \neq 0$. For $n = 1$,

$$E = 16^m - \tfrac{5}{2}4^m - 2 - 9^m.$$

This equals -5 at $m=1$, is positive at $m=2$, and stays positive because every term on the right divided by 16^m decreases with m. Thus $v_{126} = 0$. Row 3 now gives $Hv_{24} = 0$, where the previous bound proves $H \neq 0$. Finally rows 67 and 148 have the previously proved nonzero determinant on v_{128}, v_{158}. The kernel is zero.

For $n = 0$, put $X = x^2 = 4^m$, $Y = y^2 = 9^m$ and

$$P_1 = by+X+Y+1, \quad P_2 = X^2-X-by-2, \quad A = P_1+P_2 = X^2+Y-1 > 0, \quad C = X^2-3X-Y-1.$$

Here $C = -6$ at $m = 1$ and $C > 0$ for $m \geq 2$, by the same normalized-term argument. Rows 150, 159 and 41 read

$$P_1 v_{42} = 0, \qquad (-A + 2P_1)v_{42} - P_2 v_{126} = 0, \qquad (C + P_1)v_{42} + C v_{126} + Y(1 - z^{-1})v_{140} = 0.$$

Since $|z| \geq 5$, the last coefficient of v_{140} is nonzero.

If $P_1 P_2 \neq 0$, these three equations immediately kill v_{42}, v_{126}, v_{140}. If $P_2 = 0$, then $P_1 = A \neq 0$ and $v_{42} = 0$. Rows 41 and 220 on v_{126}, v_{140} have determinant

$$\frac{C(Xz - X - w - 1)}{wY} \quad \text{or} \quad -\frac{C(Xz - X + w + 1)}{wY}$$

in the negative or positive sign class, respectively. Both are nonzero, since $20^m > 4^m + 7^m + 1$ for all $m \geq 1$.

It remains to consider $P_1 = 0$. Since $P_2 = A \neq 0$, we obtain $v_{126} = -v_{42}$ and $v_{140} = 0$. The coefficient of v_{24} in row 3 is

$$H = -X^2 + X - 2 \neq 0.$$

Solve for v_{24} and substitute into row 0. Up to a nonzero rational scalar and a nonzero denominator, its coefficient of v_{42} is

$$-(X^2 + 1)(X^2 - 2X - 1)(z - w - 1) \quad \text{or} \quad (X^2 + 1)(X^2 - 2X - 1)(z + w - 3),$$

respectively. The first two factors are nonzero for $X \geq 4$. The negative-class last factor is nonzero because $7^m > 5^m + 1$. In the positive class, $7^m > 5^m + 3$ for $m \geq 2$; for $m = 1$, none of $\pm 5 \pm 7$ equals 3. Hence $v_{42} = 0$, and then $v_{126} = v_{140} = v_{24} = 0$.

In every branch, row 3 kills any remaining v_{24} and the final 2×2 determinant kills v_{128}, v_{158}. Indeed the determinant formula from the shifted case is still valid at $r = 1$; its possible zeros have $|x| \leq 1$, whereas $x = 2^m \geq 2$. Thus the leading matrix has full column rank for every allowed weight, gap, eigenvalue b, and sign triple. No grid pole can have rightmost point $s_0 = k$.

Conclusion of the quadratic-three pole argument. A hypothetical quadratic pole may be reflected to the right half of its functional-equation center. Its rightmost grid pole lies either at a positive integer shift, at the quadratic pole itself, or at the exceptional weight point. The preceding arguments exclude all three possibilities for every sign and eigenvalue. Thus $F(s, \chi_3)$ is entire. The quotient formula gives rapid decay in fixed vertical strips away from periodic denominator zeros, and the maximum principle gives it inside small disks around those zeros. Combining with the entire Ramanujan-three sum yields the two individual denominator-three functional equations and, by Mellin inversion, the invariances under $C_+ = (3, -1; -35, 12)$ and $C_- = (3, 1; 35, 12)$ used below.

44. Completion of the level-35 converse theorem

This section uses the preceding proof that the quadratic twist modulo 3 is entire. Write $F(s, \alpha)$ for the Mellin transform of $f(z + \alpha)$, $h = s - k/2$, $x = 2^h$, $y = 3^h$, $z = 5^h$, $w = 7^h$, $a = \sqrt{2} f_2$, $b = \sqrt{3} f_3$, and $\sigma_p = \sqrt{p} f_p \in \{\pm 1\}$ for $p = 5, 7$. All slash operators are determinant-normalized. Matrices differing by a nonzero scalar act identically because k is even.

The original hypotheses, modulus-six argument, and quadratic-3 pole removal give invariance under

$$T = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad U = \begin{pmatrix} 1 & 0 \\ 35 & 1 \end{pmatrix}, \quad E = \begin{pmatrix} 2 & -1 \\ -35 & 18 \end{pmatrix}, \quad B = \begin{pmatrix} 6 & 1 \\ 35 & 6 \end{pmatrix},$$

$$C_+ = \begin{pmatrix} 3 & -1 \\ -35 & 12 \end{pmatrix}, \qquad C_- = \begin{pmatrix} 3 & 1 \\ 35 & 12 \end{pmatrix},$$

and $f | W_{35} = \eta f$, where $\eta = (-1)^{k/2} \epsilon_f$.

Four individual denominator-twelve equations. Put $M_0 = \left(\begin{smallmatrix}1&0\\0&2\end{smallmatrix}\right)$, $M_1 = \left(\begin{smallmatrix}1&1\\0&2\end{smallmatrix}\right)$ and $M_\infty = \left(\begin{smallmatrix}2&0\\0&1\end{smallmatrix}\right)$. Exact multiplication gives

$$M_0 B M_\infty^{-1} = C_-, \qquad M_\infty B M_0^{-1} = C_+^{-1}, \qquad M_1 B M_1^{-1} = H = \begin{pmatrix} 41 & -17 \\ 70 & -29 \end{pmatrix}.$$

Right-multiply $f|(M_0 + M_1 + M_\infty) = af$ by B and subtract the original Hecke identity. The first and last summands cancel after interchange, so $f|M_1 B = f|M_1$, hence $f|H = f$. This argument never divides by a and includes $a = 0$.

Set $E_- = UET = \left(\begin{smallmatrix}2&1\\35&18\end{smallmatrix}\right)$ and $H_- = T^{-1}H^{-1}T = \left(\begin{smallmatrix}41&17\\-70&-29\end{smallmatrix}\right)$ projectively. The known invariant matrices

$$G_5 = EH = \begin{pmatrix} 12 & -5 \\ -175 & 73 \end{pmatrix}, \qquad G_{-5} = E_- H_- = \begin{pmatrix} 12 & 5 \\ 175 & 73 \end{pmatrix},$$

together with $G_1 = C_-^{-1}$ and $G_{-1} = C_+^{-1}$, satisfy

$$G_j^{-1} W_{35} = \frac{1}{12}\, t_{j/12} W_{5040} t_{-j/12} \qquad (j = \pm 1, \pm 5).$$

Thus each $F(s, j/12)$ is entire, rapidly decreasing in fixed vertical strips, and satisfies the self-functional-equation

$$F(s, j/12) = \epsilon_f 5040^{k/2-s} F(k-s, j/12).$$

Indeed the translated Fourier series decays at infinity, and the Fricke identity transfers that decay to zero.

The quadratic twist modulo four. Put

$$A_{12}(s) = F(s,1/12) - F(s,-1/12) + F(s,5/12) - F(s,-5/12), \quad C_4(s) = F(s,1/4) - F(s,-1/4).$$

The Hecke-3 equations at $\pm 1/4$ give

$$A_{12}(s) = (1 + b/y + y^{-2}) C_4(s), \qquad C_4(s) = \frac{y A_{12}(s)}{y + y^{-1} + b}. \tag{35.4}$$

Consequently $C_4 = 2iF(s, \chi_4)$ is meromorphic and satisfies

$$C_4(s) = \epsilon_f 560^{k/2-s} C_4(k-s). \tag{35.5}$$

There is a particularly small leading-system certificate for removing its poles. Take the full additive grid $j/420$, $0 \le j < 420$. Use the leading equations from the known matrices $T, U, E, W_{35}, B^{-1}, C_+, C_-, H$ and their words of length at most two, the Ramanujan–Fricke equations whose arguments stay on this grid, and the Hecke equations at $2, 3, 5, 7$. The exact program `level35_modularity_chi4.py` checks each of the eight sign triples. It performs 419 pivots, each a Laurent monomial in x, y, z, w independent of a, b, leaving only coordinate 265. Its nonempty residual system consists precisely of scalar multiples with sign ± 1 of

$$\frac{zw - \sigma_5\sigma_7}{zw}\, v_{265} = 0. \tag{35.6}$$

Moreover the target $v_{105} - v_{315}$ reduces identically to zero using the same unit pivots. All row supports are checked to remain inside the 420-grid. These are rational-function identities, valid also when either good-prime Hecke eigenvalue vanishes.

First, the coefficient in (35.6) is not identically zero as a function of s for any fixed eigenvalues. Thus the leading matrix has a meromorphic inverse on a selected full-rank subsystem. The usual Taylor–Fricke strip induction proves meromorphic continuation of all 420 additive transforms, exactly as for the earlier 210-grid. Every additive transform is holomorphic in a common right half-plane.

Suppose C_4 has a pole. By (35.5), choose one, s_χ, with $\Re(s_\chi - k/2) \ge 0$, and choose the rightmost grid pole $s_0 = s_\chi + n$ in its integer class, where $n \ge 0$. At s_0, every higher positive integer translate is holomorphic. The full leading equations therefore hold modulo holomorphic germs; the same is true

after each unit pivot. Applied to the highest nonzero Laurent vector, (35.6) forces $35^{s_0-k/2} = \sigma_5\sigma_7$, hence $\Re(s_0 - k/2) = 0$. It follows that $n = 0$ and $\Re(s_\chi - k/2) = 0$. But the exact target reduction then says that $F(s, 1/4) - F(s, -1/4)$ is holomorphic at $s_0 = s_\chi$, a contradiction. This target assertion is an identity of meromorphic germs modulo holomorphic germs, so it removes all Laurent orders at once. Therefore C_4, and hence $F(s, \chi_4)$, is entire.

As in the quadratic-3 case, (35.4) gives rapid decay in fixed vertical strips: use the rapidly decreasing numerator away from uniform small disks about the periodic denominator zeros, and the maximum principle inside those disks. The entire Ramanujan-4 sum and (35.5) now give the two individual denominator-four functional equations. Mellin inversion yields

$$f|D_+ = f, \quad f|D_- = f, \qquad D_+ = \begin{pmatrix} 4 & -1 \\ -35 & 9 \end{pmatrix}, \quad D_- = \begin{pmatrix} 4 & 1 \\ 35 & 9 \end{pmatrix}.$$

Generation and cuspidality. The exact verifier `level35_modularity.py` checks the Hecke matrix identities and all four denominator-twelve centers above. It then takes $T, U, E, B^{-1}, C_+, C_-, H, D_+, D_-$ and their Fricke conjugates. Each fixes f, is integral of determinant one, and has lower-left entry divisible by 35. The verifier expresses each as a word in the standard generators of $\mathrm{PSL}_2(\mathbb{Z})$, checks those words by multiplication, and completes a coset enumeration of index 48. Since

$$[\mathrm{PSL}_2(\mathbb{Z}) : \overline{\Gamma_0(35)}] = 35(1 + 1/5)(1 + 1/7) = 48,$$

these transformations generate $\Gamma_0(35)$ projectively. Even weight handles the central sign. Thus f is modular.

The polynomial bound on the original Fourier coefficients bounds f polynomially as the imaginary part tends to zero, uniformly in the real part. After any integral scaling matrix, this gives polynomial growth at infinity. Periodicity at a cusp then excludes negative Fourier exponents, so f is holomorphic at every cusp.

The squarefree level 35 has four cusp classes, represented by $\infty, 0, 1/5, 1/7$. The given Fourier series and original Fricke identity give vanishing at the first two. For $d = 5$ or 7, let c_d be the constant term of $f|\sigma_a$, where $\sigma_a = \left(\begin{smallmatrix} a & b_a \\ d & e_a \end{smallmatrix}\right) \in \mathrm{SL}_2(\mathbb{Z})$ and $(a, d) = 1$. This constant is independent of a. Indeed there is an integer n such that $\sigma_a T^n \sigma_1^{-1} \in \Gamma_0(35)$; the verifier checks this for every reduced residue a mod d. Right translation preserves the constant term.

For $t \downarrow 0$ one has

$$f(a/d + it) = (-idt)^{-k}\big(c_d + O(e^{-c/t})\big),$$

since $\sigma_a^{-1}(a/d + it) = -e_a/d + i/(d^2 t)$. Consequently the residue at $s = k$ of the Ramanujan-d Mellin sum is

$$\varphi(d)(-id)^{-k} c_d.$$

That sum is entire by hypothesis, so $c_d = 0$. Thus f vanishes at every cusp, and the level-35 conclusion is

$$f \in S_k(\Gamma_0(35)).$$

45. A converse theorem at level 32

Zhang [7] proves twist-free converse theorems at levels 32 and 36 under the Conrey–Farmer hypotheses. We include the arguments below to give a complete account of the small-level result.

Theorem 45.1 (Converse at level 32)**.** *Let k be a positive even integer and let (f_n) satisfy precisely the Euler-product, ramified-coefficient, entire finite-order continuation, and functional-equation assumptions defining $\mathcal{E}(32, k)$ in* [3]*. Then*

$$f(z) = \sum_{n \geq 1} a_n e^{2\pi i n z}, \qquad a_n = n^{(k-1)/2} f_n, \quad \textit{belongs to } S_k(\Gamma_0(32)).$$

The proof includes every complex value of f_3, including zero, and both root numbers. No additional twisted functional equation is assumed.

We use the determinant-normalized right slash action

$$(F|_k\gamma)(z) = (\det\gamma)^{k/2}(cz+d)^{-k}F(\gamma z), \qquad \gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{GL}_2^+(\mathbb{R}).$$

We suppress the weight subscript and extend the action linearly to finite sums of matrices. Positive scalar matrices act trivially; negative scalar matrices also act trivially because k is even. Put

$$T = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad W_L = \begin{pmatrix} 0 & -1 \\ L & 0 \end{pmatrix},$$
$$D_m = \begin{pmatrix} 1 & 0 \\ 0 & m \end{pmatrix}, \quad t_r = \begin{pmatrix} 1 & r \\ 0 & 1 \end{pmatrix}.$$

The usual Mellin inversion argument from the hypotheses gives

$$f|W_{32} = \eta f, \qquad \eta = i^k\epsilon_f \in \{1,-1\}. \tag{1}$$

The coefficients a_n have polynomial growth, so f is holomorphic on $\mathbb{H}$ and has polynomial growth as $\operatorname{Im} z \downarrow 0$, uniformly in $\operatorname{Re} z$. These are the standard preliminary consequences used throughout this reconstruction.

45.1. **The odd Fourier support and the Hecke relation.** Since $2^2 \mid 32$, the ramified Euler factor is 1. Thus $a_{2n} = 0$ and

$$f|t_{1/2} = -f. \tag{2}$$

At the good prime 3, the Euler recurrence gives

$$f\left|\left(\sum_{j=0}^{2}\begin{pmatrix} 1 & j \\ 0 & 3 \end{pmatrix} + \begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix}\right) = \lambda f, \qquad \lambda = \sqrt{3}f_3.\right. \tag{3}$$

For completeness, the sum of the first three slashes is $3^{1-k/2}\sum a_{3n}e^{2\pi inz}$; the recurrence $a_{3n} = a_3a_n - 3^{k-1}a_{n/3}$ makes its second term cancel the last slash in (3). No division by λ will occur.

Set $h = f|D_4$. Equations (1)–(2), and $D_4^{-1}W_{32}D_4 = 4W_2$, give

$$h|T^2 = -h, \qquad h|W_2 = \eta h. \tag{4}$$

Write $M_j = \left(\begin{smallmatrix} 1 & j \\ 0 & 3 \end{smallmatrix}\right)$ and $M_\infty = \left(\begin{smallmatrix} 3 & 0 \\ 0 & 1 \end{smallmatrix}\right)$. Conjugating (3) by D_4 gives

$$h|(M_0 + M_4 + M_8 + M_\infty) = \lambda h.$$

But $M_4 = T^2M_{-2}$ and $M_8 = T^2M_2$, so (4) rewrites this as

$$h|\mathcal{T} = \lambda h, \qquad \mathcal{T} = M_0 - M_2 - M_{-2} + M_\infty. \tag{5}$$

45.2. **An infinite elliptic transformation kills the defect.** Put $W = W_2$ and $L_j = \left(\begin{smallmatrix} 3 & 0 \\ j & 1 \end{smallmatrix}\right)$. Direct multiplication gives

$$W^{-1}M_0W = M_\infty, \quad W^{-1}M_\infty W = M_0,$$
$$W^{-1}M_2W = L_{-4}, \quad W^{-1}M_{-2}W = L_4.$$

Since $h|W = \eta h$, equation (5) also gives $h|W^{-1}\mathcal{T}W = \lambda h$. Subtracting these two eigenrelations therefore yields

$$h|(M_2 + M_{-2} - L_{-4} - L_4) = 0. \tag{6}$$

Now set

$$A = \begin{pmatrix} 3 & 2 \\ 4 & 3 \end{pmatrix}.$$

One has $L_4 = AM_{-2}$ and $L_{-4} = A^{-1}M_2$. Thus, for the holomorphic defect $d = h|(1 - A)$, equation (6) becomes

$$d|(M_{-2} - A^{-1}M_2) = 0.$$

Multiplication on the right by $M_2^{-1}A$ gives

$$d|E = d, \qquad E = M_{-2}M_2^{-1}A = \begin{pmatrix} -7/3 & -2 \\ 4 & 3 \end{pmatrix}. \tag{7}$$

Here $\det E = 1$ and $\operatorname{tr} E = 2/3$. Hence E is elliptic and has infinite order: a finite-order determinant-one matrix has roots of unity as eigenvalues, making its trace an algebraic integer; the rational number $2/3$ is not an integer.

We recall the elementary positive-weight elliptic lemma. If a holomorphic function on $\mathbb{H}$ satisfies $d|_k E = d$ for an infinite-order elliptic E and $k > 0$, then $d = 0$. Indeed, conjugate its fixed point to the centre of the disk and absorb the standard weight factor into the holomorphic function. If the first nonzero Taylor coefficient has degree $m \geq 0$, invariance forces a rotation multiplier with exponent $k+2m$ to be 1. The rotation angle is irrational modulo π, while $k+2m > 0$, a contradiction. Applying this lemma to (7) proves

$$h|A = h. \tag{8}$$

45.3. **A finite-character modular form at level eight.** Set $g = f|D_2$, so $h = g|D_2$. From (1)–(2) and (8),

$$g|T = -g, \qquad g|W_8 = \eta g, \qquad g|B = g, \quad B = D_2AD_2^{-1} = \begin{pmatrix} 3 & 1 \\ 8 & 3 \end{pmatrix}. \tag{9}$$

Conjugating the first equality by W_8 also gives

$$g|U = -g, \qquad U = \begin{pmatrix} 1 & 0 \\ 8 & 1 \end{pmatrix}. \tag{10}$$

The matrices T, U, B generate $\Gamma_0(8)$ modulo its central $-I$. An explicit Schreier certificate is supplied below, so this generation claim requires no numerical or probabilistic group computation.

For $\gamma = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \Gamma_0(8)$ define

$$\xi(\gamma) = (-1)^{b+c/8}.$$

This is a character: the diagonal entries are odd, so the two exponents b and $c/8$ add modulo 2 under matrix multiplication. It is trivial on $-I$, and $\xi(T) = \xi(U) = -1$, $\xi(B) = 1$. Equations (9)–(10) and the generation certificate prove

$$g|\gamma = \xi(\gamma)g \qquad (\gamma \in \Gamma_0(8)). \tag{11}$$

If $\gamma = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \Gamma_0(32)$, then

$$D_2^{-1}\gamma D_2 = \begin{pmatrix} a & 2b \\ c/2 & d \end{pmatrix} \in \Gamma_0(8), \qquad \xi(D_2^{-1}\gamma D_2) = (-1)^{2b+c/16} = 1.$$

Consequently (11) implies $f|\gamma = f$.

45.4. **Holomorphy and vanishing at every cusp.** The four cusp classes of $\Gamma_0(8)$ are $\infty, 0, 1/2, 1/4$. The standard denominator classification gives exactly four classes: $\sum_{d|8} \varphi(\gcd(d, 8/d)) = 4$. Primitive positive-width stabilizers can be chosen as

cusp	width	stabilizer
∞	1	$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$
0	8	$\begin{pmatrix} 1 & 0 \\ -8 & 1 \end{pmatrix}$
$1/2$	2	$\begin{pmatrix} -3 & 2 \\ -8 & 5 \end{pmatrix}$
$1/4$	1	$\begin{pmatrix} -3 & 1 \\ -16 & 5 \end{pmatrix}$

Each has character $\xi = -1$. Thus at every cusp the corresponding local slash of g is anti-periodic with its indicated width. Its Fourier exponents are half-integers, so a constant term is impossible. The polynomial growth inherited from the coefficients of f excludes negative exponents: after any cusp change of variable the local slash is at most polynomially growing at infinity, whereas a negative Fourier exponent grows exponentially. Hence all the local exponents are positive, and g vanishes at every cusp.

The subgroup $D_2^{-1}\Gamma_0(32)D_2$ has finite index in $\Gamma_0(8)$ and lies in $\ker \xi$. Vanishing at cusps persists on passing to this subgroup and on undoing the positive diagonal change of variable. Therefore $f \in S_k(\Gamma_0(32))$, completing the proof.

45.5. **Explicit generation certificate.** Put $S = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$ and work projectively. The bottom rows of the representatives r_i in the table below give the 12 distinct elements of $\mathbb{P}^1(\mathbb{Z}/8\mathbb{Z})$, hence the 12 left cosets of $\Gamma_0(8)$ in $\mathrm{PSL}_2(\mathbb{Z})$. The same table records $r_i S = \pm r_{j_S}$ and $r_i T = \pm \kappa_i r_{j_T}$:

i	r_i	j_S	j_T	κ_i
0	1	1	0	T
1	S	0	2	1
2	ST	3	4	1
3	STS	2	1	T^{-1}
4	ST^2	5	6	1
5	ST^2S	4	7	1
6	ST^3	8	9	1
7	ST^2ST	10	5	$T^{-1}B$
8	ST^3S	6	10	$B^{-1}T$
9	ST^4	11	8	$U^{-1}B$
10	ST^2STS	7	3	1
11	ST^4S	9	11	$B^{-1}U$

All entries follow by integer matrix multiplication. Every Schreier generator is therefore a word in T, U, B, proving the generation claim.

The standalone verifier `level32_continued_certificate.py` checks all Hecke and elliptic matrix identities, reconstructs and checks these Schreier words using exact rational arithmetic, and checks the four cusp signs. Its success marker is `PASS_LEVEL32_EXACT_CERTIFICATE`. No sampled rank, large polynomial computation, or assumption on λ enters this proof.

46. A converse theorem at level 36

The converse theorem at this level is also proved by Zhang [7].

Theorem 46.1 (Converse at level 36)**.** *Let k be a positive even integer and let (f_n) satisfy precisely the Euler-product, ramified-coefficient, entire finite-order continuation, and functional-equation hypotheses defining $\mathscr{E}(36,k)$ in* [3]*. Then*

$$f(z)=\sum_{n\geq 1} f_n n^{(k-1)/2} e^{2\pi i n z} \in S_k(\Gamma_0(36)).$$

Both root numbers and every complex unramified Hecke eigenvalue are included. No twisted functional equation is assumed.

Proof. Throughout, the slash action is determinant-normalized and the weight k is positive and even. Put $D=\operatorname{diag}(1,6)$, $g=f|D$, $S=\left(\begin{smallmatrix}0&-1\\1&0\end{smallmatrix}\right)$, and $T=\left(\begin{smallmatrix}1&1\\0&1\end{smallmatrix}\right)$. The Euler factors at $2,3$ give Fourier support $n\equiv\pm1 \pmod 6$, so $g(T^2-T+1)=0$. Write $h=g|TS$. Since $D^{-1}W_{36}D=6S$, the untwisted functional equation gives $g|S=\eta g$, where $\eta=i^k\epsilon_f\in\{1,-1\}$, and the identity $TST=ST^{-1}S$ in $\mathrm{PSL}_2(\mathbb{Z})$ yields

$$g|T=gT,\quad (gT)|T=gT-g,\quad h|T=g-\eta h,\qquad g|S=\eta g,\quad (gT)|S=h,\quad h|S=gT.$$

These equations hold even when the three functions are dependent.

If $\eta=-1$, the function

$$H=gT+h$$

is invariant under both S and T. It is holomorphic on the upper half-plane. Polynomial growth of the original Fourier coefficients implies a uniform estimate $f(x+iy)=O(y^{-M})$ as $y\downarrow 0$ for some M, and exponential decay as $y\to\infty$. Consequently g, gT, gTS, and hence H, have polynomial growth as $y\to\infty$, uniformly on bounded real intervals. The Fourier expansion of the one-periodic H therefore has no negative powers:

$$H(z)=\sum_{n\geq 0} b_n e^{2\pi i n z}.$$

This proves holomorphy at the level-one cusp but does not presume that $b_0=0$.

Fourier-filter lemma. Suppose there is an identity

$$\sum_{j=0}^{4} c_j H\left|\begin{pmatrix}1&6j\\0&5\end{pmatrix}+c_\infty H\right|\begin{pmatrix}5&0\\0&1\end{pmatrix}+c_* H=0, \tag{F}$$

where $c_j\in\mathbb{Q}$ are not all equal (the last two coefficients may be arbitrary complex numbers). Then $H=0$.

Indeed, in the coefficient of $e^{2\pi i n z/5}$ for $5\nmid n$, the last two terms of (F) contribute nothing. The first five terms give

$$5^{-k/2} b_n \sum_{j=0}^{4} c_j \zeta_5^{jn}=0,$$

because $6\equiv 1 \pmod 5$. A rational polynomial of degree at most four that vanishes at a primitive fifth root is a multiple of $1+X+X^2+X^3+X^4$. The assumption on the c_j therefore gives $b_n=0$ for every $5\nmid n$. It follows that $H(z+1/5)=H(z)$, including its possible constant term.

Together with $H|S=H$, this implies invariance under

$$E=S\begin{pmatrix}1&1/5\\0&1\end{pmatrix}=\begin{pmatrix}0&-1\\1&1/5\end{pmatrix}.$$

The matrix has determinant one and trace $1/5$, so is elliptic. It has infinite projective order: finite projective order would make its eigenvalues roots of unity and its rational trace an integer. A holomorphic positive-weight function invariant under an infinite-order elliptic matrix vanishes. For completeness, conjugate its fixed point to the origin of the disk and absorb the weight automorphy

factor. The Taylor coefficient of degree n is then multiplied by $\exp(i(k+2n)\theta)$, with θ/π irrational. Since $k+2n>0$, every coefficient must vanish. This proves the lemma, without an Eisenstein or cuspidality assumption.

The exact Hecke identity for the negative sign. This supplies the algebraic input to the Fourier-filter lemma. Put $\alpha_j = \left(\begin{smallmatrix} 1 & 6j \\ 0 & 5 \end{smallmatrix}\right)$ for $0 \le j \le 4$ and $\alpha_\infty = \left(\begin{smallmatrix} 5 & 0 \\ 0 & 1 \end{smallmatrix}\right)$. The original Euler product gives $\sum_j g|\alpha_j + g|\alpha_\infty = \Lambda_5 g$ with normalized eigenvalue $\Lambda_5 = \sqrt{5} f_5$. Take

$$\gamma = \begin{pmatrix} 11 & 30 \\ -150 & -409 \end{pmatrix}, \quad \gamma_j = \alpha_j \gamma \alpha_j^{-1} = \begin{pmatrix} 11 - 900j & 6 - 504j + 1080j^2 \\ -750 & -409 + 900j \end{pmatrix}, \quad \gamma_\infty = \begin{pmatrix} 11 & 150 \\ -30 & -409 \end{pmatrix}.$$

All these are integral determinant-one matrices, congruent to $-I$ modulo six. Using the three-dimensional transformation matrices for $\eta = -1$, their first rows give the exact identities

$$g|\gamma = g - 2H, \qquad g|\gamma_j = g + c_j H, \qquad (c_0, c_1, c_2, c_3, c_4) = (0, -4, 2, -2, 2), \qquad g|\gamma_\infty = g + 2H.$$

These are verified by Euclidean decomposition into S, T in `level36_hecke_check.py`; no congruence invariance for g is assumed in this calculation. Right-translate the original Hecke identity by γ, use $\alpha_j \gamma = \gamma_j \alpha_j$, and subtract the original identity. Dividing the result by two gives

$$H|\alpha_\infty - 2H|\alpha_1 + H|\alpha_2 - H|\alpha_3 + H|\alpha_4 + \Lambda_5 H = 0.$$

Its finite-branch coefficients $(0, -2, 1, -1, 1)$ are rational and not all equal. The Fourier-filter lemma proves $H = 0$, for every complex Λ_5, including zero. It remains to establish the congruence and cusp claims below. The finite congruence claim is independently checked in `level36_congruence_check.py`, which enumerates all 144 matrices of $\mathrm{SL}_2(\mathbb{Z}/6\mathbb{Z})$ and checks both generator edges at every matrix. The positive-sign representation has image of order 24; the negative-sign representation after $H = 0$ has image of order six. In both cases $-I$ acts trivially and T^3 acts as minus the identity.

From vanishing of the extension to cuspidality. For $\eta = +1$, direct use of the displayed three-dimensional transformation laws gives $T^3 = -I$ on the whole orbit space $V = \mathrm{span}(g, gT, h)$. For $\eta = -1$, once $H = 0$, we have $h = -gT$, and $T^3 = -I$ again holds on V. In either case twist the action by the sign character of $\mathrm{SL}_2(\mathbb{F}_2)$, which takes both S, T to -1. The twisted generators satisfy

$$S'^2 = T'^3 = (S'T')^3 = 1.$$

The presentation of the tetrahedral group identifies this quotient with $A_4 = \mathrm{PSL}_2(\mathbb{F}_3)$. Its generator images are precisely the reductions of S, T modulo 3, so the twisted action is trivial on $\Gamma(3)$. The twisting character is trivial on $\Gamma(2)$; hence the original action is trivial on $\Gamma(6)$.

For $\gamma = \left(\begin{smallmatrix} a & b \\ 36c & d \end{smallmatrix}\right)$ in $\Gamma_0(36)$,

$$D^{-1}\gamma D = \begin{pmatrix} a & 6b \\ 6c & d \end{pmatrix} \equiv \pm I \pmod 6.$$

The last congruence follows from $ad \equiv 1 \pmod 6$ and the fact that the only units modulo 6 are ± 1. Since k is even, g is invariant under this matrix, so $f|\gamma = f$.

Finally, each transform $g|\gamma$, $\gamma \in \mathrm{SL}_2(\mathbb{Z})$, belongs to V, has polynomial growth at infinity, is six-periodic, and is negated by translation through three. Its Fourier expansion in $e^{2\pi i z/6}$ therefore has no negative powers and no constant term. Thus g vanishes at every rational cusp. Rational rescaling by D transports this conclusion to every cusp of $\Gamma_0(36)$. Consequently $f \in S_k(\Gamma_0(36))$.

□

47. The converse theorem at level 27

The missing functional equation for the quadratic twist modulo five is not needed. The decisive additional identity is an affine Hecke relation for the obstruction to modularity. Continuation supplies its asymptotic at infinity, and repeated dilation then forces it to vanish.

Theorem 47.1. *Under exactly the original hypotheses defining $\mathscr{E}(27,k)$, for every positive even k, every Fricke sign, and every permitted Hecke eigenvalue, one has $f \in S_k(\Gamma_0(27))$.*

47.1. **The precise analytic input and its certificates.** Use the unnormalized Mellin transform

$$F(s,\alpha) = \Gamma(s)(2\pi)^{-s} \sum_{n\ge 1} f_n n^{(k-1)/2-s} e(n\alpha).$$

The following previously established continuation result is the input to the new finishing argument. In particular, its possible pole is retained; entireness of every additive twist is not assumed.

Lemma 47.2. *The finite source-closed system on 607 rational coordinates contains $F(s,1/9)$ among its continued observations. Every continued constant source observation is holomorphic away from $s=k$ and has at most a simple pole there. All these observations decrease faster than every power of $|\Im s|$ at large imaginary height on every fixed finite vertical strip.*

Here is the certificate chain for the meromorphy and pole assertions; the strip assertion is proved below. This specifies the full observable family used in the proof, including its exceptional parameter strata. Put

$$h = s - k/2, \quad (x,y,z,w) = (2^h, 3^h, 5^h, 7^h), \quad (a,b,d) = (\sqrt{2}f_2, \sqrt{5}f_5, \sqrt{7}f_7), \quad u = x^2,\ t = z^2.$$

Glue the original 270- and 378-grids along their common 54-grid and adjoin the 112 selected original mixed-prime rows specified in `level27_defects_crossprime_rows.json`. This gives 607 distinct coordinates. Every row is an original known-matrix, Euler, or Taylor–Fricke row. The positive-order source supports are constant rows; their span, together with the 54 core coordinate rows, is denoted by $\mathcal{O}$.

The exact continuation charts have rank 550 and contain $\mathcal{O}$ in their leading row spaces. They are regenerated for both signs with a generic and separately $a = 0, 1, -1$. The only spectral pivot divisors are $u-1$ and x^2+ax-1, with $u+1$ also occurring at $a=0$. The constant divisors a and a^2-1 are covered by the separate charts; there are no exclusions on b or d. Source closure, not just a rank calculation, gives meromorphic continuation of the actual observations.

For completeness, the successive leading pole constraints use

$$\begin{aligned} P &= -(u^2-u-1)(t+zb) + 4u^2 - 7u + w^2 - wd, \\ K &= u^3 - 4u^2 + 6u - 1, \\ Q_* &= K(t+zb) - 3t - 4u^3 + 7u^2 - 3u + 1. \end{aligned}$$

On $a = x^{-1} - x$ the ranks and the remaining pivot divisors are

additional equations	rank	divisors
none	550	$u-1, P$
$P=0$	550	$u-1, Q_*$
$P = Q_* = 0$	549	$u-1$

The last chart uses $K \neq 0$. On its kernel every source observation is a scalar multiple of the anchor $F(s,1/9)$. Augmenting the leading matrix by $\mathcal{O}$ restores rank 550, so an actual observable pole admits a local nuisance lift of the same order. No pole of an uncontinued auxiliary coordinate is silently used.

Here is the local lifting argument. Let $A(s)$ be the leading matrix and let the rows of the constant full-row-rank matrix M span $\mathcal{O}$. Write $d = \operatorname{rank} M$ and choose constant coordinates (U,V) with $U = MF$. In these coordinates write $A = (A_U, A_V)$. Generically A and its augmentation by M have rank $r = 550$, so A_V has rank $r-d$. The same augmentation rank at the point under consideration implies that A_V still has rank $r-d$ there. An invertible minor therefore gives an analytic right inverse on its image in a neighborhood of that point. Apply it to $H - A_U U$, where H is the holomorphic

lower-argument term in the original row equations. These equations are consistent on a punctured neighborhood by the continued observable identities; the remaining equations follow by analytic continuation. Thus one can choose a meromorphic V whose pole order is no greater than that of U. This lift preserves every positive Taylor term, since each such term is a scalar multiple of an observation in $\mathcal{O}$.

The two lower-argument first-jet certificates give, apart from their explicit localization divisors and the common factor $s-k$, $R(u,t)=S(u,t)=0$, where

$$\begin{aligned}R(u,t)&=7u^5-98u^4+546u^3-1467u^2+1622u-240+(24u-88)t,\\ S(u,t)&=35u^7-490u^6+2450u^5-9175u^4+29470u^3-40120u^2+4560u\\&\quad+(-264u^3+456u^2+2176u+1344)t.\end{aligned}$$

Their exact resultant is

$$\operatorname{Res}_t(R,S)=448(u-6)(u-4)K(u)(6u^3-14u^2-u-30).$$

The actual values satisfy $|t|=|u|^{\log_2 5}$. We give the root exclusions explicitly. Put $\alpha=\log_2 5$, so $9/4<\alpha<5/2$ and $|t|=|u|^\alpha$. The lower inequality follows from $5^4>2^9$, and the upper one from $5^2<2^5$. If $u=6$, then $R=0$ gives $t=-255/7$, whereas $(255/7)^4<6^9$ contradicts the lower bound for α.

Put $C(u)=6u^3-14u^2-u-30$. Its discriminant is -1249220, so it has one real root r and two conjugate nonreal roots. The signs at the rational endpoints give $74/25<r<2961/1000$. Reduction of $R=0$ modulo C gives

$$t=\frac{A(u)}{B(u)},\qquad A(u)=\frac{16565}{27}u^2-\frac{137495}{108}u-\frac{27775}{18},\qquad B(u)=24u-88.$$

On this real interval $B<0$, and both $A+3B$ and $A+4B$ are increasing. Their endpoint values give $A+3B>0$ and $A+4B<0$; hence $-4<t<-3$. But $r^2>8$ and $\alpha>2$, a contradiction. For either nonreal root, the product of the three roots gives $|u|^2=5/r\in(1,7/4)$. Eliminating u gives

$$1741824t^3-39678080t^2+1318180025t+5683863750=0.$$

The product of its three conjugate values is $-5683863750/1741824$. Since the real conjugate has absolute value less than 4, each nonreal conjugate has absolute value greater than 20. On the other hand, $|t|=|u|^\alpha<(7/4)^{5/4}<3$, where the last inequality follows from $(7/4)^5<3^4$. This excludes $C(u)=0$.

The factor $K(u)=u^3-4u^2+6u-1$ is treated directly with the leading relation $Q_*=0$, which gives $t=-3u^2+7u-1$ when $K=0$. The discriminant of K is -139, and its real root lies in $(3/16,19/100)$. On this interval $r^2<-3r^2+7r-1<9/40$. This contradicts $|t|=r^\alpha<r^2$. For the nonreal roots, $1<|u|^2=1/r<16/3$. Their corresponding t values satisfy

$$t^3-13t^2+76t-16=0,$$

so $|t|^2=16/t_{\mathrm{real}}>640/9$. Consequently

$$|t|^4>(640/9)^2>(16/3)^5>|u|^{10},$$

contrary to $|t|^4=|u|^{4\alpha}<|u|^{10}$.

The second first-jet chart also excludes the divisor $D=u^2-24u+112$. On this divisor the first chart is regular and still gives $R=0$. At its roots the values are

$$u=12\pm4\sqrt{2},\qquad t=\frac{-2551121\mp1790547\sqrt{2}}{337}.$$

For the smaller root, $u > 63/10$ and $|t| < 57$; thus $|t|^4 < 57^4 < (63/10)^9 < u^{4\alpha}$. For the larger root, $u < 18$ and $|t| > 15000$; thus $|t|^2 > 15000^2 > 18^5 > u^{2\alpha}$. The stated bounds follow by substituting $141421/100000 < \sqrt{2} < 141422/100000$.

At the remaining first-jet resonance $u = -2$, the dedicated regular chart gives $R(-2, t) = 0$, hence $t = -1939/17$. This contradicts $|t| = 2^\alpha = 5$. Therefore the remaining nonresonant possibility is $u = 4, t = 25$. It follows that $\operatorname{Re} h = 1$. The two exponential equalities also give $\operatorname{Im} h \log 2, \operatorname{Im} h \log 5 \in \pi\mathbb{Z}$; unique factorization implies $\log 2/\log 5 \notin \mathbb{Q}$, so $\operatorname{Im} h = 0$ and $h = 1$.

The boundary $u = 1, a \neq 0$ is removed by original-row identities, with separate charts at $a = \pm 1$. At $a = 0, u = 1$, the two regular charts have possible factors $-2bz + z^2 - 1$ and then $32z^2 - 185$; at $a = 0, u = -1$ they have $2bz + 3z^2 + 1$ and then $2768z^2 + 129405$. Since $|z| = 1$ on these boundary strata, the second factors do not vanish. Every chart includes every row of $\mathcal{O}$.

At $h = 1$ the forced eigenvalues are $(a, b, d) = (-3/2, -3, -25/7)$. The prescribed whole-germ fourth-jet elimination has 557 regular pivots for each sign and final scalar

$$\frac{636669859}{11796480000} \neq 0$$

on the visible pole direction. It removes this point for $k \geq 4$. At $k = 2$, clearing its layer normalizations introduces at most a simple pole and therefore bounds the pole order by one. More generally, at $s = k$, $k \geq 4$, the first-jet coefficient is $(s - k)$ times a nonzero scalar: positivity of $R(2^k, 5^k)$ supplies nonvanishing. Thus at $s = k$ in every weight the pole is at most simple.

Finally $\binom{s+m-k}{m}$ cancels the possible simple source pole at $s + m = k$. Descending from a right half-plane of holomorphy therefore produces no poles at lower integer shifts. This proves precisely the pole assertion of Lemma 47.2. The consolidated localization, full-source, and fourth-jet records are, respectively:

```
level27_defects_uniform_pole_audit.json
level27_defects_fullsource_jets_audit.json
level27_root_progress_audit.json
```

The last record contains the independent replays; all its 23 recorded source hashes were checked unchanged before applying the new argument.

47.2. **Vertical bounds and sectorial cusp asymptotics.** We now prove the strip assertion, rather than assuming that meromorphic continuation supplies bounds for Mellin inversion.

In the Taylor–Fricke construction use a smooth cutoff near zero. On the original inversion line $\Re w = B$, logarithmic derivatives of the order-M Taylor remainder of $(1 + iy/r)^{w-k}$ satisfy

$$\left|(y\partial_y)^j \mathcal{R}_M(w, y)\right| \ll_{M,j} y^M (1 + |\Im w|)^{M+j} e^{Cy|\Im w|}.$$

The cutoff can be chosen small enough for the original gamma decay to dominate the exponential. Hence the integrated remainder and all its logarithmic derivatives are $O_{M,j}(y^{M-B})$. Smooth cutoff errors and the removed tails are smooth and decrease exponentially at infinity. Repeated Mellin integration by parts shows that each holomorphic remainder $H_M(s)$ is rapidly decreasing on compact substrips of $\Re s > B - M$. Entire Ramanujan rows and known-matrix rows sending the centre to infinity have the same rapid bounds, since their exact transformation laws give exponential decay at both ends of the Mellin integral. Exact Euler rows have zero remainder. Thus every original row is covered.

We also need a division observation. For a nonzero exponential polynomial $D(s) = \sum c_\nu e^{\lambda_\nu s}$ with distinct real λ_ν, the family $D(s + it)$ has compact closure on compact sets, and no limit is identically zero. Around every horizontal centre in a compact interval one can choose a circle of radius between r and $2r$ avoiding the zeros of a given limit. Compactness supplies finitely many choices and a uniform positive lower bound on an available circle. If H is holomorphic and DH is rapidly decreasing on an enlarged strip, the maximum principle on that circle proves the same

bound for H on the smaller strip. Finitely many poles of H do not affect this argument at large imaginary height.

Choose independent original leading rows $B(s)$ and put $Y(s) = B(s)\mathbf{F}(s)$ in a valid fixed-eigenvalue chart. Each component of Y is entire: its original Taylor identity has a holomorphic remainder, and each positive source pole is canceled by $\binom{s+m-k}{m}$. The source-closure identities give

$$Y(s) + \sum_{m=1}^{M-1} C_m(s) Y(s+m) = H_M(s).$$

Every entry of C_m is a polynomial in s times a rational function of positive real exponentials. Clearing denominators, then using the division observation, propagates rapid decay from the original right half-plane to successive strips of width $1/2$. All positive shifts of a slightly enlarged new strip are already controlled. The separate eigenvalue charts guarantee that no denominator is identically zero. Every desired constant observation is a rational-exponential combination of Y, so the same argument proves its rapid decay at large imaginary height. This finishes Lemma 47.2.

Write $r = \operatorname{Res}_{s=k} F(s, 1/9)$, allowing $r = 0$. Mellin inversion and a contour shift now give

$$f(1/9 + iy) = r y^{-k} + O_L(y^L) \qquad (y \downarrow 0)$$

for every L. To extend this to nonvertical approaches, put $H(\zeta) = f(1/9 + \zeta) - r(-i\zeta)^{-k}$. The original coefficient bound gives polynomial growth for H in each compact angular sector. Fix a compact set $K \subset \mathbb{H}$ and a bounded domain Ω compactly inside $\mathbb{H}$ containing K and the segment $I = [i, 2i]$. For $H(\rho w)$ the outer boundary bound is $O(\rho^{-B_1})$, whereas the bound on I is $O_N(\rho^N)$ for every N. The harmonic measure of the slit I in $\Omega \setminus I$ has a positive lower bound on K away from the slit; on the slit the stronger direct estimate holds. The two-constants theorem, with N arbitrarily large, therefore gives

$$\sup_{w \in K} |H(\rho w)| = O_{K,L}(\rho^L).$$

This argument uses the original polynomial bound. It does not insert complex arguments into Mellin inversion using only Schwartz bounds.

47.3. **The obstruction and its exact affine Hecke relation.** Set

$$g = f|\operatorname{diag}(1,3), \quad R = (1,1;0,1), \quad S = (1,0;3,1), \quad W = (0,-1;3,0), \quad \eta = i^k \epsilon_f.$$

The ramified coefficient condition, its Fricke conjugate, and the automatic Hecke–2 transformation give

$$g|(1 + R + R^2) = g|(1 + S + S^2) = g|(SR - RS) = 0, \qquad g|W = \eta g.$$

For example, the known Hecke–2 matrix after rescaling is $(2, -3; -9, 14) = -S^{-2}RS^{-1}R^{-1}$ projectively; $g|S^{-2} = g|S$ then gives the displayed commutator relation. Put

$$A = g|R(SR - RS) = g|S(1 + R + R^2), \quad B = g|S(SR - RS), \quad U = R^{-1}S.$$

The exact identity $SR^{-1}S = RS^{-1}R$ and $U^3 = 1$ give $B = -A|U$. Indeed, expansion gives $A|(1 + U) + B|(1 + U^{-1}) = 0$; multiply by $(1 - U + U^2)/2$. Fricke conjugation then gives

$$A|J = -\eta A, \qquad J = WSR = (-3, -4; 3, 3).$$

These relations are proved at the cyclic vector and its explicit right translates, not imposed at arbitrary translates.

Let $\lambda = \sqrt{2} f_2$ and

$$H_2 = (1,0;0,2) + (1,3;0,2) + (2,0;0,1).$$

The original Hecke equation is $g|H_2 = \lambda g$. Its exact consequence for A is

$$\lambda A = A|\big((1,-1;0,2) - (2,0;0,1) + (2,1;0,1) - (2,2;0,1)\big). \tag{20}$$

This is a derived equation, not an assertion that an arbitrary defect inherits the Hecke eigenvalue. Here are reproducible details of the derivation. For the states (g, gR, gS, gRS) the right actions are

$$v|R = (v_1, -v_0 - v_1, v_3, -v_2 - v_3 + A), \qquad v|S = (v_2, v_3, -v_0 - v_2, -v_1 - v_3 + A|U).$$

Write $r_0 = R^{-1}$, $s_0 = S^{-1}$ for letters in the following table. Let $(\alpha_0, \alpha_1, \alpha_2)$ be the three representatives in H_2. The entry (j, γ) means $\alpha_i S R^t = \gamma \alpha_j$:

$t\backslash i$	0	1	2
0	$(0, S^2)$	$(2, R^2 s_0 R)$	$(1, R s_0 r_0)$
1	$(1, S^2 r_0)$	$(2, R^2 s_0 R^3)$	$(0, R s_0 R)$
2	$(0, S^2 R)$	$(2, R^2 s_0 R^5)$	$(1, R s_0)$

Expansion with the state actions and their inverses cancels every state coefficient separately for each α_j. The remaining A-terms are exactly the four terms in (20). The exported 45-term certificate expresses the difference in the original right ideal generated by $1 + R + R^2$, $1 + S + S^2$, $SR - RS$, and $H_2 - \lambda$, over $\mathbb{Q}[\lambda]$. `level27_root_affine_check.py` independently checks both the nine matrix factorizations and this certificate. There is no division by λ or an eigenvalue polynomial.

In ordinary function notation, replacing the argument in (20) by $2z + 1$ gives

$$\boxed{A(z) = 2^{k/2}\lambda A(2z+1) + 2^k\big(A(4z+2) - A(4z+3) + A(4z+4)\big).} \tag{21}$$

47.4. **Dilation rigidity and cuspidality.** The sectorial expansion at $1/9$ has a particularly simple consequence:

$$(g|S)(z) = 3^{-k/2}(3z+1)^{-k} f\left(\frac{1}{9} - \frac{1}{9(3z+1)}\right) = c + O_L((\Im z)^{-L}), \quad c = i^{-k}3^{3k/2}r.$$

This holds as $\Im z \to \infty$ in every fixed cone $|\Re z| \le C\Im z$. Hence $A(z) = c_A + O_L((\Im z)^{-L})$, where $c_A = 3c$. Taking the limit in (21) gives

$$c_A = (2^{k/2}\lambda + 2^k)c_A.$$

Thus $D = A - c_A$ satisfies the same homogeneous recurrence, even at the exceptional eigenvalue $\lambda = 2^{-k/2} - 2^{k/2}$.

Fix $z = x + iy \in \mathbb{H}$. Every composition of n affine maps in (21) is $z \mapsto mz + b$, with $m \ge 2^n$ and $0 \le b/m \le 2$. All images therefore remain in one fixed cone at infinity. Put $C_0 = 2^{k/2}|\lambda| + 3 \cdot 2^k$. Iteration gives

$$|D(z)| \le C_0^n C_{z,L}(2^n y)^{-L}.$$

Choose L so that $2^L > C_0$ and let $n \to \infty$. This proves $D = 0$. The obstruction A is constant. Since $k > 0$ and J has nonzero lower-left entry, the relation $A|J = -\eta A$ forces that constant to be zero.

It remains to spell out the modularity conclusion. With $A = B = 0$, the two state actions above commute and have order three. The group $\Gamma_0(3)/\{\pm I\} = \langle R, S\rangle$ acts on their span through $\mathbb{F}_3^2$. For $\gamma = (a, b; c, d)$ choose $\varepsilon = \pm 1$ with $a \equiv d \equiv \varepsilon \pmod 3$. The homomorphism

$$\gamma \longmapsto (\varepsilon b, \varepsilon c/3) \pmod 3$$

has kernel $\Gamma^0(3) \cap \Gamma_0(9)$ projectively, and this kernel fixes g. Undoing the rescaling gives $\Gamma_0(27)$ invariance of f.

Every vector in the four-state span has period three and zero trace $1 + R + R^2$. The span is also stable under W; the two cusp classes of $\Gamma_0(3)$ are represented by infinity and zero. At every cusp the corresponding transform of g lies in this span. Polynomial growth excludes negative Fourier powers there, and the zero trace excludes the constant coefficient. Thus all cusp expansions vanish. Conjugation proves $f \in S_k(\Gamma_0(27))$, completing Theorem 47.1.

What replaces the quadratic-twist functional equation. The finishing argument uses the already established continuation and simple-pole bound at the additive cusp 1/9, not an additional functional equation for f_{χ_5}. It does not assert that abstract entireness of a character twist, without the original hypotheses and this source system, implies a functional equation. In the present problem the missing quadratic-five equation now follows a posteriori from $A = 0$ and the previously checked identity $g|Z_5 = A|(M - N)$.

48. A finite-pole-chain consequence of Taylor–Fricke

Use the unnormalized Mellin variable throughout this note. Write

$$f(z) = \sum_{n\geq 1} a_n e(nz), \qquad D_\alpha(s) = (2\pi)^{-s}\Gamma(s)\sum_{n\geq 1} a_n e(n\alpha) n^{-s}.$$

Assume polynomial growth of a_n, and the exact Fricke identity

$$f(-1/(Nz)) = \omega N^{k/2} z^k f(z), \qquad \omega \in \{1, -1\},$$

where $k > 0$ is even. The hypotheses of [3] imply this with $a_n = f_n n^{(k-1)/2}$ and $\omega = \epsilon_f i^k$.

Lemma 48.1 (Bounded strips at a Fricke pair)**.** *Let $\alpha \in \mathbb{Q}^\times$ and $\beta = -1/(N\alpha)$. Suppose that both D_α and D_β have meromorphic continuations to $\mathbb{C}$, with all their poles in some bounded vertical strip. Then every pole of either function is simple and belongs to $\{k, k+1, k+2, \dots\}$.*

Proof. Fix a class $\rho + \mathbb{Z}$ in $\mathbb{C}/\mathbb{Z}$. There are only finitely many poles in that class, by the strip hypothesis. Form its finite formal singular expansion

$$P_\alpha(y) = \sum_{s_0 \in \rho+\mathbb{Z}} \operatorname*{Res}_{s=s_0} \big(D_\alpha(s) y^{-s}\big),$$

and similarly P_β. Thus each P is a finite sum of y^{-s_0} times polynomials in $\log y$. This definition concerns only principal parts; no assertion about a global inverse Mellin contour, or growth of all poles together, is necessary.

For $z = \beta + iy$ the Fricke identity gives exactly

$$f(\beta + iy) = C(1 + cy)^{-k} f\left(\alpha + i\frac{ry}{1+cy}\right), \quad C = \omega N^{-k/2}\beta^{-k}, \quad c = i/\beta, \quad r = 1/(N\beta^2).$$

Its Taylor–Mellin expansion, to arbitrarily high finite order, therefore gives the identity of formal logarithmic Laurent expansions

$$P_\beta(y) = C(1 + cy)^{-k} P_\alpha\left(\frac{ry}{1+cy}\right). \tag{22}$$

For completeness, this is the principal-part version of the Taylor–Fricke relation in Booker–Krishnamurthy [2, Lemma 2.3]. Expand the right-hand side to order M; its Mellin transform is a finite combination of $D_\alpha(s+m)$, $0 \leq m < M$, and a remainder holomorphic on $\Re s > K - M$, for a fixed K determined by coefficient growth. Taking principal parts at any fixed $s \in \rho + \mathbb{Z}$ and then taking M sufficiently large gives (22). Terms from another class modulo $\mathbb{Z}$ cannot contribute. A pole of order h gives precisely a polynomial of degree $h - 1$ in $\log y$, so this argument keeps all pole orders, not just residues.

Both sides of (22), as now expressed by finite sums, have convergent expansions in a sufficiently small punctured sector at 0. Consequently their formal equality is an analytic equality there. Continue that equality to a punctured neighborhood of $y_0 = -1/c$, avoiding 0 and choosing local branches. The left-hand side is holomorphic at $y_0 \neq 0$. On the right, put $w = 1 + cy$. A term with exponent s_0 has the form

$$C(ry)^{-s_0} w^{s_0-k} Q_{s_0}(\log(ry) - \log w),$$

where Q_{s_0} is a polynomial. All exponents in our fixed class differ by integers. Starting with their smallest exponent, ordinary local monodromy and Laurent expansion show that holomorphy at $w = 0$ is possible only when every surviving exponent $s_0 - k$ is a nonnegative integer and every Q_{s_0} is constant. One can also see this by looking at the leading term at infinity of the finite sum $P_\alpha(x)$: it must be a Laurent polynomial in x^{-1} vanishing to order at least k at infinity. Distinct exponents and logarithmic degrees cannot cancel its leading term. Iteration gives the assertion for every term. Thus poles of D_α are simple integers at least k. Interchange α and β to obtain the same conclusion for D_β. □

Corollary 48.2 (Pole cancellation after reflection)**.** *Let $T(s)$ be a finite linear combination of additive twists $D_{\alpha_j}(s)$, possibly multiplied by nowhere-vanishing entire exponentials in s. Suppose that each D_{α_j} and its Fricke partner $D_{-1/(N\alpha_j)}$ satisfy the bounded-strip hypothesis of the lemma. If*

$$T(s) = E(s)\widetilde{T}(k - s)$$

where E is entire and nowhere zero and $\widetilde{T}$ has the same kind of representation and hypotheses, then T and $\widetilde{T}$ are entire.

Proof. The lemma puts every possible pole on the integers at least k. The functional equation would simultaneously put its reflection at an integer at most 0. Since $k > 0$, no pole is possible. □

Scope of this auxiliary criterion. This lemma does not establish bounded-strip meromorphy of the individual additive twists. In particular, knowing only that $D_\alpha + D_{-\alpha}$ is meromorphic in a bounded strip is insufficient for this proof: odd Taylor terms involve $D_\alpha - D_{-\alpha}$. Therefore the meromorphic quadratic χ_5 twist obtained at level 19 cannot simply be inserted in this corollary. Recovering the required individual twists, or proving a counterpart that controls the missing odd parts, would be necessary to use this auxiliary corollary alone. The level-19 proof above instead establishes individual grid continuation and uses polynomial identities between whole meromorphic germs. The level-25 proof above uses its enlarged source-free quotient and retains possible cusp constants. Neither completed proof relies on applying this corollary without its hypotheses.

49. The general modular identification and exact count

This section applies at every level, independently of the individual small-level converse arguments. We use the normalized coefficients and critical line $\operatorname{Re} s = 1/2$ of the accompanying paper [3]. Write $\mathscr{E}(N, k)$ for its solution set and

$$\iota((f_n))(z) = \sum_{n\geq 1} f_n n^{(k-1)/2} e^{2\pi i n z}.$$

Here $M_k^\circ(\Gamma_0(N))$ consists of the modular forms whose traces, and Fricke-conjugate traces, to every level N/p vanish, and which vanish at 0 and ∞. An eigenform is always normalized to have first Fourier coefficient 1. Hecke operators include U_p at $p \mid N$.

Lemma 49.1 (An admissible normalized Hecke eigenbasis)**.** *For every $N \geq 1$ and positive even k, there is a basis $\mathcal{B}_{N,k}$ of $M_k^\circ(\Gamma_0(N))$ consisting of normalized simultaneous Hecke eigenforms. Its joint Hecke eigenspaces are one-dimensional. The normalized coefficient sequence of every member of $\mathcal{B}_{N,k}$ belongs to $\mathscr{E}(N, k)$.*

Proof. On the cuspidal summand the two trace kernels give the usual newspace, and the assertion is the classical normalized newform theorem. The Eisenstein summand requires the trace-kernel definition of newness; in particular one must retain its additional weight-two lines. The packetwise

trace calculation below gives the following explicit list. For a primitive Dirichlet character χ of conductor $c > 1$, put

$$E_\chi(z) = \sum_{n\geq 1} \left(\sum_{d|n} \chi(n/d)\overline{\chi}(d) d^{k-1} \right) e^{2\pi i n z}.$$

For $k \geq 4$ the surviving Eisenstein eigenforms are E_χ with $c^2 = N$. For $k = 2$ they are

$$E_{\chi,t} = \prod_{p|t} (1 - \chi(p) p V_p) E_\chi, \qquad N = c^2 t,$$

where t is squarefree, $(t, c) = 1$, $\chi(p)^2 = 1$ for every $p \mid t$, and $V_p F(z) = F(pz)$. Each displayed form has first coefficient 1. The principal packet contributes nothing: any surviving holomorphic line has nonzero constant term at infinity, while in weight two at level one the principal packet has no holomorphic vector.

Here is the packetwise trace calculation. Write $A_d = \mathrm{diag}(d, 1)$ and use determinant-normalized slash operators, so that $H|_k A_d = d^{k/2} H(dz)$. Fix a primitive character χ of conductor $c > 1$. Its packet at level $N = c^2 D$ has basis $E_\chi|A_d$, $d \mid D$, and

$$(E_\chi|A_d)|W_N = \eta_\chi E_\chi|A_{D/d}.$$

Thus Fricke reverses each local degeneracy basis. At a prime dividing c, the traces of the primitive vector to the preceding level vanish by character orthogonality in the trace sum. All vectors in this packet have zero constants at 0 and ∞.

First let $p \nmid c$, freeze the other prime factors, and put $v_j = E_\chi|A_{p^j}$ for $0 \leq j \leq e$. In the paper's normalization set

$$f_p = p^{-(k-1)/2}\chi(p) + p^{(k-1)/2}\overline{\chi}(p), \qquad t_p = \sqrt{p}\, f_p.$$

At $e = 1$, the two trace equations have matrix $\left(\begin{smallmatrix} p+1 & t_p \\ t_p & p+1 \end{smallmatrix}\right)$. It is singular precisely when $t_p = \pm(p+1)$. Writing $\chi(p) = e^{i\theta}$, the imaginary part of t_p is $(p^{1-k/2} - p^{k/2}) \sin\theta$. Hence singularity forces $\chi(p) = \pm 1$, and then $|t_p| = p^{1-k/2} + p^{k/2} = p + 1$ forces $k = 2$. The common kernel is therefore the line

$$v_0 - \chi(p) v_1 = E_\chi - \chi(p) p E_\chi(pz).$$

For $e \geq 2$, the trace to the preceding level satisfies

$$\mathrm{Tr}(v_j) = p v_j \quad (j < e), \qquad \mathrm{Tr}(v_e) = t_p v_{e-1} - v_{e-2}.$$

These formulas follow from the coset identity

$$\mathrm{Tr}_L^{Lp} H = p^{1-k/2} \big((H|W_{Lp})|U_p \big) |W_L^{-1} \quad (p \mid L),$$

where U_p is the usual, unnormalized coefficient operator, together with

$$U_p E_\chi = p^{(k-1)/2} f_p E_\chi - p^{k/2-1} E_\chi|A_p.$$

The first trace kernel is the line generated by

$$w_e = v_e - (t_p/p) v_{e-1} + p^{-1} v_{e-2}.$$

The second kernel is its Fricke reflection. For $e \geq 3$ their endpoint supports differ. For $e = 2$, the ratios of the coefficients of v_2 and v_0 are respectively p and p^{-1}. Thus their intersection is zero for every $e \geq 2$.

If $p \mid c$ and $e > 0$ is an exponent added to the primitive level, then $U_p E_\chi = 0$. The same coset formula gives $\mathrm{Tr}(v_j) = pv_j$ for $j < e$ and $\mathrm{Tr}(v_e) = 0$. The two kernels are now $\mathbb{C}v_e$ and $\mathbb{C}v_0$, with zero intersection.

The local operators act on separate tensor factors of the divisor-indexed packet. Their joint kernel is consequently the tensor product of these local kernels. This proves exactly the list of E_χ and $E_{\chi,t}$ above, including the exclusion of repeated added primes and of increases at primes dividing c.

For the principal packet in weight two, use the almost-holomorphic completion $E_2^* = E_2 - C/y$. Every $E_2^*|A_d$ has the same nonholomorphic term $-C/y$, so a combination is holomorphic precisely when its coefficients sum to zero. The same trace calculation leaves a joint kernel only at squarefree N, generated by

$$\prod_{p|N}(1 - A_p)E_2^* = \prod_{p|N}(1 - pV_p)E_2 \qquad (N > 1).$$

Its constant at infinity is $c_0(E_2)\prod_{p|N}(1-p) \neq 0$, so it does not belong to M_2°. At $N = 1$ the sole vector E_2^* is not holomorphic. For even $k \geq 4$, the principal packet has a joint trace kernel only at level one, where its nonzero constant term again excludes it.

The displayed divisor sums give the unramified Euler factors directly. At $p \mid c$ their local factor is 1. At an added prime $p \mid t$ in weight two, the multiplier $1 - \chi(p)pV_p$ removes the root $\chi(p)p$ of the classical Euler factor, leaving classical U_p eigenvalue $\chi(p)$ and hence normalized coefficient $f_p = \chi(p)p^{-1/2}$. Thus the required ramified conditions hold. The primitive Eisenstein form is a Fricke eigenfunction; on each added weight-two line Fricke contributes the sign $-\chi(p)$. Consequently every displayed form is a Fricke eigenfunction. If its Fricke scalar is η, then $\eta^2 = 1$ and the completed functional equation has sign $\epsilon = (-1)^{k/2}\eta \in \{\pm 1\}$. Its constants at both 0 and ∞ vanish. The Mellin integral split at the Fricke height therefore gives an entire finite-order completion. Polynomial Fourier growth gives convergence on a right half-plane.

Finally, for simultaneous Hecke eigenforms in M_k° the coefficient identity $a_n(F) = a_1(T_n F)$, with the usual U_p convention at ramified primes, shows that all coefficients are determined by the eigenvalues and $a_1(F)$. Hence every nonzero simultaneous eigenform has $a_1 \neq 0$, and a joint eigenspace has dimension at most one. This also proves that the normalized members of the basis are canonically determined and have distinct coefficient sequences. □

Lemma 49.2 (The modular members of the Euler solution set)**.** *At every level and positive even weight,*

$$\iota(\mathscr{E}(N,k)) \cap M_k(\Gamma_0(N)) = \mathcal{B}_{N,k}.$$

In particular, any converse theorem whose conclusion is merely $\iota(\mathscr{E}(N,k)) \subseteq M_k(\Gamma_0(N))$ implies

$$\iota(\mathscr{E}(N,k)) = \mathcal{B}_{N,k}, \qquad \#\mathscr{E}(N,k) = \dim M_k^\circ(\Gamma_0(N)).$$

Proof. Let $F = \iota((f_n))$ be modular. The Euler product gives $f_1 = 1$, multiplicativity and the prime-power recurrences. Comparing Fourier coefficients in the Hecke formulas makes F a simultaneous eigenform. The constant term of F is zero, and its original Fricke relation makes the constant term of $F|W_N$ zero as well.

We recall the trace argument to make the newness assertion explicit. Write $N = pM$ and put $R_a = \left(\begin{smallmatrix} 1 & a \\ 0 & p \end{smallmatrix}\right)$. With determinant-normalized slash operators, the ramified recurrence is

$$\sum_{a \bmod p} F|_k R_a = \delta_p F, \qquad \delta_p = \sqrt{p}\, f_p.$$

If $p \nmid M$, the Fricke relation and this recurrence, together with $\Gamma_0(N)$-invariance, give the usual local Atkin–Lehner relation $F|W_p = -\delta_p F$. The $p+1$ trace representatives are the identity and $W_p R_a$,

up to $\Gamma_0(N)$ and central factors. Thus

$$\operatorname{Tr}_M^N F = F + \sum_{a \bmod p} F|W_p R_a = (1-\delta_p^2)F = 0.$$

If $p \mid M$, then $U_p F = 0$, and the trace-coset formula is

$$\operatorname{Tr}_M^N F = p^{1-k/2}\big((F|W_N)|U_p\big)|W_M^{-1} = 0.$$

Since $F|W_N$ is a scalar multiple of F, its trace also vanishes. These identities establish $F \in M_k^\circ(\Gamma_0(N))$ without assuming cuspidality. The one-dimensional joint eigenspaces in Lemma 49.1, and $a_1(F) = 1$, now give $F \in \mathcal{B}_{N,k}$. The reverse inclusion is exactly the last assertion of that lemma. Fourier uniqueness makes ι injective, so counting the basis proves the final equality. □

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